

STANDARD OPERATING PROCEDURE

DOCUMENT NAME	EQUIPMENT CLEANING AND VALIDATION OF CLEANING PROCEDURES		
DOCUMENT NO.	1035-A-0072		
VERSION NO.	10.6		
DATE OF ISSUE	24-MAY-2025	EFFECTIVE DATE	

Reference Change Request No.	CR-1035-D-2026-00380
PR No.	PR# 842807

Change History:

Page	Point	Change
18	2.16.4	Point newly incorporated "Common ECR shall be designed for equipment with a similar design irrespective of their capacities. For example, reactor of different capacities shall be cleaned with the same ECR using similar cleaning techniques."
18	2.16.5	Point newly incorporated, "In such cases solvent quantity chart shall be attached to the ECR to accommodate different capacities of equipment".
20, 21	2.18.12	At Point 2.18.12, Limits for Rinse and swab updated as, $\text{Rinse Limit (ppm) for equipment} = \frac{\text{MACO (mg)} \times \text{Area Sampled by Rinse Sampling (inch}^2\text{)} \times 1000}{\text{Surface Area of Equipment Chain (inch}^2\text{)} \times \text{Rinse sampling volume (ml)}}$ $\text{Swab limit (ppm)} = \frac{\text{MACO} \times \text{Swab Surface Area} \times 1000}{\text{Surface Area of Equipment Chain} \times 10 \text{ (ml)}}$ OR $\text{Swab limit (}\mu\text{g/swab)} = \frac{\text{MACO} \times \text{Swab surface area} \times 1000}{\text{Surface Area of Equipment Chain}}$ Same is updated in annexure 1035-A-0072/A3 "Cleaning matrix for API".
24	2.19.9	Point incorporated as, "Based on the calculation, if same RPN score is obtained for more than one product than lower PDE value API shall be considered as a worst-case."
NA	NA	Point (SOP 1035-A-0072 version 10.0) "Grouping is classifying different pieces of equipment together. If equipment is to be grouped, it must be cleaned with the same cleaning SOP" removed.
NA	NA	Point (SOP 1035-A-0072 version 10.0) Grouping shall be performed on pieces of equipment with a similar design having either same or different capacity. Three-cleaning validation runs shall be performed on the equipment group / optional equipment, by ensuring at least one cleaning run involves the largest size and at least one run involves the smallest size" removed.

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31	2.28	Point updated to incorporate, "If ECR is same and solvent quantities are different depending on capacity: Three runs each shall be performed on lowest and highest capacity of equipment. For example, in case there are three blenders of different capacities 500 Liter, 1250 Liter and 2250 Liter (ECR is same and solvent quantities are different) then cleaning validation 3 runs each shall be performed on the lower capacity blender 500 Liter and higher capacity blender 2250 Liter. If ECR is same and solvent quantities are also same. Total three runs shall be performed on any of the optional equipment. For example, in case there are two reactors of capacities 1 KL and 1.5KL (ECR is same and solvent quantities are same) then three runs of cleaning validation shall be performed on any equipment combination" by removing "Three runs cleaning validation to be performed on both capacity equipment (lower and higher)".
32	2.29.4	Point updated to include term "block".
101 to 104	Annexure 1035-A-0072/A15	Annexure 1035-A-0072/A15 "Assessment for cleaning requirement for product containing genotoxic / nitrosamine compound" updated to revise action plan against the low risk category to evaluate the requirement of the controls stated in the cleaning for the impurities based on the control strategy of those impurities in the compounds.
105 to 108	Annexure 1035-A-0072/A16	Annexure 1035-A-0072/A16 "Protocol for Dirty Equipment Hold Time Study" updated to revise objective, procedure, acceptance criteria, non-compliance and frequency like in procedure "the equipment under prevailing condition for at least 3 days/ 72 hours" instead of "the equipment under prevailing condition for at least 4 days/ 96 hours".
109 to 112	Annexure 1035-A-0072/A17	Annexure 1035-A-0072/A17 "Report for Dirty Equipment Hold Time Study" updated to revise 'standard operating procedure (SOP) to be followed' and recommendation subsequently.
131 to 134	Annexure 1035-A-0072/A22	Annexure 1035-A-0072/A22 "Protocol for verification of cleaning solvent traces" procedure point 9.7 updated as, "Collect the rinse in amber coloured glass bottle, label it and send to Quality Control to determine the content of the solvent in the rinse which shall be analysed as per General Analytical Method for Residual Solvent (Reference document no. 1035-GAM-054-INH)" instead of Collect the rinse in amber coloured glass bottle, label it and send to Quality Control to determine the content of the solvent in the rinse as per the specific product GC residual solvent method. And point "If specific product GC solvent method does not have the solvent which are used for cleaning of equipment, then solvent traces shall be analysed as per General Analytical Method for Residual Solvent (Reference document no. 1035-GAM-054-INH)" deleted.

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PURPOSE :

- To describe procedure for Cleaning of equipment and accessories used in Manufacturing of drug substances and its intermediate and various cleaning approaches.
- To describe procedure for establishing acceptance criteria for cleaning and Validation of cleaning procedure to assure that cleaning procedures can consistently clean the equipment to a pre-determined level of cleanliness from the previous product and cleaning agent used in the cleaning.
- To describe the procedures for worst case selection, cleaning verification & cleaning re-validation.
- To describe the procedures to carry out dirty & cleaned equipment hold time study and Campaign cleaning studies.

SCOPE :

Applicable to equipment cleaning procedures used in manufacturing of Active Pharmaceutical Ingredients and intermediates at Cipla and Cipla subsidiary.

RESPONSIBILITY :

Production:

- To describe / develop the cleaning procedures.
- To be part of validation / verification team.
- To ensure and perform the cleaning activity as per the respective procedure for batch to batch and product changeover.
- To ensure dismantling and assembling of the equipment / parts.
- Cleaning shall be done by production operators with oversight by production officer.
- To ensure the effectiveness of cleaning by visual inspection.
- To ensure rinse / swab results as per sampling plan within acceptance criteria for product changeover.
- To ensure the overall cleanliness of the equipment prior product changeover / batch start up.

Quality Control:

- To be part of validation/ verification team.
- To ensure availability of validated analytical methods for measurement of product in swabs and rinses.
- To ensure availability of specifications for testing of rinse and swab samples.
- To analyse the cleaning validation / verification and product changeover samples as per TI sheet.

Quality Assurance:

- To be part of validation/ verification team.
- To coordinate with Production and Quality Control to prepare sampling plan and collection of samples in case of verification / validation.
- To establish worst case products.
- To collect validation samples as per approved sampling plan of equipment.
- To perform cleaning validation as per approved validation protocol and ensure completion of report.
- To ensure the Training details of the personnel involved in the cleaning of equipment, collecting of samples, testing of samples, deciding the sample plan, preparing, reviewing, approving the reports.
- To ensure training details of the personnel involved in validation /verification are available and included in the validation protocol / report.

R & D and MSTG

- R&D shall provide the Cleaning Master Process Guideline (CMPG) describing the outlines of cleaning process to manufacturing site during technology transfer of the API/Intermediate.

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- MSTG shall be involve in the cleaning related failure investigations to identify the scientific root cause.

ADL

- ADL shall develop and validate the analytical method based on residue to be cleaned prior to batch initiation.

Engineering

- Dismantling and assembling of the equipment / parts as per production request.

DEFINITION

:

Equipment Cleaning: This is a process to remove unwanted residues or remnants of previous product from manufacturing equipment.

Clean: *Product residue, process residue and environmental contaminants removed to an acceptable level.*

Cleanliness: *It is the state of being clean and free from product residue, and environmental contaminant and the habit of achieving and maintaining that state.*

Contaminant: It can be chemical or microbial contamination arising due to previous products (API / Intermediate / Chemical reagents), degradation products, residues of cleaning agents, airborne materials (e.g., dust and particulate matter), lubricants, disinfectants etc.

Cleaning Validation: A documented evidence that a cleaning procedure is consistent in reducing product residue and removal of cleaning agents, bioburden from equipment and accessories within the limits of acceptance.

Worst Case / worst case product: The product or condition that poses the highest risk of contamination or is most difficult to clean, selected to challenge the cleaning process to ensure its robustness and effectiveness across all routine operations. The product is selected based on its poor solubility, cleanability, potency or a combination of these factors.

Acceptance criteria: The specific limits and conditions that a cleaning procedure shall meet to demonstrate that residue of product, cleaning agent, and microbial contaminants are effectively removed to an acceptable level. The established limit must be calculated based on sound scientific rationale.

Revalidation: *After change, validation with three runs providing the assurance that the changes do not affect the effectiveness of cleaning process and produced results meets the predetermined acceptance criteria.*

Risk Assessment: A systematic process of organizing information to support a risk decision to be made within a risk management process. It consists of the identification of hazards and the analysis and evaluation of risks associated with exposure to those hazards.

Cleaning Verification: *The periodic repetition of initial validation with one batch / run provides the assurance that time do not affect the cleaning effectiveness by ensuring that specific equipment has been properly cleaned for product residue and removal of cleaning agent, bioburden from equipment and accessories.*

Permitted Daily Exposure (PDE): The PDE represents a substance and route-specific dose that is unlikely to cause an adverse effect if an individual is exposed at or below this dose, through the specified route (as an example oral, intravenous, dermal etc.), every day for a lifetime

Cleanability: *The ability of a cleaning procedure to effectively remove material, cleaning agent residue and microbial contamination.*

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Maximum Carryover (MACO): The maximum amount of carryover of a residual process residue (API, cleaning agent, degradant, and so forth) into the next product manufactured without presenting an appreciable health risk to patients.

Cleaning Agent: - A cleaning agent is a substance used to remove dirt, stains, Odors, and other contaminants from surfaces of equipment and objects.

Critical Cleaning Parameters: - A critical cleaning parameter which variability impact on critical cleaning attribute and therefore shall be monitored for control to ensure that cleaning process ensures cleaning to acceptable level.

Grouping strategy: A strategy for establishing similar cleaning processes, usually based on similar products or similar equipment, and to validate the cleaning process based primarily on validation data for a representative of the group.

Open type equipment/Portable Equipment: A equipment of which all accessories, sub parts and spare parts can be dismantled completely and moved during equipment cleaning to cleaning area are considered as open type equipment/Portable equipment. Some of the examples are equipment like Multi mill, Sifter, sparkler filter, Jet mill etc. falls in this category. This equipment typically is cleaned out of place (COP)

Closed type equipment/Fixed equipment: A equipment of which all accessories, sub parts and spares parts cannot be dismantled completely as the equipment is fixed at place during equipment cleaning can be considered as Closed type equipment/Fixed equipment. Equipment like Reactor, Blender, RCVD etc. falls in this category. This equipment typically is cleaned in place (CIP)

Therapeutic dose: It is the dose at which the drug shows pharmacological action based on the label claim.

Highly hazardous drug: Highly hazardous drugs typically are those with PDE values of 10 mcg / day or less and drugs that are known to cause harm, which may or may not include genotoxicity (The ability to cause a change or mutation in genetic material). Geno toxicity might involve carcinogenicity, the ability to cause cancer in animal models, humans or both; teratogenicity, which is the ability to cause defects on foetal development or foetal malformation; and lastly hazardous drugs are known to have the potential to cause fertility impairment, which is a major concern. These drugs can be classified as antineoplastic, cytotoxic agents, biologic agents, antiviral agents, hormones and immunosuppressive agents.

Periodic monitoring: Collection and analysis of cleaning process data to ensure that the validated cleaning process remains in a state of control and continues to effectively remove residues and contaminants.

Dirty equipment hold time:

The period between the completing of batch processing (equipment) till the start of cleaning is commonly referred to as the Dirty equipment hold time (DEHT) or Dirty hold time (DHT).

Clean equipment hold time:

The time between cleaning and equipment re-use, prior to additional cleaning, commonly referred to as the clean equipment hold time (CEHT). The time from the end of the cleaning process until the equipment is used again for manufacturing.

Campaign Length (no of batches):

A campaign length is a series of batches of the same product manufactured consecutively with type A cleaning between each batch. i.e. one batch immediately followed by another batch without intervening type B cleaning between the serial batches. Defined as number of batches.

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Identical equipment:

Equipment of the same working principle, same MOC and similar design (Capacities, internal Geometry etc), are considered as identical equipment. A single common cleaning procedure can be used for identical equipment.

1.0 ENVIRONMENT, HEALTH AND SAFETY :

- 1.1 Ensure personal protective equipment like nose masks, goggles, apron, hand gloves, shoe covers (where applicable) are worn during the cleaning, sampling, and testing operation.
- 1.2 There shall be safe disposal of wastes, washings as per the established procedures and compliance to environment, health, and safety norms.
- 1.3 Ensure proper earthing and bonding is done while flushing by solvents.
- 1.4 Ensure AHU is 'ON' before cleaning.
- 1.5 Ensure relevant permits are provided for equipment parts / accessories dismantling and refixing.

2.0 PROCEDURE :
2.1 GENERAL REQUIREMENTS:

- 2.1.1 Operators and manufacturing staff involved in the cleaning of equipment shall be trained on required cleaning procedures. Training shall involve SOP training followed by demonstration of cleaning process and practical evaluation of training. *Effectiveness of training shall be periodically verified.* Refer SOP 1035-A-0083 for details of training requirement.
- 2.1.2 Production and Quality assurance staff who is involved in visual cleanliness check need to be qualified for the same. Qualification shall involve, medical fitness (eyesight evaluation), observations of various photographs, evaluation on actual templates created with product residues. Refer SOP 1035-A-0083 for details of Visual cleanliness requirement.
- 2.1.3 Production and Quality assurance staff involved in rinse sampling shall be trained in rinse sampling procedures with actual demonstration and practical evaluation.
- 2.1.4 Quality assurance staff involved in Swab sampling shall be trained and qualified for swab sampling procedures with actual demonstration. Evaluation shall be done with actual recovery study by spiking product on templates. Refer SOP 1035-A-0082 for details of rinse and swab sampling, and SOP 1035-A-0083 for training on swab sampling.
- 2.1.5 Quality control staff involved in analytical method validation and method transfer of cleaning methods shall be trained in Rinse and swab sampling procedure Refer SOP 1035-A-0082.
- 2.1.6 Analytical methods used for testing of cleaning samples shall be validated as per respective procedures.
- 2.1.7 If stain is observed during visual inspection, then it shall be evaluated as per Stain SOP 1035-A-0080. For permanent stains, stain evaluation annexure shall be available and same shall be verified during visual inspection. If new stain is observed then, same shall be evaluated as per SOP 'Handling of Stain on process equipment' .

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2.1.8 Process Equipment shall be evaluated for Surface Anomalies like dent and scratches etc periodically as per applicable procedures.

2.2 **CLEANING PROCEDURE DESIGN AND LIFE CYCLE CONSIST OF FOLLOWING MAJOR ELEMENTS.**

Cleaning Process Design • New product residue characterization • Selection of cleaning solvent / agents • Development of Analytical Methods • Development and Optimization of cleaning processes • Selection of cleaning procedure. (Auto/Manual/CIP/COP) • Cleaning Procedure development for new type of equipment
Cleaning Validation • Cleaning validation approach • Selection of worst-case product • Risk assessment. • Establishment of carryover limits and acceptance criteria • Validation of Analytical Method & Recovery studies • Sampling plan preparation and sampling techniques • Preparation and approval of cleaning validation protocol • Execution of protocol and report compilation • Establishment of Dirty and Clean equipment hold time study.
Other Life cycle elements for cleaning process • Evaluation of changes for impact • Periodical cleaning verification. • Revalidation of cleaning processes • Routine monitoring with sampling and Visual inspection • Establishment of Campaign length and campaign length verification

2.3 **CLEANING PROCEDURE DEVELOPMENT:**

2.3.1 R&D shall develop and provide initial cleaning procedures for new products to site based on experience gained during development stage and pilot scale equipment cleaning. ADL shall develop analytical method based on residue to be cleaned. R & D shall provide brief cleaning procedure involving major steps in cleaning as part of product tech transfer to site.

2.4 **SELECTION OF CLEANING AGENT / CLEANING SOLVENT:**

2.4.1 For Solubility based cleaning, inorganic or organic solvents are used like Methanol, IPA, Water, Dilute Alkali, Dilute Acid etc.

2.4.2 Following are the criteria needs to be considered while selection of cleaning solvent:

- Solubility. Product residue shall be soluble in selected cleaning solvent at room temperature or temperature used during cleaning. Solvents in which product is sparingly soluble shall be avoided.
- It shall not degrade the product if also used in processing.
- It shall not retain any undue residue on equipment surface.
- It shall be compatible with the MOC of the equipment.
- Hazardous chemicals / Class1 solvents (e.g., Benzene etc.) shall be avoided.

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- Traces of cleaning agent shall be easily removable (It shall be free rinsing).

- 2.4.3 Cleaning with Surfactants or aqueous detergents, specially formulated cleaners (e.g., Teepol) etc.
- Cleaning agent shall be compatible with the equipment MOC.
 - Cleaning agent shall have good surfactant properties to reduce surface tension and facilitate easy removal of residue.
 - Cleaning agent shall be easily cleanable with water and shall not produce any adhering residue.
 - Cleaning agent shall be nontoxic or shall have very low toxicity.
 - *Documented rationale shall be available for choice of cleaning agent.*
- 2.4.4 Cleaning with destructive cleaning agent
- Sometimes cleaning agents used degrades the product during cleaning procedure. Such procedures shall be used where it is important to neutralise the potent chemicals.
 - In such cases analytical method used for cleaning shall be able to quantify the degradation product. Alternately nonspecific methods like TOC, Conductivity etc shall be used based in suitability.
- 2.5 **NEW PRODUCT INTRODUCTION:**
- 2.5.1 At the time of introduction of new product, the change request initiated must ensure below prerequisites.
- 2.5.2 An evaluation shall be performed based on the product development report received from R &D to determine the characteristics of the residues to be cleaned (product, location, cleaning agents etc.).
- 2.5.3 Solubility, concentration, cleanability, and conditions for degradation or deactivation of residues shall be considered in the assessment.
- 2.5.4 An assessment shall be performed while introducing a new product at site. Assessment shall determine cleaning and analytical testing requirement based on various materials charged into equipment, the products formed, and the major residue expected to remain in the equipment etc.
- 2.5.5 *Assessment shall also consider practical considerations in cleanability / cleaning difficulty such as residues having tendency to become very hard (scaling) or forming gummy mass etc.* The experience gained from trial cleaning procedure after taking a batch at plant scale shall be used to recommend final cleaning procedure for routine cleaning.
- 2.5.6 Ensure cleaning method validation, including recovery are performed and available as a prerequisite prior to introduction of new product into commercial multi-product equipment.
- 2.5.7 Sometimes batch to batch cleaning is also very important for quality of next batch of same product. For example, in some cases product residues after batch changes polymorphic form and if not cleaned properly same can act as seeding of wrong form. In such cases batch to batch cleaning shall be as good as product changeover cleaning.
- 2.5.8 *To record details of assessment for introduction of new product in site use annexure 1035-A-0072/A1.*
- 2.5.9 Refer below table for general recommendation for specific and non-specific testing requirements for various type of residues:

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Type of residue	Preferable testing requirements
Previous product (API / isolated Intermediate)	- These residues shall be tested in equipment, wherever the molecule of API / Intermediate is formed. e.g., Equipment used for processing of reaction stage onwards. - If the isolated intermediate is a base/ different salt of an API, then it shall be considered as API and same shall be part of API product and equipment matrix. - Ensure equipment is adequately protected from contamination after cleaning and before use.
Key starting materials (KSM)	If the equipment / vessel is used for preparation and/or addition of KSM solution, then such equipment shall be tested for content of respective KSM instead of intermediate/API that is produced with the KSM.
In-situ intermediate	The equipment which are used for manufacturing of an in-situ intermediate (non-isolated) which does not undergo further conversion in same equipment / reactor shall be tested for content of respective in-situ intermediate.
Microbial contamination	Whenever water is used as cleaning solvent, recommended to rinse the equipment with suitable organic solvent (miscible with water). This helps for effective removal of water as well as helps in faster drying of the equipment surfaces. Equipment cleaning record shall describe all these steps. Microbial testing shall be performed only for equipment operating in class ISO 8 area and CNC area (controlled but not classified) area.
Acid / Base content	- If the equipment is used for preparation and/or addition of acidic / basic solutions such as Hydrochloric acid solution, Sodium Hydroxide solution, then such equipment shall be flushed with water. - Rinse samples of such equipment shall be tested for pH and the pH of rinse sample shall be within the acceptance criteria for pH of specific water (Potable water / Purified water/ Water of injection) used for cleaning.
Charcoal	- The vessels which are used for preparation of Charcoal slurry shall be exclusively used only for Charcoal slurry. - The reactors which are used for Charcoalization of product shall be sampled and tested for following contents / aspects: - Content of respective API / intermediate - Rinse sample shall be free from any foreign/black colour particles. - Above checks shall be part of sampling plan / equipment cleaning record.
Cleaning solvent	- After completion of cleaning, equipment shall be dried with evaporation technique (through steam / hot water circulated in jacket) or dry with compressed air and lint free cloth. Accordingly, testing of cleaning solvent content during every product changeover cleaning is not necessary where cleaning solvent dried easily. - In case cleaning solvent is high boiling solvent like Toluene or DMF etc. in such cases cleaning solvent shall be tested in rinse sample during cleaning validation as one time study.
Debris/ dust	After civil work or any maintenance work cleaning shall be done as per unit specific SOP and visually check the rinse there shall not be any particle/grease/oil present.
Cleaning agent (e.g., Teepol)	Wherever, cleaning agent (e.g., Teepol) is used for equipment cleaning, the rinse samples shall be tested for presence of Teepol content.

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- 2.5.10 When new products are introduced in facility, a system must be in place to ensure that the previously used cleaning process shall remain effective.
- 2.5.11 All action items shall be well defined while initiating the Change Request related to cleaning.
- 2.5.12 Action items related to cleaning change request shall include MACO calculation, cleaning matrix updation, rinse and swab limit establishment, method availability and validation for the limits calculated, recovery on applicable surfaces (MOC), impact on other product limits and accordingly suitability, worst case selection criteria, cleaning validation with three runs, CEHT and DEHT assessment, campaign length cleaning validation.
- 2.5.13 Risk assessment for cleaning process shall be performed as per SOP 1035-G-0024, It shall cover the solubility of product, equipment, impact on existing MACO, worst case criteria, etc. and record the observations in Annexure 1035-A-0072/A1.
- 2.5.14 Risk assessment for cleaning process aspect shall be performed as per SOP no. 1035-G-0024, It shall cover the physical characteristics such as viscosity, processes creating sticky or caked on solids, materials reacting with water (cleaning solvent) to form solid or sticky gel like material, etc. and record the observations in annexure 1035-A-0072/A1.
- 2.5.15 Following are basic considerations for new product and details shall be recorded in annexure 1035-A-0072/A1
- Is the new product to be cleaned by the same cleaning process as used for the existing group?
 - Is the new product having impact over the calculated residue limits for the products in the existing group?
 - If included in the existing group, is the new product be a new “most difficult to clean” product in the group?
 - Will the new product (that is not more difficult to clean than the “most difficult to clean” product in the group) constitute a new “lowest limit” of any product in the proposed group?
- 2.6 **Introduction of new equipment:**
- 2.6.1 Following are basic considerations for new equipment and details shall be recorded in annexure 1035-A-0072/A2.
- 2.6.2 If the new equipment is identical or equivalent with minor changes and is cleaned with the same cleaning SOP, then the new equipment may be added to an already validated equipment group by performing a verification protocol run on the initial manufacture of any product in the group using that new equipment, with the sole acceptance criterion being that the equipment is visually clean.
- 2.6.3 If the new equipment is equivalent with significant changes like MOC, Dimension, Capacity etc, then that equipment shall be evaluated to determine whether the significant changes constitute a new worst-case equipment and whether the new equipment changes the calculated limits for API and cleaning agent for the products using that equipment.
- 2.6.4 If the new equipment does not constitute a new worst case equipment and does not lower the calculated limits for the group (such as by a higher surface area), then the new equipment may be added to an already validated equipment group by performing a verification protocol run on the initial manufacture of any product in the group using that new equipment, with the sole acceptance criterion being that the equipment is visually clean.

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- 2.6.5 If the new equipment does not constitute a new worst-case equipment but does lower the calculated API and cleaning agent limits for the group (such as by increasing the shared surface area), the prior validation of the worst-case equipment shall be assessed to determine the relationship between the analytical values achieved in the existing validation and the new “lower limit”. If the actual analytical values from the existing validation are significantly below the new “lower limit”, then that existing validation may still be considered acceptable at the new “lower limit”. An example of what constitutes “significantly below” is as follows: “If the previous validation results are no more than 50% of the new lower limit, then that previous validation is still applicable at the new lower limit”.
- 2.6.6 If the previous validation results are more than 50% of the new lowest limit, then validation shall be performed on the “most difficult to clean” product at the new lowest limit (As if it were a new group and validation performed for the first time).
- 2.6.7 *If the new equipment does constitute a new worst-case equipment but does not lower the calculated API and cleaning agent limits for the group (such as by increasing the shared surface area), then the new equipment may be added to an already validated equipment group by performing validation with three runs on manufacture of the worst-case product for that equipment group using that new equipment. If that worst product is not the first product manufactured on that new equipment, then validation with three runs shall be performed on the first product manufactured on that equipment; in such a case, an additional verification run shall be performed on first manufacture of the worst-case product on that new equipment. The acceptance criteria shall be the same as the acceptance criteria used for the original validation for that equipment.*
- 2.6.8 **Assessment and Development of cleaning procedure for new type of equipment:**
- Prior to / during qualification of equipment, a team from user, QA and engineering shall discuss and performed the dismantling / assembling assessment of equipment based on equipment user manual and check the equipment for its various parts.
 - During dismantling / assembling assessment, accumulation of powder residue, oil leakage etc. shall be checked. All related equipment parts such as ducts, panel, cabinets, shall be considered and covered during assessment, and extent of cleaning shall be defined.
 - Based on equipment user manual, check the dismantling ability (manual of these parts) determined the extent of which each part of equipment further to be dismantled.
 - Identify the parts of equipment which will be come in product contact / non-contact.
 - Identify the areas / parts which are difficult / hard to reach / clean for each part of equipment.
 - Against each identified parts of equipment, determine the type of cleaning required and approximate quantity of cleaning solvent for each part i.e. pressure cleaning, rinsing, scrubbing, soaking, refluxing etc.
 - Conduct the studies / trial cleaning batch and verify the effectiveness of cleaning.
 - Based on above assessment, sampling plan and justification for swab location shall be prepared, the details of assessment shall be captured in annexure 1035-A-0072/A2
- 2.7 **Selection of cleaning methods and cleaning aids:**
- Based on the equipment design and feasibility for cleaning of entire surface of the equipment, one or combination of the following cleaning methods shall be employed:
- 2.7.1 **Manual Clean-In-Place (CIP) method:**
It is the process of cleaning the equipment in its place through the manual operations such as rinsing / refluxing etc (without transferring to washing area). This is the most widely used cleaning method for

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fixed equipment. It may still involve disassembling of smaller parts (such as flanges, gaskets, smaller connecting lines, dismantled lines, valves etc) for cleaning to the same in washing area.

- 2.7.2 Automated Clean-In-Place (CIP) method:
The CIP system is a packaged system of properly integrated components typically including tank(s), pump(s), heat exchanger(s), cleaning agent feed equipment, valves, instruments, and system controls like HMI/DCS/SCADA.
- 2.7.3 To implemented automated CIP system, respective equipment shall meet specific criteria for design, MOC, installation, inspection, and maintenance that are suitable for CIP cleaning.
- 2.7.4 In such systems, following parameters (but not limited to) shall be established to ensure reproducible effectiveness of cleaning:
- CIP solution flow rate in the piping
 - Spray ball design and its coverage for entire equipment surface including hard to clean areas.
 - Time duration of rinse
- 2.7.5 Clean Out of Place (COP) method:
It is the cleaning of the equipment and/or parts that cannot be cleaned in their process position by CIP are removed from the equipment for COP according to procedure.
- 2.7.6 This cleaning of disassembled equipment / parts / accessories, in a designated washing area.
- 2.7.7 General considerations for cleaning methods:
Following instructions shall be considered as part of cleaning process based on equipment design.
- Pre-cleaning treatments with cleaning aids (e.g., Scrapping, dry mop)
 - List of parts to be disassembled during cleaning.
 - Pressurized washing with water/ solvent.
 - Manual rinsing / flushing with cleaning Solvent / agent.
 - Soaking / immersion in cleaning solvent / agents, sonication
 - Using cleaning aid like brush etc to reach difficult to reach small parts.
 - Rubbing of equipment surfaces (mesh) with wet mop etc.
- 2.8 **Major steps of typical product changeover cleaning procedure for equipment and accessories.**
- Following are the major steps based on product and facility requirement which shall be part of typical Equipment cleaning procedure (ECR).
 - The sequence of cleaning steps shall be scientific to remove the residues on surfaces from gross accumulation to the level less than swab/rinse limits.
 - ECR shall contain all below steps and shall facilitate recording of important steps in details.
 - Based on applicability use of computerised systems like Scada / DCS etc shall be made to record the cleaning operations.
- 2.8.1 Pre-cleaning: This step includes removal of gross residue with simpler methods.
- Dry mopping/ dusting / scrapping of residue from equipment surfaces.
 - Washing with water or a pressurised water gun (even if the product does not dissolve in water).
 - Soaking / wetting of equipment parts.
 - Disassembling the equipment parts for ease of cleaning.

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- 2.8.2 Cleaning with cleaning agent/solvent: This step is intended to remove all residues down to acceptable levels.
- The quantity of cleaning agent shall be specified for each stage. Quantity shall be based on the equipment capacity / surface area and design of equipment.
 - When surfactants are used, the concentration of the cleaning agent shall be defined. Use of surfactants only for powder processing equipment in open system such as Multimill and sifters & not for closed equipment such as Rotacone vacuum dryers (RCVD).
 - Depending on design of equipment, cleaning methodology shall be defined.
 - Number of reflux / rinse cycles shall be established during cleaning process development and same shall be concluded during pre-validation run.
 - There shall not be any optional reflux / rinse in equipment cleaning record post pre-validation run.
 - In ECR the number of reflux / rinses shall be clearly mentioned in equipment cleaning record in the same sequence that is confirmed during pre-validation run.
- 2.8.3 Reflux with Solvents:
- Equipment like reactors, certain vessels have capability to heat the cleaning solvent till boiling point and reflux the same through condenser. Heating of solvent not only increases the solubility of solute, but also solvent vapours reach every nuke and corner of equipment facilitating better cleaning.
 - In this operation quantity of solvent, time and reflux temperature range shall be defined. These cleaning parameters are important step and shall be recorded in ECR.
 - Number of refluxes depending on the amount of residue and size of equipment shall be defined. This is preferred method as there is less possibility of operator-to-operator variation.
- 2.8.4 Heating with Solvents
- Some equipment has the capability to heat the solvent although reflux may not be possible (equipment without proper condenser) e.g., ANFD (depending on design), RCVD etc.
 - Similar to reflux, quantity, temperature and time range for operation shall be fixed along with number of cycles.
- 2.8.5 Cleaning with Spray ball
- Cleaning done by use of spray balls wherever equipment design facilitates this.
 - Effectiveness of spray ball system to reach entire surface of equipment shall be established during development run to avoid blind spots.
- 2.8.6 Cleaning of rotary Equipment
- Rotary equipment depending on design (e.g., Blender) shall be cleaned by charging cleaning solvent/agent and rotating the equipment for fixed time.
 - Quantity of solvent, time and RPM shall be fixed.
- 2.8.7 Rinsing with Solvents
- Equipment designs which do not facilitate heating of solvents shall be cleaned with rinsing by solvent.
 - Rinsing shall be done by use of spray balls, peristaltic pump nozzle or manually by using mug.
 - Operator shall ensure that entire product contact surface area is covered during solvent rinsing.
 - For small equipment where surfaces are visible, rinsing is the best method.
 - The number of rinsing cycles and the quantity of rinse shall be fixed.
 - Rinsing shall be combined with reflux or other cleaning method as final step to give finishing to overall cleaning.
- 2.8.8 Drying of cleaned equipment:
- It is important to dry the equipment after cleaning.

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- Drying shall be done by various techniques like heating, applying vacuum, blowing compressed air, dry mopping, leaving open to air dry or combination of above.
- When low boiling solvents like Methanol, Acetone etc are used for cleaning drying is easier.
- If high boiling solvents are used like toluene, Dimethyl Formamide etc are used in such cases use of miscible low boiling solvent can remove the high boilers effectively.
- In case water is used as solvent, then it is preferred to rinse equipment with Methanol or other suitable solvent. This reduces the possibility of microbial growth due to stagnation of water.
- For products which are sensitive to microbial growth 70% IPA or suitable disinfectant shall be used for wet mopping the external surfaces.
- Post drying, the equipment surfaces shall be inspected as clean and dry prior to covering and storing.

2.8.9 For Sampling refer SOP # 1035-A-0082 (EQUIPMENT CLEANING AND VALIDATION OF CLEANING PROCEDURES).

2.8.10 General instruction for sampling

- Cleaning procedure shall contain sampling instructions at respective steps.
- Rinse sampling shall be done as separate step and shall not be done along with final rinsing step which is done as part of cleaning step.
- Rinsing solvent shall be same solvent which is used for main cleaning step.
- Solvent used for facilitating drying of previous cleaning solvent shall not be considered as sampling solvent.
- ECR shall have instruction of drying of equipment post rinse sampling.

2.9 **Major steps for typical batch to batch (A Type) cleaning procedure:**

2.9.1 A type cleaning procedure shall be designed, based on impact of cleaning procedure on sub-sequent batch processing activities.

2.9.2 Cleaning procedure shall remove any gross accumulation of residue on equipment.

2.9.3 Rinsing / flushing with solvent to remove grossly adhered product.

2.9.4 Dedusting of powder processing equipment.

2.9.5 For sensitive reactions like water sensitive reaction where any residual water may impact on reaction completion then reflux with suitable solvent to remove water traces. For example, if reactor being used in crystallisation area for purification activity and previous batch traces may have chances of degradation or may impact on crystallisation activities of next batch, then during A type cleaning, reflux shall be done with respective stage product solvent or product cleaning solvent.

2.10 **Major steps for general cleaning procedure:**

2.10.1 General cleaning procedure is not specific to any product.

2.10.2 This procedure shall be used for cleaning of new equipment or after the maintenance activities of equipment.

2.10.3 Purpose of general cleaning procedure is to remove dust, oil, grease, blur, or any other residue which may be present on new equipment or post maintenance of the equipment and area.

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- 2.10.4 Understand the type of residue e.g. Dust, traces of oil etc which is expected in equipment.
- 2.10.5 Clean the equipment with plenty of water to remove obviously adhering material.
- 2.10.6 Clean the equipment with surfactant like Teepol etc... followed by water. After cleaning check, the Teepol content in rinse sample.
- 2.10.7 Clean equipment with solvent, take reflux if required.
- 2.10.8 In case equipment is new or buffing is done, separate cleaning procedure (passivation) shall be followed.
- 2.10.9 After cleaning, equipment shall be visually clean, and rinse shall not have any suspended particles.
- 2.11 **Microbial aspects of cleaning process:**
- 2.11.1 Whenever water is used as cleaning solvent, it is recommended to rinse the equipment with suitable organic solvent (miscible with water). This helps for effective removal of water as well as helps in faster drying of the equipment surfaces.
- 2.11.2 Equipment shall be dried properly.
- 2.11.3 These steps help to restrict microbial growth.
- 2.12 **Cleaning of accessories:**
- 2.12.1 Accessories are small tools which are used for material handling or other miscellaneous activities, e.g. Sampler, Scoops etc.
- Accessories used for handling API/Intermediates/ raw materials shall be cleaned by using respective product cleaning solvent.
 - Alternately, if cleaning agent is used for cleaning of accessories, use of same shall be validated.
 - Cleaning procedure shall be defined in unit cleaning procedures.
 - Post cleaning all accessories shall be stored in way to avoid any contamination and shall be labelled.
 - If stains observed on accessories (which comes in contact with product) after cleaning, then same shall be evaluated as per stain SOP 1035-A-0080.
- 2.13 **Cleaning of Gaskets, Rubber Parts, Gloves etc:**
- 2.13.1 Gaskets, rubber parts, isolator gloves come in different MOCs depending on application. These parts may or may not be in direct contact of the product.
- 2.13.2 Examples of direct product contact parts are process pipeline gasket, multimill gasket, reactor manhole gasket etc. Example of indirect product contact part is Isolator glass gasket, VTD door gasket, Isolator gloves etc.
- 2.13.3 These parts/gaskets depending on its MOC, cleaned with process solvent if it is compatible with solvent.
- 2.13.4 Parts/gaskets which are not compatible with organic solvents shall be cleaned using cleaning agents like detergents or surfactants.
- 2.13.5 If stain or discolouration of gasket observed, then gasket shall be replaced.

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2.13.6 Equipment cleaning procedure shall describe the cleaning procedure for Gaskets, gloves etc.

2.14 Optimization of Cleaning Process (Pre-validation run/ Trial run)

2.14.1 Developed cleaning procedure shall be evaluated for their feasibility and effectiveness as a pre-validation run. (Note: Pre-validation run may not be necessary if the validated cleaning procedures are already in place for similar nature of material or same base of drug substance with different salt having same solubility). Protocol base study to be performed for pre-validation / trial run.

2.14.2 During this optimization, if additional cleaning steps (such as additional reflux/rinse) used, same shall be recorded as part of cleaning record.

2.14.3 Validated analytical methods shall only be used for analysing the samples collected during such optimization runs.

2.14.4 Sampling procedure, locations, number of samples shall be same as intended for upcoming cleaning validation.

2.14.5 If the results obtained are within the acceptable limit, the cleaning procedures shall be used for routine cleaning procedure.

2.14.6 If the results obtained are not within the acceptance criteria, the cleaning procedure shall be modified to get the desired result.

2.15 Cleaning for portable equipment and fixed equipment in powder processing area:

2.15.1 Movable/ portable equipment in powder processing area after cleaning shall be stored in clean equipment hold room.

2.15.2 Post cleaning and drying, portable equipment shall be covered with polyethene bag, shrink wrap to cover them to extent possible.

2.15.3 The equipment shall not be moved to other processing areas without type B cleaning within same processing area or washing area.

2.15.4 *For fixed equipment in processing area which are not used, but other equipment is used for processing, such equipment shall be cleaned with Type B cleaning procedure of product which was processed in the area.*

2.16 Design of Equipment Cleaning Procedure and Record (ECR):

2.16.1 Each equipment shall have a cleaning ECR for A, B type cleaning procedure. Validity of DEHT, CEHT and campaign length shall be specified in cleaning procedure.

2.16.2 Cleaning procedure shall be elaborated for better understanding to operating persons.

2.16.3 *ECR shall have a traceability of below content,*

- General instruction related to cleaning operation. Below points shall be considered but not limited,
 - Safety instruction
 - Different tools used for cleaning

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- Trouble shooting actions
- Precautions to be taken during cleaning
- Usage of cleaning ECR in which scenario this ECR shall be used for equipment cleaning.
- Equipment code, Previous product name, and batch number
- Cleaning aid used for cleaning i.e., solvent name / solution name / cleaning agent name with dilution / water and minimum quantity of solvent used for cleaning etc.
- Provision for recording of quantities used for cleaning operation.
- Provision for recording of SAP document number.
- Provision for recording of date and time for cleaning activity.
- Provision for recording of observations.
- Critical cleaning parameter shall be identified e.g. reflux temperature and time, soaking time, parts scrubbing time. CCP may vary based on the nature of product and equipment cleaning, CCP either one or combination of multiple factors.
- Review check for Critical Cleaning Parameters (CCP).
- Ensure the equipment is appropriately disassembled small parts such as flanges, gaskets, dismantled lines, or valves to the designated wash area for cleaning. After cleaning reassemble the parts & verify by second check.
- Minimum time for critical cleaning parameters shall be established during cleaning validation and document for routine cleaning.
- Review check for cleanliness verification after completion of cleaning.
- In case of product-to-product change over cleaning procedure (B type), recording provision of last rinse or swab sample results.
- Review of completed ECR by quality assurance.

2.16.4 Common/separate ECR shall be designed for equipment with a similar design irrespective of their capacities.
For example, reactor of different capacities shall be cleaned with the same ECR using similar cleaning techniques.

2.16.5 In such cases solvent quantity chart shall be attached to the ECR to accommodate different capacities of equipment.

2.17 Types of Cleaning and their Definitions & Scenarios:

Cleaning processes are classified into following types:

- Batch to Batch Cleaning for same product (Type A cleaning)
- Product to Product changeover cleaning and Campaign cleaning procedure (Type B cleaning).

2.17.1 Batch to Batch Cleaning (Type A cleaning):

- Batch to Batch cleaning or Type A cleaning is the cleaning done between the batches of same product in a campaign.
- After completion of one batch, equipment shall be cleaned as per A type cleaning before use of equipment for same product at same stage of next batch.
- **Validity:** The validity period for 'Type A' cleaning is limited to 3 days. After this period, the equipment must be cleaned again with 'Type A' cleaning and shall be used for the same product within the next 3 days. If the subsequent 3-days period expires, the equipment shall be cleaned with 'Type B' cleaning.

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- Cleaning validation / challenging not essentially be established for non-critical cleaning, e.g., between batches of the same product or different lots of the same intermediate in a bulk process.

2.17.2 Type B cleaning shall be performed in following scenarios.

- Type B cleaning is the thorough cleaning of equipment which is done for product change over.
- Type B cleaning shall also be done after completion of the campaign length even though same product is proposed to manufacture after cleaning.
- If equipment is cleaned as per type A, cleaning and proposed to be used for another product, Type B cleaning shall be done.
- Before handover of area / equipment for modification / maintenance (e.g., turn around / planned shutdown / major maintenance of area), Type B cleaning shall be done.
- After completion of cleaning validity of type A cleaning, Type B cleaning shall be done.
- Prior to area fumigation, equipment shall be cleaned as per type B cleaning procedure, the reactor may be excluded from this process as it is fully closed and external wiping of equipment after fumigation.
- In powder processing area if equipment is not used in manufacturing process, however other equipment in the room is used in manufacturing, then at the time of product change over all equipment need to be cleaned with Type B cleaning procedure of the product which is manufactured in the area and samples (Rinse / swab) to be tested for manufactured product content.
- In case, if equipment is proposed to be used for same product post maintenance, it shall be cleaned with Type B procedure without complete dismantling and handed over to maintenance. In such case line clearance and sampling is not applicable.
- **Validity:** The validity period for Type B cleaned equipment is not more than 6 days.

2.17.3 Cleaning After expiration of Validity

Type A	Type B
Type B	Idle (If after Type B cleaning equipment goes in idle state, it shall be flush with suitable solvent before use to next product).

2.18 Acceptance Criteria and Establishment of Carryover Limits:

A. For Active pharmaceutical ingredient.

- 2.18.1 For API, equipment in which active moiety is formed that step onward equipment shall be considered for calculation. For purpose of this SOP base, different salt, different polymorph etc shall be considered as API.
- 2.18.2 For deriving acceptance criteria for API manufacture preceded by another API, health-based limits shall be used.
- 2.18.3 Toxicologist shall provide the permitted daily exposure (PDE) report with PDE value for different age groups (Adult, child, new-born & premature baby) and different route of administration (Oral, parenteral, inhalation, dermal, ophthalmic etc). Copy of all toxicological reports shall be maintained by site.
- 2.18.4 PDE value of previous product shall be selected as per next product stringent age group and next product route of administration for MACO calculation.

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- 2.18.5 If one molecule is being manufactured for more than one age group (Adult, child, new-born & premature baby) then select the previous product PDE as per next product worst age group. i.e., minimum age group. Applicability of age group shall be same for previous as well as next product.
- 2.18.6 Based on previous products PDE age group, subsequent product therapeutic daily dose of the same age group shall be used. E.g., If previous product PDE selected as child among adult and child, Therapeutic daily dose (Subsequent product) of same age group i.e., child only to be used for calculation of MACO based on PDE.
- 2.18.7 Similarly, if drug substance is being manufacturing at site for oral as well as for injectable, route of administration then selects the stringent PDE value from PDE report provided by toxicological department.
- 2.18.8 For some molecule same (therapeutical daily dose) shall be collected from various web sites like www.drugs.com, www.Medscape.com, www.pdr.net etc.
- 2.18.9 Few molecules exist in more than one salt with same category of use. In such cases, the type shall be aligned to the available base information. E.g.: Amlodipine Besylate data available, while for Amlodipine Maleate and Mesylate & Amlodipine Mesylate Monohydrate data not available, hence the application information for Amlodipine Besylate shall be considered (w.r.t adult / child / new-born / premature) and vice - versa.
- 2.18.10 Few molecules having different route of administration. Those available ROA data for patient population / age group considered for that molecule which is not having other ROA data. E.g., Artemether oral ROA age group data is available from Cipla site and Artemether Intramuscular ROA data is not available hence same shall be considered the age group data of Artemether oral ROA to Artemether Intramuscular ROA.
- 2.18.11 **Maximum allowable carryover (MACO) For API shall be calculated as below:**

MACO calculation based on permitted daily exposure (PDE) by using below formula by:

$$\text{MACO} = \frac{\text{PDE (previous product)} \times \text{MBS (next product)}}{\text{TDD (next product)}}$$

MACO : Maximum allowable carryover (mg)
PDE (previous product) : Permitted daily exposure (mg)
MBS (Next Product) : Minimum batch size of next product (mg)
TDD (Next Product) : Maximum therapeutic daily dose of next product (mg)

- 2.18.12 Upon deriving the MACO limits as above, limits for Rinse and swab samples shall be calculated by using the applicable MACO as follows:

$$\text{Rinse Limit (ppm) for equipment} = \frac{\text{MACO (mg)} \times \text{Area Sampled by Rinse Sampling (inch}^2\text{)} \times 1000}{\text{Surface Area of Equipment Chain (inch}^2\text{)} \times \text{Rinse sampling volume (ml)}}$$

$$\text{Swab limit (ppm)} = \frac{\text{MACO} \times \text{Swab Surface Area} \times 1000}{\text{Surface Area of Equipment Chain} \times 10 \text{ (ml)}}$$

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OR

$$\text{Swab limit } (\mu\text{g/swab}) = \frac{\text{MACO} \times \text{Swab surface area} \times 1000}{\text{Surface Area of Equipment Chain}}$$

1000 Conversion factor from mg to mcg. for swab.

- 2.18.13 The carryover of residue of product content from chain of equipment shall be less than the MACO calculated by Permitted daily exposure (PDE).
- 2.18.14 In case limits of residue of previous product obtained from MACO calculations are high a lower limit of Not more than 10 ppm, shall be used for individual rinse and swab results.
- 2.18.15 If only one product manufactured in new / dedicated block the rinse and swab limit for cleaning validation shall be applied NMT 10 ppm.

B. For key starting materials, intermediate stage equipment's:

- 2.18.16 For non-active stages like Intermediate or Key starting material common limit of Not more than 10 PPM shall be used for individual rinse and swabs samples and for equipment chain it shall be less than 100 ppm.

C. Nitrosamine and Genotoxic compounds:

- 2.18.17 While introducing new product, assessment of potential carryover of Genotoxic and Nitrosamine impurities shall be done. Assessment shall involve evaluation of Raw materials, starting material, intermediates, known impurities, solvents, recovered solvents etc. Based on the assessment if Nitrosamine or genotoxic compound found to have carryover potential recommendations shall be made for improve cleaning procedure. Strategy for one time or routine testing of rinse, swab samples for potential carryover shall be decided and documented.
- 2.18.18 Health based Acceptance criteria shall be used for Nitrosamine and genotoxic impurities which are expected to be present in final API. Help of Toxicology expert shall be taken to decide the limit. In absence of data to decide health-based limit refer TTC (Thresh hold of Toxicological concern) criteria.

D. Change over scenarios and limits for API and intermediates.

- 2.18.19 Following table provides general guidelines for different types of change over scenarios.

Changeover Scenario	Limits (Acceptance Criteria)
Intermediate to intermediate	10 ppm
Intermediate to API	10 ppm
API to Intermediate	Limits calculated by MACO
API to API	Limits calculated by MACO

- 2.18.20 For change over involving intermediate and API stages of same product (one stage to next stage) considering one stage is input to next stage risk assessment shall be done to decide extent of cleaning requirement and acceptance criteria.
- 2.18.21 In case of product names are different but its same molecule having similar manufacturing process and same cleaning procedure, site can club such product in cleaning matrix based on thorough assessment.

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Alternately for such products after change over cleaning line clearance shall be based on visual inspection. The examples are Albuterol Sulphate and Levo salbutamol Sulphate etc.

- 2.18.22 If API crude is manufactured in final step reactor and in the same reactor API stage to be manufactured, then product change over cleaning shall be done and line clearance shall be based on visual inspection. Site shall have product wise assessment report for such exceptions.

E. Limits for Cleaning agent

- 2.18.23 **Surfactants or Detergent based Cleaning Agents:** Acceptance criteria for MACO of cleaning agents shall be calculated based on PDE value. Limits for cleaning agent for rinse, swabs samples shall meet the defined limits of MACO. If limit based on MACO is high general limit of Not more than 10 PPM shall be used for individual Rinse, Swab samples. Use formula provided in section 2.18 for calculation of API with PDE value of cleaning Agent.
- 2.18.24 **Solvent based Cleaning:** Acceptance criteria for Rinse, swab sample shall be 1/10th of respective ICH limit for the solvent.
- 2.18.25 **Cleaning with Acid, Alkalis:** In case equipment is cleaned with Acids or alkalis, it shall be further cleaned with water to remove residues of Acids, alkalis. Acceptance criteria for Rinse samples is based on pH of type of water used for rinsing.
- 2.18.26 Acceptance criteria for pH of Purified water (pH 5.0 to 7.0) and potable water (pH 6.5 to 8.5)

F. Microbial Contamination limits

- 2.18.27 Bioburden on equipment shall comply with the limits as specified in SOP 1035-L-0086, "Microbiological monitoring of environment, surfaces and personnel", which are established in reference to the SOP. 1035-L-0089 "Establishment of Alert and Action levels".
- 2.18.28 Microbial testing shall be performed only for equipment operating in class ISO 8 area and CNC area (controlled but not classified) area. Limits shall be established based on area swabbed and class of area.

Visual Verification requirement for different types of cleaning

- 2.18.29 **Type A Cleaning** – Production operator shall visually inspect equipment, there shall not be any gross accumulation of powder / material residue of previous batch of same product on contact surfaces.
- 2.18.30 **Type B cleaning** – Equipment shall be inspected visually for presence of any product residues. It shall be visually clean, no product residues, traces etc shall be visible. Refer SOP 1035-A-0083 for Visual inspection of equipment.
- 2.18.31 Normal human naked eye can detect approximately 100 µg per 4 Inch² of residue. Below this level the residue is not visible to human naked eye.

2.19 Selection of worst-case product for cleaning validation

- 2.19.1 Site can choose to perform cleaning validation for each product instead of worst-case approach.

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2.19.2 For Worst Case approach, worst case product shall be selected for cleaning validation based on solubility Toxicity (Least PDE) and cleanability for each solvent group within each manufacturing block. For example, there are 15 products in a manufacturing block and two solvent groups then minimum two worst case products need to be validated, one from each solvent group.

2.19.3 **Step 1: Ranking based on solubility profile in particular solvent group:**

2.19.3.1 Rating shall be assigned to each product based upon the solubility profile as per below table. **Rating:** Rating is a valuation or rank on a scale based on solubility, 1 is best and 5 is worst.

2.19.3.2 Ranking based on solubility profile:

Solubility	Approximate quantities of solvent by volume for 1 part of solute by weight (at 25°C).	Rating
Very soluble	Less than 1 part.	1
Freely soluble	From 1 to 10 parts	2
Soluble	From 11 to 30 parts	3
Sparingly soluble	From 31 to 100 parts	4
Slightly soluble	From 101 to 1000 parts	5
Very slightly soluble	From 1001 to 10000 parts	
Practically insoluble	More than 10000 parts	

2.19.3.3 Solvents with less solubility shall be avoided for cleaning of equipment. Based on evaluation of risk, a separate cleaning SOP shall be prepared for product having worst solubility rating (i.e. 5) and cleaning validation shall be performed for each case. For Detergent / Surfactants solubility shall be considered as Practically insoluble.

2.19.3.4 Products shall be selected based on solubility in different solvent group. API / Intermediate having least solubility in their cleaning solvent are most difficult to clean and possibility of carryover contamination of that ingredient into the next product is maximum. Hence the products having worst solubility profile in their cleaning solvent shall be selected as the worst-case product in this criterion.

2.19.4 **Step 2: Ranking based on toxicity (Least PDE value):**

2.19.4.1 Collect the PDE value of each product / API, product having lowest PDE value shall be selected as worst case in 'toxicity' criterion. PDE evaluation report shall be made by toxicologist as per SOP 1035-G-0209.

2.19.4.2 Rating shall be assigned to each product based upon the toxicity (PDE) as per below table.

Parameter	Criteria (mg/day)	Rating
PDE / Toxicity (mg / day)	More than 100	1
	10 to 99.9	2
	1 to 9.9	3
	0.01 to 0.99	4
	Less than 0.01 mg	5

2.19.5 **Step - 3: Worst Case Products on basis of difficulty to clean.**

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- 2.19.5.1 Difficult to clean Products shall be identified based on feedback from operators and shop floor staff. Refer annexure 1035-A-0072/A24 for documentation of operator interview.
- 2.19.5.2 For new products feedback of API R&D as per SOP # RD-S-006, title Cleaning Procedure Development shall also be considered while deciding the rating.
- 2.19.5.3 *Typically, product which forms hard scale on equipment surface, sticky or gummy mass etc shall be considered as difficult to clean.*
- 2.19.5.4 Guidance criteria provided below shall be used to determine the degree of difficulty in cleaning.

Parameter	Category of cleaning	Rating
Difficult to Clean	Easily cleanable	1
	Moderately Difficult to clean	2
	Difficult to clean	3

- 2.19.6 **Step – 4: Final RPN calculation:**
- 2.19.6.1 For API worst product selection: The rating of Solubility, Toxicity (PDE), and Difficult to clean shall be multiplied to get final score (Solubility Rating X Toxicity (PDE) X Difficult to clean). The highest RPN score product shall be identified as the worst-case product.
- 2.19.6.2 For Intermediate worst product selection: The rating of Solubility and Difficult to clean shall be multiplied to get final score (Solubility Rating X Difficult to clean). The highest RPN score product shall be identified as the worst-case product
- 2.19.6.3 RPN calculation shall be performed for the product based in solubility group.

Example of final RPN calculation for API worst product selection:

Product name	Based on solubility	Based on toxicity (PDE)	Based on difficult to clean	RPN calculation
Product A	1	1	1	1
Product B	2	2	1	4
Product C	3	3	2	18
Product D	4	4	2	32
Product E	4	5	3	60

- 2.19.7 Based on RPN calculation Product E is identified as worst product. Review the production schedule if worst product is not being manufactured then cleaning validation shall be performed on second or third worst case product.
- 2.19.8 Data Compilation for highest RPN score approach shall be done in validated “Microsoft Excel” / calibrated calculator.
- 2.19.9 Based on the calculation, if same RPN score is obtained for more than one product than lower PDE value API shall be considered as a worst-case.
- 2.20 **Analytical Method requirement for Rinse and Swab Sampling:**

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- 2.20.1 Analytical methods used for estimation of Residues shall be validated as per SOP 1035-L-0016, "Validation and Verification of Analytical Method".
- 2.20.2 *Recovery studies shall be performed for all material of constructions (MOC) used in manufacturing of product.*
- 2.20.3 A template of defined size (4 sq inch, 9 sq inch etc.) shall be used for recovery study. Template MOC shall be same as product contact surface MOC. Example of surfaces used are Stainless steel, Glass lined, Hastelloy, Halar, Teflon, Silicone, rubber, Neoprene etc.
- 2.20.4 QA shall provide the signed copy of list of MOCs to Analytical development / QC department for Analytical method validation. In case of introduction of any new MOC due to change in equipment QA shall inform the same to QC and QC shall plan the recovery study for new MOC.
- 2.20.5 Grouping or family approach for recovery studies: Perform one residue study on a surface which through documented evidence is equivalent (in terms of percent recovery) to other surfaces for which a formal recovery study is not performed. Equivalence for establishing such group or family shall be established based on published studies or in-house data back up with scientific risk-based justifications.
- 2.20.6 Alternatively, another approach is to exclude formal recovery studies for sampled surfaces constituting less than a small percentage i.e. 2% of the total equipment surface area (e.g. gaskets). *In such cases, based on justification the recovery value to be used for the excluded surface shall be the lowest recovery of any other surface type for which a formal sampling recovery study shall be performed, or the minimum acceptable recovery percentage required by the applicable procedures.*
- 2.20.7 % Area for Gaskets shall be calculated by adding areas of all gaskets (for which recovery study is not available) divided by total surface area for that product. Gasket area shall be considered for product contact area only.
- 2.20.8 The approach through which the recovery study is planned and proceeded shall be adequately mentioned and justified in respective Analytical method validation documentations.
- 2.20.9 The rinse and swab limit shall be higher than LOQ value of analytical method. In case the acceptance criteria are found less than the LOQ of the analytical method, one or more of the following actions may be initiated. Change control procedure shall be followed for implementation of changes.
- Analytical method shall be optimized to achieve lower limit of detection by slight modification, such as increasing injection volume in case of chromatographic method like HPLC/ GC/ UPLC etc.
 - New more sensitive analytical method shall be developed and validated.
 - Equipment used in manufacturing of Product made dedicated.
 - Larger surface area considered for swabbing (9 sq. inch instead of 4 sq. inch).
 - In case of rinse sampling, decrease the volume of rinse sample, resulting into increase in the residue concentration.
 - Increasing the batch size of next product.
 - Restricting the order of manufacture of the product.
- 2.21 **Visual inspection:**
- 2.21.1 Personnel involved in equipment cleaning, sampling and line clearance shall be trained on a SOP 1035-A-0083 "Training on cleaning of equipment and visual inspection".

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- 2.21.2 The person performing visual inspection shall be qualified as per respective procedure (1035-A-0083).
- 2.21.3 Visual inspection is the process of using the human eye, alone or in conjunction with various aids, as the sensing mechanism from which judgments may be made about the condition of the surface to be inspected.
- 2.21.4 Supplementary aids, (such as a camera, mirror, torch) shall be used to enable inspection of possible residues in hard-to-reach areas which are not in line of sight.
- 2.22 **Cleaning Matrix:**
- 2.22.1 Cleaning Matrix is calculation of MACO limit for all possible product changeover scenarios in the production block.
- 2.22.2 Cleaning Matrix shall be prepared block wise.
- 2.22.3 Cleaning matrix shall have all the data required for doing MACO calculation e.g. Products manufactured in block, PDE value, Maximum daily Dose, Cleanability details, *Surface Area of equipment included small parts and utensil* used in particular product, cleaning solvent, Solubility data, last rinse volume, Method LOD, LOQ values, etc.
- 2.22.4 Ensure rinse limit and swab limit shall be higher than LOQ value of analytical method, else either more sensitive method shall be used or that product cannot be manufactured in the block.
- 2.22.5 Select the API and intermediate worst-case product for cleaning validation based on RPN (Risk Priority Number) calculation. For the API, the MACO (Maximum Allowable Carryover) limit shall be used for cleaning validation, while for the intermediate product, a limit of 100 PPM shall be used for cleaning validation.
- 2.22.6 If the product is manufactured by two different processes, the total final rinse volume and surface area of chain of equipment shall be calculated for both the processes (if cleaning procedures are different). Least final rinse volume and surface area of chain of equipment shall be considered for MACO calculation.
- 2.22.7 QA shall prepare and approve a computerized pre-printed report of cleaning matrix for API and Intermediates as per annexure 1035-A-0072/A3 and 1035-A-0072/A4 respectively. Cleaning matrix for API shall be numbered as "Plant code/CLMX/API/API block" and for Intermediate numbered as "plant code/CLMX/INT/API block" For e.g. 1030/CLMX/API/API-I and 1030/CLMX/INT/API-I, 1030 is the plant code of Patalganga unit-I, CLMX/API is for cleaning matrix for API, CLMX/INT is for cleaning matrix for intermediates and API-I is the manufacturing block number.
- 2.22.8 If other product MACO, rinse and swab limit observed stringent than the selected worst-case product (based of RPN calculation), then same product also be selected for one batch cleaning verification run.
- 2.22.9 If the worst-case product is not planned within 3 years and cannot be validated due to limited batches or any other reason, then the review and validation of the second or subsequent worst-case product shall be considered. Cleaning validation shall be re-run for the first worst-case product when it becomes available.

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- 2.22.10 Cleaning matrix is dynamic document and shall be kept up to date based on changes which impacts the cleaning matrix, e.g. addition of new product, addition of equipment, change in dose, change in batch size change in PDE etc.
- 2.22.11 If a commercial product has not been manufactured in the last 3 years, and/or if the API was manufactured for registration purposes (trial batch) and there is no future or forecasts plan after validation, then that API shall be removed from the cleaning matrix.
- 2.22.12 After completion of pre-validation / feasibility batches, cleaning matrix shall be amended prior to take validation batches in manufacturing facility. (if required).
- 2.22.13 List of cleaning matrix amendments shall be prepared as per annexure 1035-A-0072/A5.
- 2.23 **Pre-Requisite for Cleaning Validation:**
- 2.23.1 **Cleaning Validation** is the generation of documented evidence that the cleaning procedures consistently remove residues of pharmaceutical actives and the contaminants (byproduct, in situ, unreacted material, degradant, reagent, solvent etc.) from the manufacturing facility and equipment to predetermined level to protect the follow up pharmaceutical product from cross contamination.
- 2.23.2 Prerequisites: Ensure that the prerequisites for cleaning validation as listed below are available:
- Validated analytical method for measurement of product in the rinses and swab.
 - Validated analytical method for measurement of cleaning agent in the rinses. (If cleaning agent used for cleaning).
 - Validated microbiological method for measurement of bioburden on the equipment (In case of micro sample).
 - Solubility chart, therapeutic daily dose, batch size (output), equipment surface area data.
 - Permitted daily exposure (PDE) report of API.
 - Product change over cleaning procedure with ECR checklist.
 - An approved sampling plan.
 - Specification for cleaning sample analysis.
 - Training and qualification of personnel on cleaning procedure as per SOP 1035-A-0083
 - Training and qualification of personnel on visual inspection of equipment as per SOP 1035-A-0083.
 - Training of personnel on sampling plan.
 - Training of personnel on swab sampling.
 - Training of personal validation protocol and report.
 - Check availability of sampling tools like sampling bottle for rinse collection, swab stick required for swab collection with MOC(s) details, swab type, make / manufacturing date / validity, 4/9 square inch template, swab sample collection test tube and label for collected samples.
- 2.24 **Sampling plan:**
- 2.24.1 Before initiation of cleaning validation, sampling plan shall be prepared and approved as per annexure 1035-A-0072/A6.
- 2.24.2 Sampling plan shall have provision for the below information but not limited to,
- Equipment name along with code
 - Previous product name along with batch number
 - Cleaning SOP number with version number

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- Previous product manufacturing completed on
- Cleaning started on with cleaning done by
- Cleaning solvent name
- Type, lot no., manufacturing date of swab stick
- Swabbing diluent
- Date of sampling / validation, etc.
- Rinse sample and swab samples after cleaning step.

- 2.24.3 Location of swab samples shall be selected as per SOP 1035-A-0082. Equipment parts with maximum product contact which are hard to clean shall be considered during cleaning validation.
- 2.24.4 Rinse and swab sampling checklist as per SOP 1035-A-0082 shall be filled during rinse and swab sampling.
- 2.24.5 Individual equipment specific diagram with swab location shall be prepared as per SOP 1035-A-0082 and same shall be attached with cleaning validation protocol and report.
- 2.24.6 If swab or rinse sample is missed during sampling, then deviation shall be initiated as per SOP 1035-G-0205.
- 2.25 ***Equipment Grouping / optional equipment cleaning:***
- 2.25.1 *When the grouping of equipment is performed, Justification for the group shall be documented / included in the validation documentation.*
- 2.25.2 If two different types of equipment proposed in same processing stage. (for e.g. ANFD and CF used for filtration or RCVD and TD used for drying). Three run cleaning validation shall be done with both the types of equipment.
- 2.26 **Execution of cleaning validation protocol:**
- 2.26.1 After confirming the availability of prerequisites and cleaning matrix for cleaning validation, equipment cleaning of selected product shall be planned.
- 2.26.2 The ECR (Equipment cleaning record) shall be issued by QA to production for initiation of cleaning validation as per request.
- 2.26.3 According to ECR cleaning of equipment shall be initiated by production.
- 2.26.4 Approx 100 ml of cleaning solvent shall be collected as a blank for evaluation of interference of cleaning solvents by QA.
- 2.26.5 Before rinse/ swab sample collection, TI sheet shall be raised in SAP by Quality Assurance / Production as an intimation to QC for cleaning sample and preparation of swab diluent to store the swab stick.
- 2.26.6 On rinse sample TI sheet, details of Product name, batch number, equipment name, MOC of equipment, analysis code, rinse sample number, rinse quantity, blank fresh solvent name and quantity and limit / acceptance criteria for rinse shall be included.

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- 2.26.7 On swab sample TI sheet, details of Product name, batch number, equipment name, MOC of equipment, analysis code, swab sample number, information related to MOC of equipment parts and acceptance criteria for swab shall be included.

A. Procedure for Rinse sampling:

- 2.26.8 Ensure adequate gowning and gloves wear prior to rinse sampling.
- 2.26.9 Ensure availability of cleaned bottles to avoid any potential contaminant carryover.
- 2.26.10 As per sampling plan, collect the entire volume of rinse content in suitable tank / vessel and collect the portion of rinse sample (About 100 ml) into sample bottle by quality assurance.
- 2.26.11 For rinse sample, entire volume of rinse content shall be collected in suitable tank/ vessel and portion of rinse sample (about 100ml) shall be collected into sample bottle by quality assurance.
- 2.26.12 The rinse samples collected shall represent the entire rinse volume.
- 2.26.13 After collection of rinse sample, clarity of rinse sample shall be checked by QA and recorded the same on sampling plan. *Each rinse sample shall be labelled as per annexure 1035-A-0072/A7.* After labelling, rinse sample shall be sent immediately to QC for analysis of previous product content / cleaning agent.

B. Procedure for Swab sampling:

- 2.26.14 As per quantity of swabs mention on TI sheet, swab sample collection test tube / bottle shall be labelled as per annexure 1035-A-0072/A7 and handed over to Quality control for issuance of diluent (10 ml for each swab).
- 2.26.15 After the issuance of the diluent, the sample collection test tubes / bottles shall be taken for swab collection according to the sampling plan.
- 2.26.16 Equipment / equipment part shall be dried according to the instruction specified in ECR after the final rinse. The dried equipment and its part shall be visually inspected for cleanliness by production, followed by QA.
- 2.26.17 After a satisfactory visual inspection, swab samples shall be collected by Quality Assurance as per SOP 1035-A-0082. The collected swab samples along with the sampling plan shall be sent to QC for analysis of previous product content.
- 2.26.18 Quality Control shall initiate the analysis as per cleaning specification of product.
- 2.26.19 *Analysis of cleaning validation sample shall be initiated within 24 hours after sampling, until and unless solution stability studies are available.*
- 2.26.20 If rinse or swab result observed more than limit, out of specification shall be raised reported and handled as per SOP 1035-L-0143. Corrective and preventive action shall be taken based on investigation.
- 2.26.21 If extraneous peak observed during analysis, incidence shall be raised, investigated as per SOP 1035-L-0146. Corrective and preventive action shall be taken based on investigation.

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- 2.26.22 Sampling of *Swab and Rinse* sampling may be done for cleaning validation. *Swab and rinse sampling are acceptable methods of sampling, and the choice of either or both will depend on the equipment type and accessibility to the sampling location.*
- 2.26.23 During cleaning validation, equipment shall not be used for further processing, till the results of chemical testing of previous product comply as per acceptance criteria.
- 2.26.24 Equipment which are common for intermediate stages of API and final isolation stages shall be validated as per protocol for worst case of API for cleaning validation.
- 2.27 **Reporting of Results and Conclusion of Cleaning validation:**
- 2.27.1 After completion of analysis, Quality Control shall provide the result with completed TI sheet to Quality Assurance / Production.
- 2.27.2 Compliance of all the results shall be verified by quality assurance, if non-compliance observed investigation shall be opened.
- 2.27.3 Details of critical cleaning process parameters (CPP) shall be recorded from ECR of equipment.
- 2.27.4 Calculation details with respect to limits obtained for final rinse and swabs shall meet the established acceptance criteria. Observation / non-conformance details shall be reported in validation report.
- 2.27.5 Total carryover by rinse or swab method for the equipment chain observed more than the MACO limit, a deviation shall be raised and same shall be investigated as per the respective SOP.
- 2.27.6 Results and observation of individual equipment and equipment chain shall be compiled and approved by Section Head / Head Unit Quality Assurance.
- 2.27.7 Comparison of all three chain / runs shall be done and final conclusion with recommendation shall be drawn by validation team and approved by Unit Head and Head Unit Quality Assurance.
- 2.27.8 Conclusion and recommendation of the cleaning validation report shall be discussed about the following, but not limited to:
- Number of batches that are assessed for cleaning validation.
 - Rinse and swab samples were collected as per sampling plan. The swab samples collected by using Texwipe along with the acceptance criteria for rinse and swab limits.
 - Critical cleaning process parameters like, reflux time, temperature, soaking and scrubbing time, solvent volume, details of cleaning aids (Scrubber/ brush) and jet with pressure parameter etc.
 - Clarity of last rinse sample of each equipment found visually clear.
 - Visual inspection methodologies followed along with their conclusions.
 - No change in existing cleaning procedure for all three-equipment chain / verification run
 - Conclusion on application of validated / verified cleaning procedures for routine equipment cleaning.
 - Any changes that require to existing cleaning procedure for routine application of product change over or campaign cleaning.
 - In the event, of any non-conformity, then conclusion and recommendation shall cover any change in cleaning procedure or solvent quantity, or reflux conditions or number of refluxes required, accordingly the procedure shall be revised and / or revalidate the cleaning procedure.
 - Any changes / CAPA requires to be implemented as an outcome of cleaning validation, QMS shall be initiated for the same and reference QMS document shall be mentioned in report.

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2.27.9 Attachments: Following shall be attached to the validation report:

- Approved sampling plan for the equipment.
- Technical information sheet of cleaning validation sample analysis.
- Photocopy of equipment cleaning record.

2.27.10 Cleaning validation protocol for API shall be prepared as per annexure 1035-A-0072/A8 and shall be numbered as "Plant code/APIWCCVP/API block" For e.g. 1030/APIWCCVP/API-I, 1030 is the plant code of Patalganga unit-I, APIWCCVP is for API worst case cleaning validation protocol, API-I is manufacturing block number.

2.27.11 Cleaning validation protocol for intermediate shall be prepared as per annexure 1035-A-0072/A10 and shall be numbered as "Plant code/INTWCCVP/API block" For e.g. 1030/INTWCCVP/API-I, 1030 is the plant code of Patalganga unit-I, INTWCCVP is for intermediate worst case cleaning validation protocol, API-I is manufacturing block number.

2.27.12 The computerised pre-printed report shall be prepared for reporting of cleaning validation results, conclusion and recommendations. API cleaning validation report shall be prepared as per annexure 1035-A-0072/A9 and numbered as Plant code/APIWCCVR/API block. Intermediate cleaning validation report shall be prepared as per annexure 1035-A-0073/A11 and numbered as Plant code/INTWCCVR/API block.

2.28 **Initial cleaning validation**

- *Worst-case product: Three runs for complete equipment chain.*
- *Subsequent worst-case product: Three runs for that equipment which are not covered in worst-case product.*
- *Optional equipment of worst-case product:*
- *If ECR is same and solvent quantities are different depending on capacity: Three runs each shall be performed on lowest and highest capacity of equipment.*
For example, in case there are three blenders of different capacities 500 Liter, 1250 Liter and 2250 Liter (ECR is same and solvent quantities are different) then cleaning validation 3 runs each shall be performed on the lower capacity blender 500 Liter and higher capacity blender 2250 Liter.
- *If ECR is same and solvent quantities are also same. Total three runs shall be performed on any of the optional equipment.*
For example, in case there are two reactors of capacities 1 KL and 1.5KL (ECR is same and solvent quantities are same) then three runs of cleaning validation shall be performed on any equipment combination.

2.29 **Periodic validation / verification:**

2.29.1 Verification of cleaning procedures for worst case product shall be performed every three years by checking rinse / swab sample for one batch / run. Protocol for periodic cleaning verification shall be prepared as per annexure 1035-A-0072/A8.

2.29.2 A computerised pre-printed report shall be prepared for reporting of cleaning verification results, conclusion and recommendations of periodic cleaning verification. Report shall be prepared as per annexure as per annexure 1035-A-0072/A12 and numbered as Plant code/WCCVR/API block.

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- 2.29.3 *Results observed during initial cleaning validation and periodic cleaning validation shall be compared in final summary report of periodic cleaning validation. In case verification sample results are significantly higher than the validation results (More than 75 % of the limit) consider the same as out of alert and investigate the reason for the same.*
- 2.29.4 *Periodic verification to be performed on one batch per year considering least PDE in the block.*
- 2.29.5 *Periodic cleaning validations / verifications of dedicated / non-dedicated equipment shall be tracked through validation master plan.*
- 2.29.6 Refer Annexure 1035-A-0072/A13 for Flow Diagram of validation and implementation of validated cleaning procedure.
- 2.30 **CONTINUED CLEANING PROCESS VERIFICATION: - Routine monitoring during product changeover cleaning:**
- 2.30.1 After successful completion of three cleaning validation runs. Validated cleaning procedure shall be used for product change over cleaning.
- 2.30.2 Equipment shall be verified 'visually clean' before releasing for the next use. Detailed procedure for Visual Cleanliness certification is provided in SOP 1035-A-0047.
- 2.30.3 Apart from visual inspection, Swab / Rinse sample shall be collected from individual equipment and same shall be analysed for the content of previous product. TI sheet shall be raised by production department.
- 2.30.4 *Sampling plan: Site shall make equipment wise sampling plan for routine monitoring. Sampling plan for routine monitoring shall be subset of samples taken during validation runs. Number of samples shall be less than validation and shall consist of hard to clean locations. Rinse sample shall be considered for equipment cleaned in place (large equipment having higher surface area). Documented justification shall be available for sampling plan. SOP 1035-A-0082 shall be referred for collection of samples.*
- 2.30.5 Analysis of samples shall be initiated within 24 hours after sampling, until and unless solution stability studies for more duration are available. Initial notification shall be provided to QC on anticipated samples expected to be received in QC for having the instrument ready for testing.
- 2.30.6 In case of product for which campaign cleaning validation is done and campaign length is established as per this SOP, Routine monitoring of cleaning samples shall be waived off when next product is same.
- 2.30.7 *Results of routine monitoring shall be compared with the cleaning limits and if results are observed out of limit, then it shall be investigated through OOS procedure.*
- 2.31 **Annual cleaning review report:**
- 2.31.1 Annual cleaning review report shall be prepared and documented as per annexure no. 1035-A-0072/A14. During review data shall be compiled for the period of January to December. Report shall be approved before end of February month of next year. Annual Cleaning Review report shall have following data.
- 2.31.2 Review of Training and Qualification of concerned employee on visual inspections.

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- 2.31.3 Review of Training of concerned employee on Cleaning of Equipment.
- 2.31.4 Review of Training and qualification of concerned employee on Swab sampling.
- 2.31.5 *Review of all deviations related to cleaning and line clearance. Trending of deviations shall be done and additional CAPA actions shall be derived based on Trends.*
- 2.31.6 Review and trending of OOS and OOAL (out of Alert limit) cases to verify root cause and CAPA adequacy.
- 2.31.7 Review of Changes related to cleaning procedures and cleaning matrix.
- 2.31.8 Review of cleaning validation and verifications runs.
- 2.31.9 Review of open change controls and CAPA related to cleaning process.
- 2.31.10 Based on review of data and trends conclusion shall be drawn and additional CAPA actions shall be anticipated.
- 2.32 ***Impact assessment of changes and revalidation criteria***
- Any change in cleaning procedure, analytical methods and in master data required to make cleaning matrix may potentially have impact on cleaning limits and Cleaning Validations.
 - All such changes need to be evaluated and impact on validated cleaning state shall be assessed.
 - For impact assessment of changes and revalidation criteria refer annexure 1035-A-0072/A25. Cleaning matrix shall be updated during each change.
- 2.33 **Microbiological Aspect of cleaning validation:**
- 2.33.1 Bioburden study of equipment shall be performed during worst case product cleaning validation of each solubility group. Bioload sample shall be collected by Microbiologist / QA.
- 2.33.2 Bioburden study shall be performed only for equipment used in controlled and classified or controlled but not classified (CNC) clean areas. Typically for manufacturing of non-sterile API environment is non controlled till crystallisation step hence no microbial limits applied to such equipment.
- 2.33.3 Micro swab sample shall be collected first from different location other than chemical swab sample.
- 2.33.4 Micro swab samples shall be collected from each group of equipment specified in below table. The swab sample shall be collected after cleaning, drying and visual inspection of equipment prior to chemical swab and sent to QC for complete microbiological analysis.

Group	Equipment	Runs
Group A	Crystalliser Reactor	Three runs
Group B	Closed type equipment like centrifuge, Blender, Dryers, RCVD, VTD, ANFD, RVPD, FBD etc.	Three runs
Group C	Portable equipment & equipment which shall be dismantled completely- E.g. Multi mill, Sifters, Jet mill, Cad mill, Co mill, sparkler filter, Nutsche filter, Buckner filter etc	Three runs

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- 2.33.5 Rationale for Equipment grouping: As equipment are cleaned with common cleaning procedure and based on solubility, study shall be done for at least one equipment from each group.
- 2.33.6 The micro study shall be conducted in three runs for any worst case or non-worst-case product of each solubility group.
- 2.33.7 The bioburden of equipment shall comply with the limit as specified in SOP 1035-L-0086, Microbiological environment monitoring of Air, surfaces and personnel' which are established in reference to the SOP 1035-L-0089 'Establishment of Alert Level'.
- 2.33.8 The bioburden TI sheet shall be attached with cleaning validation report of worst-case product of each solubility group.
- 2.33.9 Periodic microbiological study: Once in three years on one run from each equipment group on any worst case or non-worst-case product from each solubility group.
- 2.34 ***Nitrosamine and Genotoxic compounds evaluation during cleaning.***
- 2.34.1 Product manufactured at site shall be assessed for cleaning requirement for product containing genotoxic / Nitrosamine compound.
- 2.34.2 R &D, Tech Transfer, MSTG, Site QA, QC, Manufacturing team shall review the assessment report of genotoxic / Nitrosamine compound and risk category (i.e. High, low and no risk) shall be evaluated.
- 2.34.3 Based on evaluation of risk category, action plan shall be assigned for cleaning requirement.
- 2.34.4 Assessment for cleaning requirement of product containing genotoxic / Nitrosamine compound shall be performed as per annexure 1035-A-0072/ A15.
- 2.35 ***Dirty Equipment Hold Time (DEHT) study:***
- 2.35.1 Dirty equipment hold time is the time between the completion of batch processing step in the equipment till the start of cleaning procedure. For example, for reactor time from reactor is emptied out till start of pre cleaning / rinsing of reactor.
- 2.35.2 DEHT is important for the control of a cleaning process. The nature of the "soil" on the equipment surfaces may change over time. Those changes include drying of the soil, degradation etc. Such changes may make the soil more difficult to remove by the cleaning process. Since the cleaning process shall be designed to adequately clean the equipment under the worst cases of normal operating conditions, a demonstration of effectiveness under the worst-case (usually the longest) DEHT shall be documented.
- 2.35.3 DEHT shall be done by holding the equipment for cleaning for certain time before commencing the type B cleaning.
- 2.35.4 Rationale for selection of product: DEHT study shall be conducted for worst case product selected based on RPN calculation from each solvent group when common cleaning procedures are used based on solubility. Priority shall be given for the product where the water base reaction equipment studied for DEHT.

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2.35.5 Rationale for selection of equipment: As equipment is cleaned with common cleaning procedure based on solubility, study shall be done for at least one equipment from each group.

Group A : Crystalliser Reactor

Group B : Closed type equipment – E.g., Blender, Dryers, RCVD, VTD, ANFD, RVPD etc.

Group C : Portable equipment & equipment which shall be dismantled completely- E.g. Multi mill, Sifters, Jet mill, Cad mill, Co mill etc.

2.35.6 *The study shall be conducted in three runs.*

2.35.7 Based on routine working condition DEHT target time shall be at least 72 hours. Site may decide to have higher DEHT.

2.35.8 *During cleaning validation of worst-case product out of three runs at least one run shall be done with dirty hold time.* DEHT study can also be combined with periodic cleaning verification or campaign cleaning validation.

2.35.9 During dirty equipment hold time study, if operator have any observations like hardening of residue, formation of sticky or gummy mass, etc. then same shall be documented before start of cleaning.

2.35.10 Procedure for DEHT Study: After completion of processing, equipment shall be stored under prevailing condition. The equipment which is identified shall not be cleaned by Type A or Type B till holding period completed. Post that Type B cleaning shall be performed, and cleaning validation sample shall be taken in same way as per regular cleaning validation study. Time until equipment was hold dirty shall be recorded.

2.35.11 DEHT study shall be part of cleaning validation protocol, sampling of DEHT study shall be same as regular cleaning validation. Refer Annexure 1035-A-0072/A16 for DEHT study protocol and shall be numbered as “SP/Plant code/DEHT” For e.g. SP/1030/DEHT, where SP is for study protocol, 1030 is the plant code of Patalganga unit-I. and DEHT is Dirty Equipment Hold Time.

2.35.12 A computerized pre-printed report shall be prepared for DEHT study and shall be numbered as SR/1030/DEHT as per specimen annexure 1035-A-0072/17.

2.35.13 If observations from visual cleanliness and swab rinse studies for DEHT run are within acceptance criteria, Dirty hold time shall be concluded. Dirty hold time shall be included in respective cleaning procedures.

2.35.14 In case results of DEHT study is fail, then dirty hold time shall be reduced, and study shall be planned again. Failure shall be handled as per respective deviation / OOS procedures.

2.35.15 *Deviation shall be initiated if equipment exceeds the validated dirty equipment hold time. Further, equipment shall be cleaned with product changeover cleaning (type B) followed by cleaning verification.*

2.36 **Clean Equipment Hold Time (CEHT) Study:**

2.36.1 A clean equipment hold time study shall be conducted to ensure that cleaned equipment, when stored under prescribed conditions, does not exceed the microbial contamination levels of freshly cleaned equipment.

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- 2.36.2 The study shall be performed only for equipment used in controlled and classified or controlled but not classified (CNC) clean areas. Typically for manufacturing of non-sterile API environment is non controlled till crystallisation step hence no microbial limits applied to such equipment.
- 2.36.3 Rationale for selection of product: CEHT study shall be done for any one product from each solvent group when common cleaning procedures are used based on solubility. Priority shall be given for the product where cleaning solvent is water.
- 2.36.4 Rationale for selection of equipment: As equipment are cleaned with common cleaning procedure and exposed to same environment, study shall be done for at least one equipment from each group.
- Group A : Crystalliser Reactor
Group B : Closed type equipment – E.g., Blender, Dryers, RCVD, VTD, ANFD, RVPD etc.
Group C : Portable equipment & equipment which shall be dismantled completely- E.g. Multi mill, Sifters, Jet mill, Cad mill, Co mill etc.
- 2.36.5 *The study shall be conducted in three runs.*
- 2.36.6 *Target for CEHT study shall be maximum 7 days.*
- 2.36.7 Procedure for CEHT Study:
- Equipment shall be cleaned and dried according to the Type B cleaning procedure (product changeover cleaning).
 - The equipment designated for the study shall not be used for manufacturing until the study is completed.
 - After cleaning, equipment shall be kept in closed conditions as per procedure.
 - Equipment surfaces shall be visually inspected to ensure cleanliness before swab sampling.
 - Swab sampling locations for study equipment shall be identified.
 - Swab samples shall be collected by Microbiologist / QA from identified product contact surface after cleaning and drying (Day 1).
 - Equipment shall be closed and protected as prescribed in the plant SOP.
 - Product contact surfaces shall be periodically sampled for microbiological proliferation on Day 4, Day 6 and Day 7.
 - Swab samples shall be collected from adjacent locations identified for swabbing each time.
 - Swab samples shall be sent to the microbial lab along with the TI sheet.
 - Based on the results, determine the maximum number of days the equipment shall be kept after cleaning before re-cleaning is required.
 - If Day 7 results are meeting the acceptance criteria CEHT shall be defined as 6 days.
- 2.36.8 CEHT study shall be performed based on an approved protocol, the template for protocol is provided as Annexure 1035-A-0072/A18 and shall be numbered as “SP/Plant code/CEHT” For e.g. SP/1030/CEHT, where SP is for study protocol, 1030 is the plant code of Patalganga unit-I and CEHT is for Clean Equipment Hold Time.
- 2.36.9 A computerized pre-printed report shall be prepared for CEHT study and shall be numbered as SR/1030/CEHT as per specimen annexure 1035-A-0072/19.
- 2.36.10 The CEHT study report shall be summarized by QA after the study's completion for the selected equipment and product. Supporting data shall be attached to the report.

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2.36.11 Based on CEHT report clean equipment hold time shall be included in routine cleaning procedures.

2.36.12 In case equipment is idle after defined clean equipment hold time it must be cleaned again before use.

2.37 **Campaign Length Cleaning Validation:**

2.37.1 **Campaign cleaning validation** refers to the process of ensuring that cleaning procedures are effective in removing residues from equipment used in the production of multiple batches of the same product, manufactured consecutively.

2.37.2 In a campaign, batches are produced one after another without a full cleaning between each batch. Instead, a more thorough cleaning, "Product changeover cleaning" or "Type B cleaning," is performed after the final batch of the campaign. This ensures that any residues from the previous batches are adequately removed before starting a new campaign or switching to a different product.

2.37.3 **Approach for deciding campaign length & campaign time.**

- During initial cleaning validation, cleaning is typically performed after every batch. For deciding campaign length, a step wise approach of increasing campaign length shall be followed. Typical example is given below.
- Short campaign length of 3 to 5 batches shall be started and cleaning verification shall be completed after the campaign. If results of cleaning are comparable with initial cleaning validation and quality of the batches manufactured is as per trend, either same campaign length shall be validated with two more runs or if product volumes are high step wise increase shall be done.
- Risk assessment shall be performed considering experience of previous campaign, extend of batch-to-batch cleaning etc to decide the number of batches in campaign. Typically target of 10-15 batches shall be taken. If results of cleaning are comparable with initial cleaning validation and previous short campaign run and quality of the batches manufactured is as per trend, then campaign length shall be validated with two more runs.
- *Campaign length shall not be extended beyond 30 batches or three months whichever is earlier.*
- For product with low cycle time and high volumes, number of batches in campaign could be more than 30, however such cases shall be justified based on risk assessment and good batch to batch cleaning procedure which avoids build-up of residue in equipment.
- Campaign based cleaning validation shall be executed on three campaigns possibly having same number of batches of each group of worst-case product.
- In the campaign cleaning validation, one out of three campaign runs shall be extended to establish uncleaned equipment hold time.

2.37.4 **Selection of Product:**

- API : Campaign length cleaning validation shall be performed on the worst-case product of each group, which is selected based on the RPN calculation of less solubility, less PDE and cleanability of each solvent group. If API having highest campaign length than selected worst-case product then same API highest campaign length also be validated.
- Intermediate: Campaign length cleaning validation shall be performed on the worst-case product of each group, which is selected based on the RPN calculation of less solubility and cleanability of each solvent group. If intermediate having highest campaign length than selected worst-case intermediate, then same intermediate highest campaign length also be validated.
- For example: if product "A" is selected as a worst-case product from methanol soluble group having campaign length 7 to 8 batches and product "B" from methanol soluble group is having campaign length 14 to 15 batches, then both the product shall be validated for campaign length.

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- If the sufficient number of batches of identified worst case product are not in manufacturing plan, then campaign-based cleaning validation shall be performed on subsequent worst-case product.
- Campaign cleaning validation shall be performed considering total equipment chain used for the worst-case product. If any type of equipment is not part of the worst-case product equipment chain, then such equipment campaign length validation shall be performed on subsequent worst-case product.

- 2.37.5 **Campaign Cleaning Validation Procedure:** Campaign based cleaning validation study shall be executed based on the study protocol designed for campaign length establishment as annexure 1035-A-0072/A20. Campaign cleaning validation study protocol shall be numbered as SP/PLANT CODE/CAMPCL, e.g. SP/1030/CAMPCL, where SP is for study protocol, 1030 is the plant code of Patalganga unit-I and CAMPCL is for Campaign cleaning.
- 2.37.6 A computerized pre-printed report shall be prepared for Campaign cleaning validation study and shall be numbered as SR/PLANT CODE/CAMPCL as per specimen annexure 1035-A-0072/21.
- 2.37.7 Batch to batch cleaning (Type A) shall be carried out during campaign length cleaning. At the end of campaign, product change over cleaning (Type B) shall be carried out.
- 2.37.8 Campaign length cleaning validation shall meet below three objectives.
- A) Type B cleaning procedure shall be able to clean the equipment as per defined acceptance criteria. Results of campaign cleaning shall be compared with routine cleaning validation.
- B) *Impurity profile (RRT wise) of Batches manufactured in campaign shall be compared with each other. There shall not be increasing trend of degradation impurities.*
- C) The last batch of campaign shall meet the microbiological acceptance criteria.
- 2.37.9 In case of any failure in cleaning, investigation shall be done and if failure is attributed to inability of cleaning procedure, then length of campaign shall be reduced accordingly. Impact assessment on batches shall be done.
- 2.38 **Verification of Cleaning Solvent traces:**
- 2.38.1 *Organic solvents are used for cleaning of equipment. These solvents are generally volatile, and equipment shall be dried easily by applying heating or vacuum or by using compressed air or nitrogen. For low boiling solvents like Acetone, Methanol etc even at room temperature solvents get evaporated. In such cases routine monitoring of Solvent traces post Type B cleaning is not required.*
- 2.38.2 *However as one time exercise "Verification of Cleaning Solvent Traces" shall be done for each solvent group by following procedure.*
- 2.38.3 *Study shall be done for any product within the solvent group. Study shall be done for each solvent with three runs.*
- 2.38.4 *This is not applicable for surfactant-based cleaning agents and Acids, Bases and when only water is used as solvent.*
- 2.38.5 *In case High Boiling solvent is used for cleaning followed by other solvent to remove high boiling solvent, then both solvent traces shall be checked.*

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2.38.6 *Rationale for selection of equipment: As equipment are cleaned with common cleaning procedure and exposed to same environment, study shall be done for at least one equipment from each group.*

Group A : Crystalliser Reactor

Group B : Closed type equipment – E.g., Blender, Dryers, RCVD, VTD, ANFD, RVPD etc.

Group C : Portable equipment & equipment which shall be dismantled completely- E.g. Multi mill, Sifters, Jet mill, Cad mill, Co mill etc.

2.38.7 *This verification study shall be done through an approved protocol as per annexure 1035-A-0072/A22 and shall be numbered as SP/Plant Code/CLSOLTRA. e.g. SP/1030/CLSOLTRA, where SP is for study protocol, 1030 is the plant code of Patalganga unit-I and CLSOLTRA is for cleaning solvent traces.*

2.38.8 *Equipment shall be cleaned and dried as per product change over cleaning procedure.*

2.38.9 *Equipment shall be rinsed with Purified water and Rinse shall be analysed with validated GC or TOC method.*

2.38.10 *Acceptance criteria: Residue of solvents shall be less than 1/10th of their respective ICH limit as per ICH Q3C.*

2.38.11 *A computerized pre-printed report shall be prepared for verification of solvent traces study and shall be numbered as SR/1030/CLSOLTRA as per specimen annexure 1035-A-0072/23.*

3.0 ABBREVIATIONS :

AHU	:	Air handling unit	ml	:	Millilitre
API	:	Active pharmaceutical ingredient	mm	:	Millimetre
BLOD	:	Below limit of detection	MOC	:	Material of construction
BLOQ	:	Below limit of quantification	MSDS	:	Material safety and data sheet
cGMP	:	Current good manufacturing practices	NMT	:	Not more than
GC	:	Gas chromatography	OOS	:	Out of specification
HPLC	:	High performance liquid chromatography	PDE	:	Permitted daily exposure
inch ²	:	Square inch	PLC	:	Programmable logic control
LOD	:	Limit of detection	ppm	:	Parts per million
LOQ	:	Limit of quantification	Sq.	:	Square
Lt.	:	Litre	TI	:	Technical information sheet
MACO	:	Maximum allowable carryover	UPLC	:	Ultra-micro high performance liquid chromatography
mg	:	Milligram	UV	:	Ultraviolet
MBS	:	Minimum batch size	ICH	:	International council for harmonisation of technical requirements for pharmaceuticals for human use

4.0 REFERENCES :

1035-A-0043	:	Surface area calculations of equipment and accessories for swab / rinse sampling.
1035-A-0017	:	Labelling
1035-A-0047	:	Line clearance equipment and area

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1035-A-0080	:	Handling of stains on process equipment
1035-A-0082	:	Swab / rinse sampling of equipment and accessories
1035-A-0083	:	Training on cleaning of equipment and visual inspection
1035-G-0205	:	Deviation handling
1035-G-0226	:	Evaluation of nitrosamine impurities
1035-G-0024	:	Risk management by failure modes, effect, and criticality analysis
RD-S-006	:	Cleaning procedure development.
1035-L-0016	:	Validation and verification of analytical method
1035-L-0086	:	Microbiological monitoring of environment, surfaces and personnel
1035-L-0089	:	Establishment of alert levels".
1035-L-0143	:	Out of specification (OOS) and out of trend (OOT) investigation procedure
1035-L-0146	:	Investigation and resolution procedure for additional peaks in chromatographic analysis

5.0 ANNEXURES :

Annexure Number	Annexure Name	To be used as
1035-A-0072/A1	Assessment of cleaning procedure for introduction of new API /Intermediate.(1035-A-0072/F7)	Specimen copy
1035-A-0072/A2	Evaluation of introduction for new equipment in facility(1035-A-0072/F8)	Specimen copy
1035-A-0072/A3	Cleaning matrix for API (1035-A-0072/F9/1)	Specimen copy
1035-A-0072/A4	Cleaning matrix for intermediate (1035-A-0072/F10)	Specimen copy
1035-A-0072/A5	List of cleaning matrix amendment for API / intermediate (1035-A-0072/L1)	Specimen copy
1035-A-0072/A6	Equipment cleaning, sampling plan & sampling activity verification for cleaning validation (1035-A-0072/F11)	Specimen copy
1035-A-0072/A7	Cleaning sample label (1035-A-0072/LA1/3)	Specimen copy
1035-A-0072/A8	Protocol for API Cleaning validation / Verification / Periodic cleaning verification (1035-A-0072/P2)	Specimen copy
1035-A-0072/A9	Report for API cleaning validation / Verification (1035-A-0072/R2)	Specimen copy
1035-A-0072/A10	Protocol for intermediate cleaning validation / Verification / Periodic cleaning verification. (1035-A-0072/P3)	Specimen copy
1035-A-0072/A11	Report for Intermediate cleaning validation / Verification (1035-A-0072/R3)	Specimen copy
1035-A-0072/A12	Report for periodic cleaning verification (1035-A-0072/R7)	Specimen copy
1035-A-0072/A13	Flow Diagram of validation and implementation of validated cleaning procedure (1035-A-0072/FL1)	Specimen copy
1035-A-0072/A14	Annual cleaning review report (1035-A-0072/F12)	Specimen copy
1035-A-0072/A15	Assessment for cleaning requirement for product containing genotoxic / nitrosamine compound (1035-A-0072/F13/1)	Specimen copy
1035-A-0072/A16	Protocol for Dirty Equipment Hold Time Study (1035-A-0072/P4/1)	Specimen copy
1035-A-0072/A17	Report for Dirty Equipment Hold Time Study (1035-A-0072/R4/1)	Specimen copy

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Annexure Number	Annexure Name	To be used as
1035-A-0072/A18	Protocol for Cleaned Equipment Hold Time Study (1035-A-0072/P5)	Specimen copy
1035-A-0072/A19	Report for Cleaned Equipment Hold Time Study (1035-A-0072/R5)	Specimen copy
1035-A-0072/A20	Protocol for Campaign cleaning validation study (1035-A-0072/P1/2)	Specimen copy
1035-A-0072/A21	Report for Campaign cleaning validation study (1035-A-0072/R1/2)	Specimen copy
1035-A-0072/A22	Protocol for verification of cleaning solvent traces (1035-A-0072/P6/1)	Specimen copy
1035-A-0072/A23	Report for verification of cleaning solvent traces (1035-A-0072/R6)	Specimen copy
1035-A-0072/A24	Operator interview for product cleanability evaluation (1035-A-0072/F14)	Specimen copy
1035-A-0072/A25	Impact assessment of changes and Revalidation criteria (1035-A-0072/RF1)	Specimen copy

ASSESSMENT OF CLEANING PROCEDURE FOR INTRODUCTION OF NEW API /INTERMEDIATE

API / Intermediate Name	
R&D cleaning MPG number	

• **Assessment of Cleaning method suitability and testing requirement:**

- During assessment check the manufacturing process and list down the product residue formed, various material charged into the equipment, which residue remains in equipment after processing.
- Verify the suitability of the cleaning method for removing residue from equipment and outline the testing requirements according to the SOP for checking removal of residue on equipment surfaces.

Details of residue	Suitability of cleaning Method for removal of residue	Requirement for Testing of Residues
API		
Isolated intermediate		
In Situ Intermediate		
KSM		
Organic Reagent		
Inorganics		
Charcoal		

• **Assessment for batch-to-batch cleaning procedure**

Determine extent of cleaning required for batch-to-batch cleaning considering following factors

Factors	Remark	If Yes, Details
Is there any impact of polymorphic form?	Yes / No	
Is there any impact on Impurity profile?	Yes / No	
Safety requirement (Anhydrous reaction).	Yes / No	
Other	Yes / No	

Note: If above mention any factor answer is yes, then specify the proposal for additional cleaning requirement / Inclusion of Extensive Cleaning Steps in the batch-to-batch cleaning to minimise the risk shall be specified below.

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• **Assessment of other factors impacting cleaning performance:**

R & D / MSTG and Production team to conduct the studies / trial cleaning batch to verify below points to decide additional requirements for hard to clean product. Reference document No.:

**ANNEXURE 1035-A-0072/A1
(SPECIMEN COPY)**



ASSESSMENT OF CLEANING PROCEDURE FOR INTRODUCTION OF NEW API /INTERMEDIATE

Sr. No.	Factors identified	Answer	Score (yes -1, No-0)
i.	Process creating sticky or caked on solid which is difficult to clean.	Yes (1) / No (0)	
ii.	Highly viscous or shear thickening material resistant to cleaning.	Yes (1) / No (0)	
iii.	Material reacting with water (cleaning solvent) to form solid or sticky gel-like material.	Yes (1) / No (0)	
iv.	Product residue may rust MOC in presence of	Yes (1) / No (0)	
v.	Residue get hardened when left for longer time?	Yes (1) / No (0)	
vi.	Total Score		

Based on above study, give the rating for cleanability of the product.

Category of cleaning	Score range	Cleanability Rating	Cleanability Rating to product
Easily cleanable	0-1	1	
Moderately Difficult to clean	2-3	2	
Difficult to clean	4-5	3	

• **Assessment for applicability of existing cleaning procedure and worst-case assessment**

Sr. No.	Check point for evaluation of new product	Remark (Record Yes / No / Not Applicable)
i.	Shall the new product to be cleaned by the same cleaning process as used for the existing group?	
ii.	If yes, amend the cleaning matrix as per SOP annexure 1035-A-0072/A3 and 1035-A-0072/A4	
iii.	If no, new cleaning record for equipment cleaning shall be prepared and study to be performed on pre-validation batch.	
iv.	<u>Shall the new product change the calculated residue limits for the products in the existing group?</u>	
v.	<u>Check the availability of cleaning method validation new API / intermediates/ insitu intermediates.</u>	
vi.	<u>Check availability of recovery studies of applicable surfaces (MOC).</u>	
vii.	Check the acceptance criteria. Acceptance criteria (i.e. individual rinse and swab limit) shall be more than the LOQ value (i.e. LOQ value shall not be more than 30% acceptance limit).	
viii.	If included in the existing group, shall the new product be a new "most difficult to clean" product in the new group?	
ix.	Shall the new product (that is not more difficult to clean than the "most difficult to clean" product in the group) constitute a new lowest limit of any product in the proposed group?	
x.	Additional If any.	

Recommendation:

ANNEXURE 1035-A-0072/A1
(SPECIMEN COPY)



ASSESSMENT OF CLEANING PROCEDURE FOR INTRODUCTION OF NEW API /INTERMEDIATE

Team Approval:

R & D / MSTG
Date:

Production
Date:

Quality Control
Date:

Quality Assurance
Date:

Reviewed By:
Unit Head
Date:

Approved By:
Head Unit Quality Assurance
Date:

Note: This assessment report to be attached with cleaning matrix amendment.

EVALUATION OF INTRODUCTION FOR NEW EQUIPMENT IN FACILITY

Equipment Name : _____

Capacity : _____

Equipment ID : _____

A. Evaluation of new equipment against the worst case and existing equipment :

Sr. No.	Check point for evaluation of new equipment	Remark (circle the option)
i.	Is the new equipment is identical or equivalent with minor changes and is cleaned with the same cleaning SOP.	Yes / No / Not applicable
ii.	If the new equipment is equivalent with significant changes like MOC, Dimension, Capacity etc,	Yes / No / Not applicable
iii.	If the new equipment does not constitute a new worst-case equipment and does not lower the calculated limits for the group (such as by a higher surface area),	Yes / No / Not applicable
iv.	If the new equipment does not constitute a new worst-case equipment but does lower the calculated API (rinse and swab limit) and cleaning agent limits for the group (such as by increasing the shared surface area),	Yes / No / Not applicable
v.	If new equipment does constitute a new worst-case equipment but does not lower the calculated API and cleaning agent limits for the group. (such as by increasing the shared surface area)	Yes / No / Not applicable
vi.	Additional if any.	Yes / No / Not applicable

Conclusion:

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B. Assessment and development of cleaning procedure for new type of equipment:

- Prior to / during qualification of equipment, a team from user, QA and engineering shall discuss and performed the dismantling / assembling assessment of equipment based on equipment user manual and check the equipment for its various parts.
- Accumulation of powder residue, oil leakage etc. shall be checked. All related equipment parts shall be considered and covered during assessment, and extent of cleaning shall be defined.
- Based on user manual, check the dismantling ability (manual of these parts) determined the extent of which each part of equipment further to be dismantled.
- Identify the parts of equipment which will be come in product contact / non-contact.
- Identify the areas / parts which are difficult / hard to reach / clean for each part of equipment.
- Against each identified parts of equipment, determine the type of cleaning required and approximate quantity of cleaning solvent for each part i.e. pressure cleaning, rinsing, scrubbing, soaking, refluxing etc.
- Conduct the studies / trial cleaning batch and verify the effectiveness of cleaning.

**ANNEXURE 1035-A-0072/A2
(SPECIMEN COPY)**



EVALUATION OF INTRODUCTION FOR NEW EQUIPMENT IN FACILITY

Sr.	Equipment parts	Contact / non-contact parts	Disassembled/ dismantled (Yes / No)	Chance of high powder accumulation (Yes / No)	Easy to clean (Yes / No) (1)	Easy to access for cleaning (Yes / No) (2)	Easy for visual inspection (3)

Evaluation criteria:

- If any of the criteria mentioned in 1,2 and 3 is 'No' location shall be considered for sampling.
- If all the criteria mentioned is 'Yes' then the chance of high powder accumulation and location which are yielded the highest residue concentration during study shall be considered for sampling.

Conclusion:

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Signatory	Department	Sign	Date
Prepared By	Production		
	Quality assurance		
Reviewed By	Quality assurance		
Approved by	Head unit quality assurance		

Note : This evaluation to be attached with cleaning matrix amendment.

Document no. Plant code/CLMX/API/API block	CLEANING MATRIX FOR API	Page 1 of 7
Amendment Version no.:		

REFERENCE VALIDATION MASTER PLAN

1.0 OBJECTIVE AND SCOPE:

To decide limit for cleaning validation and worst-case product (API) manufactured in block XYZ

Limit applicable for equipment which are used for manufacturing of API where active moiety is formed

2.0 REASON FOR AMENDMENT:

Reason	To be recorded.
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3.0 DETAILS:

3.1 List down the products manufactured in the block XYZ with solubility details. Based on solubility, grouped the product as per below table.

Group A : Methanol Soluble

Sr.	Product name
1.	ABC
2.	DEF
3.	GHI

Group B : Water Soluble

Sr.	Product name
1.	JKL
2.	MNO

3.2 Check the manufacturing flow and equipment usage for listed product. Calculate the Total share surface area and total last rinse volume of equipment chain of each API. The calculation to be done by using validated excel as per below table and same to be attached as annexure-I.

Previous product :

Equipment used for manufacturing stage	Equipment	Final Rinse volume of equipment	Maximum surface area of equipment	Next Product	Next Product	Next Product
Crude preparation	Reactor -1			√		√
Filtration	Centrifuge-1				√	
Drying	Dryer 2			√	√	
Sifting	Sifter 3			√		√
Accessories	Scoop / spatula			√	√	√
Total rinse volume in Liter						
Total common surface area						

Note: Tick mark (√) to be done for applicable product against equipment and summation to be done of final rinse and common surface area.

Document no. Plant code/CLMX/API/API block	CLEANING MATRIX FOR API	Page 2 of 7
Amendment Version no.:		

REFERENCE VALIDATION MASTER PLAN

- 3.3 Tabulate the data of Minimum batch size, Permitted Daily Exposure (PDE) of all age group, limit of detection, limit of quantification and route of administration of each API as per below table and attached annexure-II.

Sr.	Product Name	Minimum batch size (kg)	Max. therapeutic daily dose (mg)	PDE value for all age group.	Limit of detection (ppm)	Limit of quantification (ppm)	Route of administration
1.	ABC						
2.	DEF						
3.	GHI						
4.	JKL						

- 3.4 Based on above data, calculate the MACO, Rinse limits and swab limits as per below formulas for each product. MACO calculation shall be done in validated "Microsoft Excel" and same is to be attached as annexure -III.

MACO CALCULATION FORMULA BY PERMITTED DAILY EXPOSURE (PDE CRITERIA):

$$\text{MACO in mg (By PDE criteria)} = \frac{\text{PDE (previous)} \times \text{MBS (next product)}}{\text{TDD (next)}}$$

$$\text{Rinse Limit (ppm) for equipment} = \frac{\text{MACO (mg)} \times \text{Area Sampled by Rinse Sampling (inch}^2\text{)} \times 1000}{\text{Surface Area of Equipment Chain (inch}^2\text{)} \times \text{Rinse sampling volume (ml)}}$$

$$\text{Swab limit (ppm)} = \frac{\text{MACO} \times \text{Swab Surface Area} \times 1000}{\text{Surface Area of Equipment Chain} \times 10 \text{ (ml)}}$$

OR

$$\text{Swab limit (}\mu\text{g/swab)} = \frac{\text{MACO} \times \text{Swab surface area} \times 1000}{\text{Surface Area of Equipment Chain}}$$

MACO	=	Maximum allowable carryover
MBS (Next Product)	=	Minimum batch size of next product (mg)
TDD (Next Product)	=	Therapeutic daily dose of next product (mg)
PDE	=	Permitted daily exposure
1000	=	Conversion factor from mg to μg

**ANNEXURE 1035-A-0072/A3
(SPECIMEN COPY)**



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REFERENCE VALIDATION MASTER PLAN

MACO Calculation sheet:

Previous product :

Product Name	MBS (kg)	(Conversion factor) KG to mg	PDE Previous	TDD next	Swab surface area
Next product		1000000			4/9
Next product		1000000			4/9
Next product		1000000			4/9

Conversion Factor (from mg to µg)	Total Rinse	Total Surface area	MACO by Permitted Daily Exposure Criteria (PDE)		
			MACO in Mg	Rinse limit in ppm	Swab limit in ppm or µg/swab
1000					
1000					
1000					

- 3.5 Summarise the MACO, Rinse and swab limits as per below table and same to be attached as annexure -IV

Product Name	Parameters	Previous product name	Previous product name	Previous product name
Next product Name	MACO in mg	NA		
	Rinse in ppm	NA		
	Swab in µg / swab or ppm	NA		
Next product Name	MACO in mg		NA	
	Rinse in ppm		NA	
	Swab in µg / swab or ppm		NA	

- 3.6 Compile the data of solubility, toxicity (Least PDE value) and cleanability (Difficult to clean) with rating for all listed products. Based on rating calculate RPN for each product as per below table. The highest RPN product to be selected as a worst-case product for cleaning validation from each solubility group.

Group 1 : Methanol soluble

Sr.	Product	Solubility of API in Solvent	API solubility Rating (A)	PDE (mcg/day)	PDE Rating (B)	Difficult to clean	Difficult to clean rating (C)	Overall RPN (AXBXC)
1.	ABC							
2.	DEF							
3.	GHI							
4.	JKL							

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REFERENCE VALIDATION MASTER PLAN

First worst Product Name considering Max. RPN:	
Second worst Product Name considering second Max. RPN:	
Third worst Product Name considering third Max. RPN:	

Group 2 : Water soluble

Sr.	Product	Solubility of API in Solvent	API solubility Rating (A)	PDE (mcg/day)	PDE Rating (B)	Category of cleaning	Cleanability rating (C)	Overall RPN (AXBXC)
1.	MNO							
2.	PQR							
3.	STU							
4.	VWX							

First worst Product Name considering Max. RPN:	
Second worst Product Name considering second Max. RPN:	
Third worst Product Name considering third Max. RPN:	

- 3.7 List down the equipment's, ancillary (small) parts, and utensil available in department along with, MOC, capacity, surface area. Compile the equipment usage details and highlight the worst-case product for each equipment as pe below table and same to be attached as annexure-V.

Sr.	Equipment name	Equipmen t code	Material of Constru ction	Surface Area (inch ² or m ²)	Capacity of Equipment	Methanol soluble		Water soluble	
						Product name	Product name	Product name	Product name
1	Reactor -1						√	√	
2	Filter-1					√			√
3	Dryer 2						√	√	√
4	Sifter 1					√			
5	Utensil						√		√

Note: Tick mark (√) to be done for applicable product against equipment.
Highlighted product is the worst product for the equipment and this product to be considered for worst case cleaning validation.

- 3.8 Summarise the limits (MACO, rinse and swab limits) used for cleaning validation of worst-case product selected based on the RPN calculation from each solubility group and equipment which are not part of worst-case product manufacturing flow.

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REFERENCE VALIDATION MASTER PLAN

Table A: For Methanol soluble product

MACO, rinse & swab Limits for worst product					
Selection Criteria	Worst case	Name of API	RPN calculation	Limits	
Based on RPN Calculation	1 st worst case product			MACO in mg	
				Rinse in ppm	
				Swab in µg / swab or ppm	

Table B : For Water soluble product

MACO, rinse & swab Limits for worst product					
Selection Criteria	Worst case	Name of API	RPN calculation	Limits	
Based on RPN Calculation	1 st worst case product			MACO in mg	
				Rinse in ppm	
				Swab in µg / swab or ppm	

Table C: For Equipment's which are not part of worst-case product manufacturing flow.

MACO, rinse & swab Limits for worst product					
Equipment number	Worst case	Name of API	RPN calculation	Limits	
Reactor	1 st worst case product			MACO in mg	
				Rinse in ppm	
				Swab in µg / swab or ppm	
Dryer	1 st worst case product			MACO in mg	
				Rinse in ppm	
				Swab in µg / swab or ppm	

4.0 SUMMARY AND CONCLUSION:

Based on overall review of cleaning matrix, product is selected as a worst product for cleaning validation. The MACO limit, rinse limit and swab limit is to be used for cleaning validation of equipment.

Cleaning validation of Individual equipment 1....., 2..... and 3..... to be done when used for subsequent worst product as these equipment's are not part of manufacturing flow of worst- case product.

E.g. Jetmill is not used for worst case product, so Jetmill to be validated with three runs for subsequent worst product.

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REFERENCE VALIDATION MASTER PLAN

Three runs equipment chain cleaning validation to be done when equipment used for manufacturing of selected worst case product.....

Three run cleaning validation of individual equipment to be done when equipment used for subsequent worst case productas this equipment is not a part of manufacturing flow of selected worst case product.

5.0 Annexures:

- Annexure-I : Total shared surface area and rinse volume of equipment chain.
- Annexure-II The data Minimum batch size, PDE of all age group, LOD, LOQ and route of administration of all API
- Annexure-III MACO calculation sheet by PDE criteria
- Annexure-IV Summarize limits of MACO, rinse and swab
- Annexure-V List of Equipment and utensil with MOC, Capacity, surface area and usages details

6.0 FREQUENCY:

Cleaning matrix shall be updated during each change specified below.

- 6.1 Whenever a new product is introduced or removal of existing molecule.
- 6.2 Change in batch size.
- 6.3 Addition of new equipment or removal of equipment from manufacturing flow.
- 6.4 If a commercial product has not been manufactured in the last 3 years, and/or if the API was manufactured for registration purposes (trial batch) and there is no future or forecasts plan after validation, then that API shall be removed from the cleaning matrix.
- 6.5 Any change in master data of cleaning matrix.

7.0 CLEANING MATRIX APPROVAL:

Quality Assurance	:	_____	Date	:	_____
Production	:	_____	Date	:	_____
Quality Control	:	_____	Date	:	_____

8.0 REVIEW AND COMMENTS

.....
.....

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REFERENCE VALIDATION MASTER PLAN

Section Head (Quality Assurance)
Date:

9.0 APPROVED BY:

Head Unit Quality Assurance
Date:

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REFERENCE VALIDATION MASTER PLAN

1.0 OBJECTIVE AND SCOPE:

To decide limit for cleaning validation and worst-case product (INT) manufactured in block XYZ

Limit applicable for equipment which are used for manufacturing of intermediates and API (before active moiety formed).

2.0 REASON FOR AMENDMENT:

Reason	To be recorded.
--------	-----------------

3.0 DETAILS:

3.1 List down the intermediates and API (before active moiety formed) manufactured in the block XYZ with solubility details. Based on solubility, grouped the product as per below table.

Group 1 : Methanol Soluble

Sr.	Product name
1.	ABC
2.	DEF
3.	GHI

Group 2 : Water Soluble

Sr.	Product name
1.	JKL
2.	MNO

3.2 Compile the data of solubility and cleanability (Difficult to clean) with rating for all listed products. Based on rating calculate RPN for each product as per below table. The highest RPN product to be selected as a worst-case product for cleaning validation from each solubility group.

Group 1 : Methanol soluble

Sr.	Product	Solubility of API in Solvent	API solubility Rating (A)	Difficult to clean	Difficult to clean rating (B)	Overall RPN (AXB)
1.	ABC					
2.	DEF					
3.	GHI					
4.	JKL					

First worst Product Name considering Max. RPN:	
Second worst Product Name considering second Max. RPN:	
Third worst Product Name considering third Max. RPN:	

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REFERENCE VALIDATION MASTER PLAN

Group 2 : Water soluble

Sr.	Product	Solubility of API in Solvent	API solubility Rating (A)	Category of cleaning	Cleanability rating (B)	Overall RPN (AXB)
1.	MNO					
2.	PQR					
3.	STU					
4.	VWX					

First worst Product Name considering Max. RPN:	
Second worst Product Name considering second Max. RPN:	
Third worst Product Name considering third Max. RPN:	

- 3.3 List down the equipment's, ancillary (small) parts, and utensil available in department along with, MOC, capacity, surface area. Compile the equipment usage details and highlight the worst-case product for each equipment as pe below table and same to be attached as annexure-I.

Sr.	Equipment name	Equipmen t code	Material of Constru ction	Surface Area (inch ² or m ²)	Capacity of Equipme nt	Methanol soluble		Water soluble	
						Product name	Product name	Product name	Product name
1	Reactor -1						√	√	
2	Filter-1					√			√
3	Dryer 2						√	√	√
4	Sifter 1					√			
5	Utensil						√		√

Note: Tick mark (√) to be done for applicable product against equipment.
Highlighted product is the worst product for the equipment and this product to be considered for worst case cleaning validation.

- 3.4 Summarise the worst-case product from each solubility group and use 100 ppm limit for chain of equipment and 10 ppm limit for Rinse and swab sample of cleaning validation. The details to be tabulated as per below table.

Table A: For Methanol soluble product

MACO, rinse & swab Limits for worst product					
Selection Criteria	Worst case	Name of API	RPN calculation	Limits	
Based on RPN Calculation	1 st worst case product			Chain of equipment (in ppm)	100 ppm
				Rinse in ppm	10 ppm
				Swab in ppm	10 ppm

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REFERENCE VALIDATION MASTER PLAN

Table B : For Water soluble product

MACO, rinse & swab Limits for worst product					
Selection Criteria	Worst case	Name of API	RPN calculation	Limits	
Based on RPN Calculation	1 st worst case product			Chain of equipment (in ppm)	100 ppm
				Rinse in ppm	10 ppm
				Swab in ppm	10 ppm

Table C: For Equipment's which are not part of worst-case product manufacturing flow.

MACO, rinse & swab Limits for worst product					
Equipment number	Worst case	Name of API	RPN calculation	Limits	
Reactor	1 st worst case product			Chain of equipment (in ppm)	100 ppm
				Rinse in ppm	10 ppm
				Swab in ppm	10 ppm
Dryer	1 st worst case product			Chain of equipment (in ppm)	100 ppm
				Rinse in ppm	10 ppm
				Swab in ppm	10 ppm

4.0 SUMMARY AND CONCLUSION:

Based on overall review of cleaning matrix, product is selected as a worst product for cleaning validation. The MACO limit 100 ppm for chain of equipment, rinse and swab limit 10 ppm for individual equipment are to be used during cleaning validation.

Cleaning validation of Individual equipment 1....., 2..... and 3..... to be done when used for subsequent worst product as these equipment's are not part of manufacturing flow of worst case product.

E.g. ANFD is not used for worst case product, so ANFD to be validated with three run for subsequent worst product.

Three runs equipment chain cleaning validation to be done when equipment used for manufacturing of selected worst case product.....

Three run cleaning validation of individual equipment to be done when equipment used for subsequent worst- case productas this equipment is not a part of manufacturing flow of selected worst-case product.

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REFERENCE VALIDATION MASTER PLAN

5.0 Annexures:

Annexure-I : List of Equipment and utensil with MOC, Capacity, surface area and equipment usages details

6.0 FREQUENCY:

Cleaning matrix shall be updated during each change specified below.

- 6.1 Whenever a new product is introduced or removal of existing molecule.
- 6.2 Addition of new equipment in worst or non-worst selected product and removal of equipment from manufacturing flow.
- 6.3 If a commercial product has not been manufactured in the last 3 years, and/or if the API was manufactured for registration purposes (trial batch) and there is no future or forecasts plan after validation, then that API shall be removed from the cleaning matrix.
- 6.4 Any change in master data of cleaning matrix.

7.0 CLEANING MATRIX APPROVAL:

Quality Assurance	:	_____	Date	:	_____
Production	:	_____	Date	:	_____
Quality Control	:	_____	Date	:	_____

8.0 REVIEW AND COMMENTS:

.....
.....

Section Head (Quality Assurance)
Date:

9.0 APPROVED BY:

Head Unit Quality Assurance
Date:

ANNEXURE 1035-A-0072/A5
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LIST OF CLEANING MATRIX AMENDMENT FOR API / INTERMEDIATE

DEPARTMENT: _____

Sr. No.	Amendment No.	Initial / Amendment preparation Date	Reason for Amendment
1	Initial		
2	Amendment No.1		
3	Amendment No.2		
4	Amendment No.3		
5	Amendment No.4		
6	Amendment No.5		
7			
8			
9			
10			
11			

Prepared by:
Date:

Checked by:
Date:

**ANNEXURE 1035-A-0072/A6
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**EQUIPMENT CLEANING, SAMPLING PLAN & SAMPLING ACTIVITY VERIFICATION FOR CLEANING
VALIDATION**

Note: Separate sampling plan to be prepared for rinse and swab sampling.

EQUIPMENT NAME: Equipment Code No.:			
Previous Product Name		Batch No.	
Cleaning SOP number with version number		Date of validation	
API manufacturing completed on	Date :..... time:	Cleaning solvent	
Cleaning started on	Date :..... time:	Swab used for sampling	Texwipe
Cleaning Completed on	Date :..... time:	Type of swab stick	
Lot No. of the Swab stick used		Manufacturing date of the Swab	
Expiry date of swab (If available)		Shelf life of swab is within specified period (Yes / No) (If available)	
Swabs discarded during sampling (In nos.)		Swabbing Solvent / diluent	

Date/ Time	Rinse/ Swab sample No.	Sample collection Instructions	Sampling qty.	Observation / Recordings	Done by QA sign & date
Sampling instruction for rinse:					
	R1	Sample to be collect at cleaning ECR step no.....	Qty_____lit	Visually clear / not clear	
	R2	Sample to be collect at cleaning ECR step no.....	Qty_____lit	Visually clear / not clear	
	R3	Sample to be collect at cleaning ECR step no.....	Qty_____lit	Visually clear / not clear	
Sampling instruction for swab:					
	S1	Swab of part	1 swab in 10 ml diluent	Visually: Cleaned / Not cleaned	
	S2	Swab of part	1 swab in 10 ml diluent	Visually: Cleaned / Not cleaned	
	S3	Swab of part	1 swab in 10 ml diluent	Visually: Cleaned / Not cleaned	

Note : After swab sampling transfer the swab head into the diluent immediately.

Prepared By: _____
Date: _____

Checked by: _____
Date: _____

**ANNEXURE 1035-A-0072/A7
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Cipla
Location

CLEANING SAMPLE LABEL

Cipla
Location

CLEANING SAMPLE: RINSE / SWAB

PRODUCT NAME:

BATCH NO.:

EQUIPMENT / PART NAME:

ITEM: PRODUCT / CLEANING AGENT

SAMPLE NO.:

SAMPLE VOLUME:

SAMPLED BY SIGNATURE

DATE:

TIME:

(1035-A-0072/LA1/3)

PROTOCOL FOR API CLEANING VALIDATION / VERIFICATION / PERIODIC CLEANING VERIFICATION		
Document no. Plant code/APIWCCVP/API block	PRODUCT NAME	Issued date:
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REFERENCE VALIDATION MASTER PLAN

1.0 OBJECTIVE:

In case of validation: To validate and ensure that the cleaning procedure..... is effective and removes all the residues of product and cleaning agent up to the MACO, rinse and swab limits decided during cleaning matrix.

In case of verification: To verify and ensure that the cleaning procedure..... is effective and removes all the residues of product and cleaning agent up to the MACO, rinse and swab limits decided during cleaning matrix.

In case periodic verification: To verify periodically, the cleaning procedure.....is effective and consistently removes all the residues of product..... and cleaning agent up to the MACO, rinse and swab limits decided during cleaning matrix.

2.0 SCOPE:

In case of validation: Protocol is applicable for validation of cleaning procedurewhen equipment's used for the manufacturing of selected worst-case API..... from soluble group.

In case of verification: Protocol is applicable for verification of cleaning procedurewhen equipment's used for the manufacturing of non-worst-case API..... from soluble group.

In case periodic verification: Protocol is applicable for periodic cleaning verification of cleaning procedures when equipment used for manufacturing of selected worst-case API..... from soluble group

3.0 REASON FOR VALIDATION: Initial / Concurrent / Verification / Periodic verification

4.0 VALIDATION TEAM:

Representatives from Production, Quality control and Quality Assurance. (Individual names shall be recorded in report section)

5.0 RESPONSIBILITIES:

5.1 Production:

- To be part of validation team.
- To ensure that the cleaning activity is performed as per the respective product changeover procedure.
- To ensure the effectiveness of cleaning by visual inspection.
- To ensure rinse / swab results as per sampling plan within acceptance criteria for product changeover.

**ANNEXURE 1035-A-0072/A8
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Location

PROTOCOL FOR API CLEANING VALIDATION / VERIFICATION / PERIODIC CLEANING VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

5.2 Quality Control:

- To be part of validation team.
- To ensure availability of validated analytical methods for measurement of product in swabs and rinses.
- To analyse the cleaning validation samples as per TI sheet

5.3 Quality Assurance:

- To be part of validation team.
- To coordinate with Production and Quality Control for cleaning validation.
- To collect validation samples as per approved sampling plan of equipment.
- To perform cleaning validation as per approved validation protocol and ensure completion of report.
- To ensure the Training details of the personnel involved in the cleaning of equipment, collecting of samples, testing of samples, deciding the sample plan, preparing, reviewing, approving the reports

6.0 REVIEW OF PRE-REQUISITE AND TRAINING DETAILS

Sr.	Item	Remark Document no./ Available /NA
i.	Product change over cleaning procedure with ECR checklist	Document no.:
ii.	Validated analytical method for measurement of product in the rinses and swab	Document no.:
iii.	Validated analytical method for measurement of cleaning agent in the rinses (In case cleaning agent used for cleaning)	Document no.:
iv.	Validated microbiological method for measurement of bioburden on the equipment (In case of micro sample)	Document no.:
v.	Approved sampling plan for cleaning validation	Available
vi.	Individual equipment specific diagram with swab location shall be prepared as per SOP 1035-A-0082.	Available
vii.	Specification for cleaning sample analysis	Document no.:
viii.	Training on cleaning validation SOP 1035-A-0072	Available / Training ID:
ix.	Training and qualification of personnel on cleaning procedure as per SOP 1035-A-0083	Available / Training ID:
x.	Training and qualification of personnel on visual inspection of equipment as per SOP 1035-A-0083	Available / Training ID:
xi.	Training of personnel on sampling plan.	Available / Training ID:
xii.	Training of personnel on swab sampling as pe SOP 1035-A-0082	Available / Training ID:

7.0 DESCRIPTION:

Prepared By
Sign / Date : _____

PROTOCOL FOR API CLEANING VALIDATION / VERIFICATION / PERIODIC CLEANING VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

Validation done on batch numbers, Equipment cleaning SOP number, cleaning solvent / cleaning agent with concentration, MACO limit, Rinse and swab limit, alert limit of rinse and swab for Periodic verification / Validation shall be recorded in report section.

8.0 PROCEDURE:

- 8.1 Proper training session on cleaning validation protocol and report shall be conducted to all validation team member prior to initiate the cleaning validation.
- 8.2 Quality Assurance shall issue the required ECR and production shall initiate the equipment cleaning.
- 8.3 TI sheet shall be raised as an intimation to QC for cleaning sample and preparation of swab diluent prior to rinse and swab sampling.
- 8.4 Quality assurance shall be arranged the sampling tools like, Sampling bottles for rinse collection, Swab stick required for swab collection, 4/9 square inch template, Swab sample collection test tube / bottle and Labels required for labelling for collected samples etc.
- 8.5 Quality Assurance shall collect 100 ml of cleaning solvent as blank for evaluation of interference of cleaning solvent.
- 8.6 Sampling plan:

The time and number of samples shall be drawn, shall be described in sampling plan. and same to be attached with cleaning validation protocol.

Sampling equipment, sampling locations and sampling intervals shall be described in this section. Rinse samples shall be drawn in cleaned sample bottle. Swab sample shall be collected by using Texwipe swab stick. Sampling details shall be recorded in validation report section.

Sampling plan for validation samples shall be elaborated in protocol for predefined sample quantity along with ECR step number and sample number. Equipment chain including small parts and utensil which considered in cleaning matrix shall be consider for validation samples.

Validation samples:

<u>Type of equipment</u>	<u>Applicable Equipment code number</u>	<u>Sampling interval</u>	<u>Number of samples</u>	<u>Sampling quantity</u>
Reactor	R1, R2, R3	Rinse sample: 100ml rinse sample to be collected after ECR step no.....	1 rinse sample	1 X about 100 ml
		Swab (chemical & Micro) samples: to be collected after ECR step no.....	Chemical Swab: Reactor body, Agitator and Bottom valve Micro: Reactor body	Chemical: 3 swabs Micro: 1 swab

**ANNEXURE 1035-A-0072/A8
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Cipla
Location

PROTOCOL FOR API CLEANING VALIDATION / VERIFICATION / PERIODIC CLEANING VERIFICATION		
Document no. Plant code/APIWCCVP/API block	PRODUCT NAME	Issued date:
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REFERENCE VALIDATION MASTER PLAN

Type of equipment	Applicable Equipment code number	Sampling interval	Number of samples	Sampling quantity
Filter	CF-1, CF-3, CF-4	Rinse sample: 100 ml rinse sample to be collected after ECR step no.....	1 rinse sample	1 X about 100 ml
		Swab (Chemical & Micro) samples to be collected after ECR step no.....	Chemical Swab: Centrifuge basket, Cone, Bag holding plate, Reaction mass feeder and MLR tank Micro: CF Basket	Chemical: 5 swabs Micro: 1 swab
Dryer	TD-1, TD-2 TD-5	Rinse sample: 100 ml rinse sample to be collected after ECR step no.....	1 rinse sample	1 X about 100 ml
		Swab (Chemical & Micro) samples: to be collected after ECR step no.....	Chemical Swab: Trays, view glass, door and inner surface Micro swab: Trays	Chemical: 4 swabs, Micro: 1 swab
Mill	MM-1, MM-4, MM-5	Rinse sample: 100 ml rinse sample to be collected after ECR step no.....	1 rinse sample	1 X about 100 ml
		Swab (Chemical & Micro) samples: to be collected after ECR step no.....	Chemical Swab: Discharge chute, feed hopper, gasket and screen. Micro swab: screen	Chemical: 4 swabs, Micro: 1 swab
Sifter	VS-3, VS-5	Rinse sample: 100 ml rinse sample to be collected after ECR step no.....	1 rinse sample	1 X about 100 ml
		Swab (Chemical & Micro) sample: to be collected after ECR step no.....	Chemical Swab: Sieve, upper plate, lower hopper and discharge chute Micro swab: lower hopper	Chemical: 4 swabs, Micro: 1 swab

8.7 According to the above defined sampling plan, Quality Assurance shall collect the rinse and swab (chemical and micro) samples as per SOP 1035-A-0082. Micro swab sample shall be collected first from different location other than chemical swab sample by microbiologist / QA.

8.8 The microbial contamination (bioload) sampling shall be drawn for equipment used in controlled and classified or controlled but not classified (CNC) clean areas. Typically for manufacturing of non-sterile API environment is non controlled till crystallisation step hence no microbial limits shall be applied to such equipment.

8.9 Micro swab samples shall be collected from each group of equipment specified in below table.

**ANNEXURE 1035-A-0072/A8
(SPECIMEN COPY)**



PROTOCOL FOR API CLEANING VALIDATION / VERIFICATION / PERIODIC CLEANING VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

Group	Equipment
Group A	Crystalliser Reactor
Group B	Closed type equipment like centrifuge, Blender, Dryers, RCVD, VTD, ANFD, RVPD, FBD etc.
Group C	Portable equipment & equipment which shall be dismantled completely- E.g. Multi mill, Sifters, Jet mill, Cad mill, Co mill, sparkler filter, Nutsche filter, Buckner filter etc.

- 8.10 The collected rinse and swab samples along with the sampling plan shall be sent to QC for analysis of previous product content, cleaning agent (If used) and microbial contamination.
- 8.11 Quality Control shall initiate the analysis as per cleaning specification of product / microbiological method.
- 8.12 *Analysis of cleaning validation sample shall be initiated within 24 hours after sampling, until and unless solution stability studies are available.*
- 8.13 Any non-compliance (OOS or additional peaks) observed, same shall be logged, investigated and CAPA shall be taken based on investigation.
- 8.14 After completion of analysis, Quality Control shall provide the result of samples analysed to Quality Assurance / production.
- 8.15 Quality Assurance shall verify the compliance of all the results (like rinse and swab results observed within the rinse and swab limit, absence of cleaning agent etc.)
- 8.16 A computerized pre-printed report shall be prepared for cleaning validation / verification / periodic cleaning verification after receiving all the TI sheet from quality control as per annexure 1035-A-0072/A9 and 1035-A-0072/A12 respectively. Report for Cleaning validation / verification / periodic cleaning verification shall be reviewed by section head of quality assurance and approved by Head unit quality assurance.
- 8.17 Interim report shall be prepared after 1st chain and 2nd chain cleaning validation.
- 8.18 Total carryover of equipment chain by rinse or swab method observed more than MACO limit, then deviation shall be raised and same shall be investigated as per respective SOP.
- 8.19 Cleaning validation shall be performed on three batches to conclude the cleaning validation. All critical cleaning steps and test results to be monitored.
- 8.20 Periodic cleaning verification shall be performed on one batch / run. To conclude the verification all critical steps and test results shall be compared with cleaning validation runs.
- 8.21 In case periodic verification sample results are significantly higher than the validation results (More than 75 % of the limit) consider the same as out of alert and investigate the reason for the same.

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9.0 ACCEPTANCE CRITERIA:

Cleaning process shall be capable of achieving the intended results on a consistent and continuing basis and thus shall be capable of consistently meets the established MACO, Rinse and swab limits. The critical process parameters of cleaning process shall meet the acceptance criteria specified in respective equipment cleaning record.

9.1 Critical Process Parameters (CPP):

Review the Critical Process parameters of cleaning procedure Observation of Critical Process parameters to be captured in validation report.

Sr.	Equipment type	ECR step number	Critical process parameters	Limits
i.	Reactor	5	Pressure jet pressure	Pressure: > 25 Kg /cm ²
			Time for pressure cleaning	Time: NLT 10 minutes
		8	Reflux the content	Time: 30+5 minutes
				Reflux Qty : 20 lt.
		11	Pressure cleaning of dismantled parts	Pressure: > 25 Kg /cm ²
			Time for pressure cleaning	Time: NLT 5 minutes
		16	Soaking of small parts	Time: NLT 10 minutes
		20	Final rinse solvent quantity	Qty: 10 lt.
ii.	Centrifuge	3	Pressure jet pressure	Vacuum: NLT 400 mmHg
			Time for pressure cleaning	Temp: NLT45°C
		7	Pressure cleaning of dismantled parts	Time: NLT 30 minutes.
			Time for pressure cleaning	Pressure: > 25 Kg /cm ²
		9	Soaking of small parts	Time: NLT 10 minutes
			Final rinse solvent quantity	Time: NLT 10 minutes
		12		Qty: 10 lt.
iii.	Dryer	5	Pressure jet pressure	Pressure: > 25 Kg /cm ²
			Time for pressure cleaning	Time: NLT 10 minutes
		12	Final rinse solvent quantity	Qty: 10 lt.
				Vacuum: NLT 400 mmHg
		15	Final drying	Temp: NLT45°C
				Time: NLT 30 minutes.
		3	Pressure jet pressure	Pressure: > 25 Kg /cm ²
			Time for pressure cleaning	Time: NLT 10 minutes
iv.	Mill	6	Soaking of small parts	Time: NLT 10 minutes
			Scrubbing time	Time: NLT 10 minutes
		10	Final rinse solvent quantity	Qty: 5 lt.
v.	Sifter	4	Pressure jet pressure	Pressure: > 25 Kg /cm ²
			Time for pressure cleaning	Time: NLT 10 minutes
		10	Final rinse solvent quantity	Qty: 5 lt.

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9.2 Sampling and testing review:

9.2.1 Equipment shall be visually clean after cleaning.

9.2.2 Final rinse shall be visually clear.

9.2.3 Limits for chemical analysis:

For Rinse sample (Rinse Limit)	NMTPPM
For swab sample (Swab Limit)	NMT..... µg / swab or PPM
Carryover from chain of equipment (MACO)	NMTmg

Note: If limits for rinse and swabs sample more than 10 ppm, then acceptance criteria shall be NMT 10 ppm and if less than 10 ppm then that value calculated based on PDE shall be considered as acceptable limit.

9.2.4 Bioload :

Total Count	NMT: 500 cfu / swab
Mold Count	NMT: 50 cfu / swab

10.0 NON – CONFORMANCE:

If non-conformance observed shall be investigated as per respective SOP. Details and outcome of non-conformance shall be recorded in report section.

11.0 FREQUENCY:

11.1 Initial validation with three runs on complete equipment chain.

11.2 Any changes in cleaning procedure and cleaning solvent quantity.

11.3 Periodic cleaning verification of cleaning procedures for worst case product after every three years by checking rinse / swab sample on one batch / run.

11.4 Addition of new equipment and type in manufacturing flow based on risk assessment.

12.0 SUMMARY OF FINDINGS AND CONCLUSION OF VALIDATION:

Summarize the findings and conclude the cleaning validation based on the interpretation of the results of the validation.

In the summary LOD / LOQ values shall be considered for calculating the contamination in the next product.

**ANNEXURE 1035-A-0072/A8
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13.0 RECOMMENDATION:

Record the final recommendation of validation report based on the interpretation of the results of the validation.

14.0 ANNEXURES:

Annexure-I	Technical information sheet along with sampling plan of rinse and swab (chemical and micro) samples and result COA.
Annexure-II	Copy of Equipment cleaning record (ECR) of type B cleaning.

15.0 VALIDATION PERFORMED BY:

The personnel who have performed the validation, supervised the validation, completed the record shall sign the validation report

16.0 REVIEW AND COMMENTS:

Quality Assurance shall review the validation report and shall include the comments with sign and date.

17.0 APPROVED BY:

Head Unit Quality Assurance shall approve the validation report .

18.0 ABBREVIATIONS:

API	Active Pharmaceutical Ingredient	lt	Litre
SOP	Standard Operating Procedure	ppm	Parts per million
QA	Quality Assurance	A.R. No.	Analytical reference number
QC	Quality Control	µg	Microgram
MACO	Maximum allowable carryover	CO	Carryover
ECR	Equipment cleaning record	NLT	Not less than

19.0 PROTOCOL APPROVAL:

Prepared by (Validation team) Sign & Date			Reviewed by Sign & Date	Approved by Sign & Date
Quality Assurance	Production	Quality Control	Quality Assurance	Head Unit Quality Assurance

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1.0 REASON FOR VALIDATION: To be mentioned.

2.0 VALIDATION TEAM:

Representatives from Production, Quality control and Quality Assurance.

Production	Individual names to be recorded
Quality control	Individual names to be recorded
Quality Assurance	Individual names to be recorded

3.0 DESCRIPTION:

Parameters	First chain	Second chain	Third chain
Batch Number			
Equipment Cleaning SOP number			
Cleaning solvent / Cleaning agent (Concentration of cleaning agent)			
MACO limit			
Rinse limit			
Swab Limit			

4.0 TRAINING DETAILS OF PERSON INVOLVED IN CLEANING VALIDATION / VERIFICATION:

Training on cleaning validation protocol and report: Class ID :

5.0 SUMMARY:

A. Details of critical cleaning process parameters (CPP):

<u>Sr. No.</u>	<u>Equipment code</u>	<u>ECR step number</u>	<u>Critical cleaning process parameters</u>	<u>Limit</u>	<u>Observation of CPP</u>		
					<u>1st chain</u>	<u>2nd chain</u>	<u>3rd chain</u>
1.	Reactor	5	Pressure jet pressure	Pressure: >25Kg /cm ²			
			Time for pressure cleaning	Time: NLT 10 min.			
		8	Reflux the content	Time: 30+5 min			
				Reflux Qty: 20 lt.			
		11	Pressure cleaning of dismantled parts	Pressure: > 25 Kg /cm ²			
			Time for pressure cleaning	Time: NLT 5 minutes			

**ANNEXURE 1035-A-0072/A9
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<u>Sr. No.</u>	<u>Equipment code</u>	<u>ECR step number</u>	<u>Critical cleaning process parameters</u>	<u>Limit</u>	<u>Observation of CPP</u>		
					<u>1st chain</u>	<u>2nd chain</u>	<u>3rd chain</u>
1.	Reactor	16	Soaking of small parts	Time: NLT 10 min.			
		20	Final rinse solvent quantity	Qty: 10 lt.			
		21	Final drying	Vacuum: NLT 400 mmHg			
				Temp: NLT 45°C			
				Time: NLT 30 min.			
2.	Centrifuge	3	Pressure jet pressure	Pressure: > 25 Kg /cm ²			
			Time for pressure cleaning	Time: NLT 10 minutes			
		7	Pressure cleaning of dismantled parts	Pressure: > 25 Kg /cm ²			
			Time for pressure cleaning	Time: NLT 10 minutes			
		9	Soaking of small parts	Time: NLT 10 min.			
		12	Final rinse solvent quantity	Qty: 10 lt.			
3.	Dryer	5	Pressure jet pressure	Pressure: > 25 Kg /cm ²			
			Time for pressure cleaning	Time: NLT 10 min.			
		12	Final rinse solvent quantity	Qty: 10 lt.			
		15	Final drying	Vacuum: NLT 400 mmHg			
				Temp: NLT 45°C			
				Time: NLT 30 min.			
4.	Mill	3	Pressure jet pressure	Pressure: > 25 Kg /cm ²			
			Time for pressure cleaning	Time: NLT 10 minutes			
		6	Soaking of small parts	Time: NLT 10 minutes			
		8	Scrubbing time	Time: NLT 10 minutes			
		10	Final rinse solvent quantity	Qty: 5 lt.			
5.	Sifter	4	Pressure jet pressure	Pressure: > 25 Kg /cm ²			
			Time for pressure cleaning	Time: NLT 10 minutes			
		10	Final rinse solvent quantity	Qty: 5 lt.			

Prepared By
Sign / Date : _____

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B. Result of visual inspection of final rinse:

Sr. No.	Equipment	Equipment code	Name of equipment parts	Visual observation of final rinse (Clear /Not Clear)		
				1 st chain	2 nd chain	3 rd chain
1.						
2.						
3.						

C. Result of visual observation:

Sr. No.	Equipment	Equipment code	Name of equipment parts	Visual observation of equipment (Clean /Not Clean)		
				1 st chain	2 nd chain	3 rd chain
1.						
2.						
3.						

D. Results of cleaning agent determination:

Cleaning agent Name: _____			Concentration: _____		
Sr. No.	Equipment	Equipment code	Carryover of cleaning agent in equipment by rinse method (mg)		
			1 st chain	2 nd chain	3 rd chain
1.					
2.					
3.					
Carryover of cleaning agent in chain of equipment $\sum$ CO = _____ mg					
MACO limit			= NMT _____ mg		

E. Results of Rinse samples:

Sr. No.	Equipment	Equipment code	Carryover of product in equipment by rinse method (CO in mg) = Last rinse value x Last rinse volume		
			1 st chain	2 nd chain	3 rd chain

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1.					
2.					
3.					
Carryover of product in chain of equipment $\sum$ CO = _____ mg					
MACO limit			= NMT _____ mg		

F. Results of Swab samples:

Sr. No.	Equipment Detail		Carryover of product in equipment by swab method (CO in mg) = In PPM: [(Max. Swab value X 10 ml / swab surface area) X Area of Equipment] / 1000 OR In µg/swab: [(Max. Swab value / swab surface area) X Area of Equipment] / 1000		
	Equipment Name	Code No.	1 st chain	2 nd chain	3 rd chain
1.					
2.					
3.					
Carryover of product in chain of equipment $\sum$ CO = _____ mg / µg					
MACO limit			= NMT _____ mg / µg		

G. Results of microbial determination:

Sr. No.	Equipment code	Name of equipment parts	Microbial Results					
			Total count (Limit :)			Fungal count (Limit :)		
			1 st chain	2 nd chain	3 rd chain	1 st chain	2 nd chain	3 rd chain
1.								
2.								

6.0 NON – CONFORMANCE:

Any non-conformance if observed during validation shall be investigated and outcome of non-conformance shall be recorded.

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Sr. No.	Document No.	Description of Non-conformance	Outcome of non-conformance	CAPA

7.0 CONCLUSION OF VALIDATION:

Equipment considered during cleaning validation are e.g., reactors, centrifuge, sparkler filter / twin bag filters, Rotacone vacuum driers, vacuum driers, fluid bed driers, Multimill, jet mill, sifters, and ancillary accessories such as scoops, scrappers etc.

Rinse and swab samples were collected as per sampling plan. The swab samples collected by using Texwipe swab.

The critical cleaning process parameters (CPP) of each equipment of all three chains found as per acceptance criteria specified in ECR.

The clarity of last rinse sample of each equipment found visually clear and equipment parts of each equipment found visually clean.

Three batches were assessed for cleaning validation as per equipment specific cleaning procedures. The results of each equipment of 3 batches met / not met within acceptance criteria for rinse (____) and swab (____) limits.

The total carryover from chain of equipment by rinse and swab method met / not met MACO (____) limit.

The precise instructions in each of the procedures for dismantling and cleaning is found adequate / not adequate. The details of cleaning aids (Scrubber/ brush/ jet) specified in the procedure is adequate / not adequate for routine cleaning application.

The methodology for visual inspection post cleaning with use of torch light / (indicate type of light source) is adequate / not adequate.

There is no change in existing cleaning procedure for all three-equipment chain. The validated cleaning procedure is robust to be applied for routine equipment cleaning.

The bioload on equipment surfaces found within / not within the acceptance criteria.

In the event, of any nonconformity, then conclusion and recommendation shall cover any changes in cleaning procedure or solvent quantity, or reflux condition or number of refluxes required, accordingly the procedure shall be revised, hence summary to include :

Based on three validation batches, improvement in cleaning procedure is suggested for-----, -----, ----- which need to be revised w.r.t reflux temperature / reflux condition/ used of scrubbing tools for component / accessories -----, -----, ----- and batch validation to be executed.

Reference QMS document: applicable / not applicable : reference number.....

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Based on the outcome of validation activity, it is certified that the defined cleaning procedure of equipment and accessories used for the manufacturing of drug substance is validate / not validated and shall be used / not used / any other remark for its intended purpose

8.0 RECOMMENDATION:

Based on overall cleaning validation results, the cleaning procedure (.....) to be followed for routine cleaning of equipment / accessories used for manufacturing ofproduct group for routine application of product change over or campaign cleaning.

The Same critical cleaning process parameters like, reflux time, temperature, soaking and scrubbing time and solvent volume to be followed for routine cleaning.

9.0 ANNEXURES:

Annexure-I	Technical information sheet along with sampling plan of rinse and swab (chemical and micro) samples and result COA.
Annexure-II	Copy of Equipment cleaning record (ECR) of type B cleaning.

10.0 VALIDATION TEAM SIGNING:

Quality Assurance : _____ Date : _____

Production : _____ Date : _____

Quality Control : _____ Date : _____

11.0 REVIEW AND COMMENTS:

REVIEWED BY:

Section Head (Quality Assurance)
Date:

12.0 APPROVED BY:

Head Unit Quality Assurance
Date:

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1.0 OBJECTIVE:

In case of validation: To validate and ensure that the cleaning procedure..... is effective and removes all the residues of intermediates /API (before active moiety formed) and cleaning agent up to the acceptance criteria decided during cleaning matrix.

In case of verification: To verify and ensure that the cleaning procedure..... is effective and removes all the residues of intermediates /API (before active moiety formed)..... and cleaning agent up to the acceptance criteria decided during cleaning matrix.

In case periodic verification: To verify periodically, the cleaning procedure.....is effective and consistently removes all the residues of intermediates /API (before active moiety formed)..... and cleaning agent up to the acceptance criteria decided during cleaning matrix.

2.0 SCOPE:

In case of validation: Protocol is applicable for validation of cleaning procedurewhen equipment's used for the manufacturing of selected worst-case intermediates / API (before active moiety formed) from soluble group.

In case of verification: Protocol is applicable for verification of cleaning procedurewhen equipment's used for the manufacturing of non-worst-case intermediates / API (before active moiety formed)..... from soluble group.

In case periodic verification: Protocol is applicable for periodic cleaning verification of cleaning procedures when equipment used for manufacturing of selected worst case intermediates / API (before active moiety formed)..... from soluble group

3.0 REASON FOR VALIDATION: Initial / Concurrent / Verification / Periodic verification

4.0 VALIDATION TEAM:

Representatives from Production, Quality control and Quality Assurance. (Individual names shall be recorded in report section)

5.0 RESPONSIBILITIES:

5.1 Production:

- To be part of validation team.
- To ensure that the cleaning activity is performed as per the respective product changeover procedure.
- To ensure the effectiveness of cleaning by visual inspection.
- To ensure rinse / swab results as per sampling plan within acceptance criteria for product changeover.

5.2 Quality Control:

- To be part of validation team.

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- To ensure availability of validated analytical methods for measurement of product in swabs and rinses.
- To analyse the cleaning validation samples as per TI sheet

5.3 Quality Assurance:

- To be part of validation team.
- To coordinate with Production and Quality Control for cleaning validation.
- To collect validation samples as per approved sampling plan of equipment.
- To perform cleaning validation as per approved validation protocol and ensure completion of report.
- To ensure the Training details of the personnel involved in the cleaning of equipment, collecting of samples, testing of samples, deciding the sample plan, preparing, reviewing, approving the reports

6.0 REVIEW OF PRE-REQUISITE AND TRAINING DETAILS:

Sr.	Item	Remark Document no./ Available /NA
i.	Product change over cleaning procedure with ECR checklist	Document no.:
ii.	Validated analytical method for measurement of product in the rinses and swab	Document no.:
iii.	Validated analytical method for measurement of cleaning agent in the rinses (in case cleaning agent used for cleaning)	Document no.:
iv.	Validated microbiological method for measurement of bioburden on the equipment (In case of micro sample)	Document no.:
v.	Approved sampling plan for cleaning validation	Available
vi.	Individual equipment specific diagram with swab location shall be prepared as per SOP 1035-A-0082.	Available
vii.	Specification for cleaning sample analysis	Document no.:
viii.	Training on cleaning validation SOP 1035-A-0072	Available / Training ID:
ix.	Training and qualification of personnel on cleaning procedure as per SOP 1035-A-0083	Available / Training ID:
x.	Training and qualification of personnel on visual inspection of equipment as per SOP 1035-A-0083	Available / Training ID:
xi.	Training of personnel on sampling plan.	Available / Training ID:
xii.	Training of personnel on swab sampling as pe SOP 1035-A-0082	Available / Training ID:

7.0 DESCRIPTION:

Validation done on batch numbers, Equipment cleaning SOP number, cleaning solvent / cleaning agent with concentration shall be recorded in report section.

8.0 PROCEDURE:

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- 8.1 Proper training session on cleaning validation protocol and report shall be conducted to all validation team member prior to initiate the cleaning validation.
- 8.2 Quality Assurance shall issue the required ECR and production shall initiate the equipment cleaning.
- 8.3 TI sheet shall be raised as an intimation to QC for cleaning sample and preparation of swab diluent prior to rinse and swab sampling.
- 8.4 Quality assurance shall be arranged the sampling tools like, Sampling bottles for rinse collection, Swab stick required for swab collection, 4/9 square inch template, Swab sample collection test tube / bottle and Labels required for labelling for collected samples etc.
- 8.5 Quality Assurance shall collect 100 ml of cleaning solvent as blank for evaluation of interference of cleaning solvent.
- 8.6 Sampling plan:

The time and number of samples shall be drawn, shall be described in sampling plan. and same to be attached with cleaning validation protocol.

Sampling equipment, sampling locations and sampling intervals shall be described in this section. Rinse samples shall be drawn in cleaned sample bottle. Swab sample shall be collected by using Texwipe swab stick after moistening in swabbing diluent and swab heads to be stored in swab sample collection test tube / bottle containing 10ml diluent. Sampling details shall be recorded in validation report section. Sampling plan for validation samples shall be elaborated in protocol for predefined sample quantity along with ECR step number and sample number. Equipment chain including small parts and utensil which considered in cleaning matrix shall be consider for validation samples.

Validation samples:

<u>Type of equipment</u>	<u>Applicable Equipment code number</u>	<u>Sampling interval</u>	<u>Number of samples</u>	<u>Sampling quantity</u>
Reactor	R1, R2, R3	Rinse sample: 100ml rinse sample to be collected after ECR step no.....	1 rinse sample	1 X about 100 ml
		Swab samples: to be collected after ECR step no.....	Swab from: Reactor body, Agitator and Bottom valve	3 swabs

**ANNEXURE 1035-A-0072/A10
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Type of equipment	Applicable Equipment code number	Sampling interval	Number of samples	Sampling quantity
Filter	CF-1, CF-3, CF-4	Rinse sample: 100 ml rinse sample to be collected after ECR step no.....	1 rinse sample	1 X about 100 ml
		Swab samples to be collected after ECR step no.....	Swab from: Centrifuge basket, Cone, Bag holding plate, Reaction mass feeder and MLR tank	5 swabs
Dryer	TD-1, TD-2, TD-5	Rinse sample: 100 ml rinse sample to be collected after ECR step no.....	1 rinse sample	1 X about 100 ml
		Swab samples to be collected after ECR step no.....	Swab from: Trays, view glass, door and inner surface	4 swabs
Mill	MM-1, MM-4, MM-5	Rinse sample: 100 ml rinse sample to be collected after ECR step no.....	1 rinse sample	1 X about 100 ml
		Swab samples to be collected after ECR step no.....	Swab from: Discharge chute, feed hopper, gasket and screen.	4 swabs
Sifter	VS-3, VS-5	Rinse sample: 100 ml rinse sample to be collected after ECR step no.....	1 rinse sample	1 X about 100 ml
		Swab sample to be collected after ECR step no.....	Swab from: Sieve, upper plate, lower hopper and discharge chute	4 swabs

- 8.7 According to the above defined sampling plan, Quality Assurance shall collect the rinse and swab samples as per SOP 1035-A-0082.
- 8.8 The collected rinse and swab samples along with the sampling plan shall be sent to QC for analysis of previous product content.
- 8.9 Quality Control shall initiate the analysis as per cleaning specification of product.
- 8.10 *Analysis of cleaning validation sample shall be initiated within 24 hours after sampling, until and unless solution stability studies are available.*
- 8.11 Any non-compliance (OOS or extraneous peaks) observed, same shall be logged, investigated and CAPA shall be taken based on investigation.

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- 8.12 After completion of analysis, Quality Control shall provide the result of samples analysed to Quality Assurance.
- 8.13 Quality Assurance shall verify the compliance of all the results (like rinse and swab results observed within the rinse and swab limit, absence of cleaning agent etc.)
- 8.14 A computerized pre-printed report shall be prepared for cleaning validation / verification / periodic cleaning verification after receiving all the TI sheet from quality control as per annexure 1035-A-0072/A11 and 1035-A-0072/A12 respectively. Report for Cleaning validation / verification / periodic cleaning verification shall be reviewed by section head of quality assurance and approved by Head unit quality assurance.
- 8.15 Interim report shall be prepared after 1st chain and 2nd chain cleaning validation.
- 8.16 Total carryover of equipment chain by rinse or swab method observed more than 100 ppm, then deviation shall be raised and same shall be investigated as per respective SOP.
- 8.17 Validation shall be performed on three batches to conclude the cleaning validation. All critical cleaning steps and test results to be monitored.
- 8.18 Periodic cleaning verification shall be performed on one batch / run and to conclude the verification all critical steps and test results shall be compared with cleaning validation runs.

9.0 ACCEPTANCE CRITERIA:

Cleaning process shall be capable of achieving the intended results on a consistent and continuing basis and thus shall be capable of consistently meets the established MACO, Rinse and swab limits. The critical process parameters of cleaning process shall meet the acceptance criteria specified in respective equipment cleaning record.

9.1 Critical Process Parameters (CPP):

Review the Critical Process parameters of cleaning procedure Observation of Critical Process parameters to be captured in validation report.

Sr.	Equipment type	ECR step number	Critical process parameters	Limits
i.	Reactor	5	Pressure jet pressure	Pressure: > 25 Kg /cm ²
			Time for pressure cleaning	Time: NLT 10 minutes
		8	Reflux the content	Time: 30+5 minutes
				Reflux Qty : 20 lt.
		11	Pressure cleaning of dismantled parts	Pressure: > 25 Kg /cm ²
			Time for pressure cleaning	Time: NLT 5 minutes
		16	Soaking of small parts	Time: NLT 10 minutes
		20	Final rinse solvent quantity	Qty: 10 lt.
		21	Final drying	Vacuum: NLT 400 mmHg
				Temp: NLT45°C
				Time: NLT 30 minutes.

Prepared By
Sign / Date : _____

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ii.	Centrifuge	3	Pressure jet pressure	Pressure: > 25 Kg /cm ²
			Time for pressure cleaning	Time: NLT 10 minutes
		7	Pressure cleaning of dismantled parts	Pressure: > 25 Kg /cm ²
			Time for pressure cleaning	Time: NLT 10 minutes
		9	Soaking of small parts	Time: NLT 10 minutes
		12	Final rinse solvent quantity	Qty: 10 lt.
iii.	Dryer	5	Pressure jet pressure	Pressure: > 25 Kg /cm ²
			Time for pressure cleaning	Time: NLT 10 minutes
		12	Final rinse solvent quantity	Qty: 10 lt.
		15	Final drying	Vacuum: NLT 400 mmHg Temp: NLT 45°C Time: NLT 30 minutes.
iv.	Mill	3	Pressure jet pressure	Pressure: > 25 Kg /cm ²
			Time for pressure cleaning	Time: NLT 10 minutes
		6	Soaking of small parts	Time: NLT 10 minutes
		8	Scrubbing time	Time: NLT 10 minutes
		10	Final rinse solvent quantity	Qty: 5 lt.

9.2 Sampling and testing review:

9.2.1 Equipment shall be visually clean after cleaning.

9.2.2 Final rinse shall be visually clear.

9.2.3 Limits for chemical analysis:

For Rinse sample (Rinse Limit)	NMT 10 PPM
For swab sample (Swab Limit)	NMT 10 PPM
Carryover from chain of equipment	NMT 100 PPM

10.0 NON – CONFORMANCE:

If non-conformance observed shall be investigated as per respective SOP. Details and outcome of non-conformance shall be recorded in report section.

11.0 FREQUENCY:

11.1 Initial validation with three runs on complete equipment chain.

11.2 Any changes in cleaning procedure and cleaning solvent quantity.

11.3 Periodic cleaning verification of cleaning procedures for worst case product after every three years by checking rinse / swab sample on one batch / run.

11.4 Addition of new equipment type in manufacturing flow.

PROTOCOL FOR INTERMEDIATE CLEANING VALIDATION / VERIFICATION / PERIODIC CLEANING VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

12.0 SUMMARY OF FINDINGS AND CONCLUSION OF VALIDATION:

Summarize the findings and conclude the cleaning validation based on the interpretation of the results of the validation.

In the summary LOD / LOQ values shall be considered for calculating the contamination in the next product.

13.0 RECOMMENDATION:

Record the final recommendation of validation report based on the interpretation of the results of the validation.

14.0 ANNEXURES:

Annexure-I	Technical information sheet along with sampling plan of rinse and swab (chemical and micro) samples and result COA.
Annexure-II	Copy of Equipment cleaning record (ECR) of type B cleaning.

15.0 VALIDATION PERFORMED BY:

The personnel who have performed the validation, supervised the validation, completed the record shall sign the validation report

16.0 REVIEW AND COMMENTS:

Quality Assurance shall review the validation report and shall include the comments with sign and date.

17.0 APPROVED BY:

Head Unit Quality Assurance shall approve the validation report.

18.0 ABBREVIATIONS:

API	Active Pharmaceutical Ingredient	lt	Litre
SOP	Standard Operating Procedure	ppm	Parts per million
QA	Quality Assurance	A.R. No.	Analytical reference number
QC	Quality Control	µg	Microgram
MACO	Maximum allowable carryover	CO	Carryover
ECR	Equipment cleaning record	NLT	Not less than

ANNEXURE 1035-A-0072/A10
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PROTOCOL FOR INTERMEDIATE CLEANING VALIDATION / VERIFICATION / PERIODIC CLEANING VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

19.0 PROTOCOL APPROVAL:

Prepared by (Validation team) Sign & Date			Reviewed by Sign & Date	Approved by Sign & Date
Quality Assurance	Production	Quality Control	Quality Assurance	Head Unit Quality Assurance

REPORT FOR INTERMEDIATE CLEANING VALIDATION / VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

1.0 REASON FOR VALIDATION: To be mentioned.

2.0 VALIDATION TEAM:

Representatives from Production, Quality control and Quality Assurance.

Production	Individual names to be recorded
Quality control	Individual names to be recorded
Quality Assurance	Individual names to be recorded

3.0 DESCRIPTION:

Parameters	First chain	Second chain	Third chain
Batch Number			
Equipment Cleaning SOP number			
Cleaning solvent / Cleaning agent (Concentration of cleaning agent)			

4.0 TRAINING DETAILS OF PERSON INVOLVED IN CLEANING VALIDATION:

Training on cleaning validation protocol and report: Class ID :

5.0 SUMMARY:

A. Details of critical cleaning process parameters (CPP):

<u>Sr. No.</u>	<u>Equipment code</u>	<u>ECR step number</u>	<u>Critical cleaning process parameters</u>	<u>Limit</u>	<u>Observation of CPP</u>		
					<u>1st chain</u>	<u>2nd chain</u>	<u>3rd chain</u>
1.	Reactor	5	Pressure jet pressure	Pressure: >25Kg /cm ²			
			Time for pressure cleaning	Time: NLT 10 min.			
		8	Reflux the content	Time: 30+5 min			
				Reflux Qty: 20 lt.			
		11	Pressure cleaning of dismantled parts	Pressure: > 25 Kg /cm ²			
			Time for pressure cleaning	Time: NLT 5 minutes			

**ANNEXURE 1035-A-0072/A11
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REPORT FOR INTERMEDIATE CLEANING VALIDATION / VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

Sr. No.	Equipment code	ECR step number	Critical cleaning process parameters	Limit	Observation of CPP		
					1 st chain	2 nd chain	3 rd chain
1.	Reactor	16	Soaking of small parts	Time: NLT 10 min.			
		20	Final rinse solvent quantity	Qty: 10 lt.			
		21	Final drying	Vacuum: NLT 400 mmHg			
				Temp: NLT45°C			
				Time: NLT 30 min.			
2.	Centrifuge	3	Pressure jet pressure	Pressure: > 25 Kg /cm ²			
			Time for pressure cleaning	Time: NLT 10 minutes			
		7	Pressure cleaning of dismantled parts	Pressure: > 25 Kg /cm ²			
			Time for pressure cleaning	Time: NLT 10 minutes			
		9	Soaking of small parts	Time: NLT 10 minutes			
		12	Final rinse solvent quantity	Qty: 10 lt.			
3.	Dryer	5	Pressure jet pressure	Pressure: > 25 Kg /cm ²			
			Time for pressure cleaning	Time: NLT 10 minutes			
		12	Final rinse solvent quantity	Qty: 10 lt.			
		15	Final drying	Vacuum: NLT 400 mmHg			
				Temp: NLT45°C			
				Time: NLT 30 minutes.			
4.	Mill	3	Pressure jet pressure	Pressure: > 25 Kg /cm ²			
			Time for pressure cleaning	Time: NLT 10 minutes			
		6	Soaking of small parts	Time: NLT 10 minutes			
		8	Scrubbing time	Time: NLT 10 minutes			
		10	Final rinse solvent quantity	Qty: 5 lt.			

REPORT FOR INTERMEDIATE CLEANING VALIDATION / VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

B. Result of visual inspection of final rinse:

Sr. No.	Equipment	Equipment code	Name of equipment parts	Visual observation of final rinse (Clear /Not Clear)		
				1 st chain	2 nd chain	3 rd chain
1.						
2.						
3.						

C. Result of visual observation:

Sr. No.	Equipment	Equipment code	Name of equipment parts	Visual observation of equipment (Clean /Not Clean)		
				1 st chain	2 nd chain	3 rd chain
1.						
2.						
3.						

D. Results of cleaning agent determination:

Cleaning agent Name: _____			Concentration: _____		
Sr. No.	Equipment	Equipment code	Carryover of cleaning agent in equipment by rinse method		
			1 st chain	2 nd chain	3 rd chain
1.					
2.					
3.					
Carryover of cleaning agent in chain of equipment $\sum CO =$ _____					
limit			= NMT _____		

E. Results of Rinse samples:

Sr. No.	Equipment	Equipment code	Carryover of product in equipment chain by rinse method (CO in ppm)		
			1 st chain	2 nd chain	3 rd chain
1.					

REPORT FOR INTERMEDIATE CLEANING VALIDATION / VERIFICATION		
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2.					
3.					
Carryover of product in chain of equipment Σ CO					
Limit			= NMT 100 ppm.		

F. Results of Swab samples:

Sr. No.	Equipment Detail		Carryover of product in equipment by swab method (CO in ppm)		
	Equipment Name	Code No.	1 st chain	2 nd chain	3 rd chain
1.					
2.					
3.					
Carryover of product in chain of equipment Σ CO = _____ in ppm					
Limit			= NMT 100 ppm		

6.0 NON – CONFORMANCE:

Any non-conformance if observed during validation shall be investigated and outcome of non-conformance shall be recorded.

Sr. No.	Document No.	Description of Non-conformance	Outcome of non-conformance	CAPA

7.0 CONCLUSION OF VALIDATION:

Equipment considered during cleaning validation are e.g., reactors, centrifuge, sparkler filter / twin bag filters, Rotacone vacuum driers, vacuum driers, fluid bed driers, Multimill, jet mill, sifters, and ancillary accessories such as scoops, scrappers etc.
Rinse and swab samples were collected as per sampling plan. The swab samples collected by using Texwipe swab.

The critical cleaning process parameters (CPP) of each equipment of all three chains found as per acceptance criteria specified in ECR.

The clarity of last rinse sample of each equipment found visually clear and equipment parts of each equipment found visually clean.

REPORT FOR INTERMEDIATE CLEANING VALIDATION / VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

Three batches were assessed for cleaning validation as per equipment specific cleaning procedures. The results of each equipment of 3 batches met / not met within acceptance criteria for rinse (____) and swab (____) limits.

The total carryover from chain of equipment by rinse and swab method met / not met MACO (____) limit

The precise instructions in each of the procedures for dismantling and cleaning is found adequate / not adequate. The details of cleaning aids (Scrubber/ brush/ jet) specified in the procedure is adequate / not adequate for routine cleaning application.

The methodology for visual inspection post cleaning with use of torch light / (indicate type of light source) is adequate / not adequate.

There is no change in existing cleaning procedure for all three-equipment chain. The validated cleaning procedure is robust to be applied for routine equipment cleaning.

In the event, of any nonconformity, then conclusion and recommendation shall cover any changes in cleaning procedure or solvent quantity, or reflux condition or number of refluxes required, accordingly the procedure shall be revised, hence summary to include :

Based on three validation batches, improvement in cleaning procedure is suggested for-----, -----, ----- which need to be revised w.r.t reflux temperature / reflux condition/ used of scrubbing tools for component / accessories -----, -----, ----- and batch validation to be executed.

Reference QMS document : applicable / not applicable : reference number.....

Based on the outcome of validation activity, it is certified that the defined cleaning procedure of equipment and accessories used for the manufacturing of drug substance is validate / not validated and shall be used / not used / any other remark for its intended purpose

8.0 RECOMMENDATION:

Based on overall cleaning validation results, the cleaning procedure (.....) to be followed for routine cleaning of equipment / accessories used for manufacturing ofproduct group for routine application of product change over or campaign cleaning.

The Same critical cleaning process parameters like, reflux time, temperature, soaking and scrubbing time and solvent volume to be followed for routine cleaning.

9.0 ANNEXURES:

Annexure-I	Technical information sheet along with sampling plan of rinse and swab (chemical and micro) samples and result COA.
Annexure-II	Copy of Equipment cleaning record (ECR) of type B cleaning.

REPORT FOR INTERMEDIATE CLEANING VALIDATION / VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

10.0 VALIDATION TEAM SIGNING:

Quality Assurance : _____ Date : _____

Production : _____ Date : _____

Quality Control : _____ Date : _____

11.0 REVIEW AND COMMENTS:

REVIEWED BY:

Section Head (Quality Assurance)
Date:

12.0 APPROVED BY:

Head Unit Quality Assurance
Date:

REPORT FOR PERIODIC CLEANING VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

1.0 VERIFICATION TEAM:

Representatives from Production, Quality control and Quality Assurance.

Production	Individual names to be recorded
Quality control	Individual names to be recorded
Quality Assurance	Individual names to be recorded

2.0 DESCRIPTION:

Parameters	Initial cleaning validation			Periodic Verification
	1 st Chain	2 nd Chain	3 rd Chain	
Batch number				
Cleaning SOP number				
Cleaning solvent				
MACO Limit				
Rinse Limit				
Swab limit				
Periodic verification / Validation rinse alert limit	75% of Rinse Limit =			
Periodic verification / Validation rinse alert limit	75% of Swab Limit =			

3.0 TRAINING DETAILS OF PERSON INVOLVED IN PERIODIC VERIFICATION:

Training on cleaning validation protocol and report: Class ID :

4.0 SUMMARY:

A. DETAILS OF CRITICAL CLEANING PROCESS PARAMETERS:

Sr. No.	Equipment code	ECR step number	Critical cleaning process parameters	Limit	Observation of CPP			
					1 st chain	2 nd chain	3 rd chain	Periodic Verification
1.	Reactor	5	Pressure jet pressure	Pressure: >25Kg /cm ²				
			Time for pressure cleaning	Time: NLT 10 min.				
		8	Reflux the content	Time: 30+5 min				
				Reflux Qty: 20 lt.				

**ANNEXURE 1035-A-0072/A12
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REPORT FOR PERIODIC CLEANING VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

Sr. No.	Equipment code	ECR step number	Critical cleaning process parameters	Limit	Observation of CPP			
					1 st chain	2 nd chain	3 rd chain	Periodic Verification
1.	Reactor							
		11	Pressure cleaning of dismantled parts	Pressure: > 25 Kg /cm ²				
			Time for pressure cleaning	Time: NLT 5 minutes				
		16	Soaking of small parts	Time: NLT 10 min.				
		20	Final rinse solvent quantity	Qty: 10 lt.				
		21	Final drying	Vacuum: NLT 400 mmHg				
				Temp: NLT45°C				
				Time: NLT 30 min.				
2.	Centrifuge	3	Pressure jet pressure	Pressure: > 25 Kg /cm ²				
			Time for pressure cleaning	Time: NLT 10 minutes				
		7	Pressure cleaning of dismantled parts	Pressure: > 25 Kg /cm ²				
			Time for pressure cleaning	Time: NLT 10 minutes				
		9	Soaking of small parts	Time: NLT 10 minutes				
		12	Final rinse solvent quantity	Qty: 10 lt.				
3.	Dryer	5	Pressure jet pressure	Pressure: > 25 Kg /cm ²				
			Time for pressure cleaning	Time: NLT 10 minutes				
		12	Final rinse solvent quantity	Qty: 10 lt.				
		15	Final drying	Vacuum: NLT 400 mmHg				
				Temp: NLT45°C				
				Time: NLT 30 minutes.				

**ANNEXURE 1035-A-0072/A12
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REPORT FOR PERIODIC CLEANING VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

Sr. No.	Equipment code	ECR step number	Critical cleaning process parameters	Limit	Observation of CPP			
					1 st chain	2 nd chain	3 rd chain	Periodic Verification
4.	Mill	3	Pressure jet pressure	Pressure: > 25 Kg /cm ²				
			Time for pressure cleaning	Time: NLT 10 minutes				
		6	Soaking of small parts	Time: NLT 10 minutes				
		8	Scrubbing time	Time: NLT 10 minutes				
		10	Final rinse solvent quantity	Qty: 5 lt.				
5.	Sifter	4	Pressure jet pressure	Pressure: > 25 Kg /cm ²				
			Time for pressure cleaning	Time: NLT 10 minutes				
		10	Final rinse solvent quantity	Qty: 5 lt.				

B. RESULT OF VISUAL INSPECTION OF FINAL RINSE:

Sr. No.	Equipment	Equipment code	Name of equipment parts	Visual observation of final rinse (Clear /Not Clear)			
				Initial Validation			Periodic Verification
				1 st Chain	2 nd Chain	3 rd Chain	
1.							
2.							
3.							
4.							

REPORT FOR PERIODIC CLEANING VERIFICATION		
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C. RESULT OF VISUAL INSPECTION OF EQUIPMENT:

Sr. No.	Equipment	Equipment code	Name of equipment parts	Visual observation of equipment (Clean /Not Clean)			
				Initial Validation			Periodic Verification
				1 st Chain	2 nd Chain	3 rd Chain	
1							
2							
3							
4							

D. RESULT OF RINSE SAMPLES

Sr. No.	Equipment	Equipment code	Carryover of product in equipment by rinse method (CO in mg / ppm)			
			Initial Validation			Periodic Verification
			1 st Chain	2 nd Chain	3 rd Chain	
1.						
2.						
3.						
4.						
Carryover of product in chain of equipment Σ CO = _____mg / ppm						
MACO limit			= NMT _____ mg /ppm			

REPORT FOR PERIODIC CLEANING VERIFICATION		
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E. RESULTS OF SWAB SAMPLES

Sr. No.	Equipment	Equipment code	Carryover of product in equipment by Swab method (CO in mg / ppm)			
			Initial Validation			Periodic Verification
			1 st Chain	2 nd Chain	3 rd Chain	
1.						
2.						
3.						
4.						
Carryover of product in chain of equipment $\sum$ CO = _____ mg / ppm						
MACO limit			= NMT _____ mg / ppm			

F. RESULTS OF CLEANING AGENT DETERMINATION:

Sr. No.	Equipment	Equipment code	Carryover of cleaning agent in equipment by rinse method (CO in mg / ppm)			
			Initial validation			Periodic Verification
			1 st Chain	2 nd Chain	3 rd Chain	
1.						
2.						
3.						
Carryover of cleaning agent in chain of equipment $\sum$ CO = _____ mg / ppm						
MACO limit			= NMT _____ mg / ppm			

REPORT FOR PERIODIC CLEANING VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

G. RESULTLS OF MICROBIAL DETERMINATION: (In case water is used as final rinse)

Sr. No.	Equipment code	Name of equipment parts	Microbial Results							
			Total count (Limit :)				Fungal count(Limit :)			
			1 st chain	2 nd chain	3 rd chain	Periodic verificat ion	1 st chain	2 nd chain	3 rd chain	Periodic verificat ion
1.										
2.										
3.										

H. REVIEW OF OUT OF ALERT RESULTS:

Above comparison data shows, out of expectations / alert results observed / Not observed as significant difference noticed / not noticed in results.

Investigation done: Yes / NA

Investigation conclusion:

5.0 NON – CONFORMANCE:

Any non-conformance if observed during periodic cleaning verification shall investigated and outcome shall be recorded

Sr. No.	Document No.	Description of Non-conformance	Outcome of non-conformance

6.0 CONCLUSION:

Equipment considered during periodic cleaning verification are e.g., reactors, centrifuge, sparkler filter / twin bag filters, Rotacone vacuum driers, vacuum driers, fluid bed driers, Multimill, jet mill, sifters, and ancillary accessories such as scoops, scrappers etc.

Rinse and swab samples were collected as per sampling plan. The swab samples collected by using Texwipe swab.

The critical cleaning process parameters (CPP) of each equipment used for periodic cleaning verification batch found as pe instruction specified in ECR and comparable with initial cleaning validation

REPORT FOR PERIODIC CLEANING VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

One batch were assessed for periodic cleaning verification as per equipment specific cleaning procedures. The results of each equipment met / not met within acceptance criteria for rinse (____) and swab (____) limits.

The clarity of last rinse sample of each equipment found visually clear and equipment parts of each equipment found visually clean.

The precise instructions in each of the procedures for dismantling and cleaning is found adequate / not adequate. The details of cleaning aids (Scrubber/ brush/ jet) specified in the procedure is adequate / not adequate for routine cleaning application.

The methodology for visual inspection post cleaning with use of torch light / (indicate type of light source) is adequate / not adequate.

There is no change in existing cleaning procedure for all three-equipment chain / verification run. The periodically verified cleaning procedure is robust to be applied for routine equipment cleaning post manufacturing.

In the event, of any nonconformity, then conclusion and recommendation shall cover any changes in cleaning procedure or solvent quantity, or reflux condition or number of refluxes required, accordingly the procedure shall be revised, hence summary to include:

Reference QMS document : applicable / not applicable : reference number.....

Based on the outcome of periodic cleaning verification activity, it is certified that the defined cleaning procedure of equipment and accessories used for the manufacturing of drug substance is stands validated and shall be used routine equipment cleaning.

7.0 RECOMMENDATION:

Based on overall periodic cleaning verification results, the cleaning procedure (.....) to be followed for routine cleaning of equipment / accessories used for manufacturing ofproduct group for routine application of product change over or campaign cleaning.

The Same critical cleaning process parameters like, reflux time, temperature, soaking and scrubbing time and solvent volume to be followed for routine cleaning.

8.0 ANNEXURES:

Annexure-I	Technical information sheet along with sampling plan of rinse and swab (chemical and micro) samples and result COA.
Annexure-II	Copy of Equipment cleaning record (ECR) of type B cleaning.

REPORT FOR PERIODIC CLEANING VERIFICATION		
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REFERENCE VALIDATION MASTER PLAN

9.0 VERIFICATION TEAM SIGNING:

Quality Assurance	:	_____	Date	:	_____
Production	:	_____	Date	:	_____
Quality Control	:	_____	Date	:	_____

10.0 REVIEW AND COMMENTS:

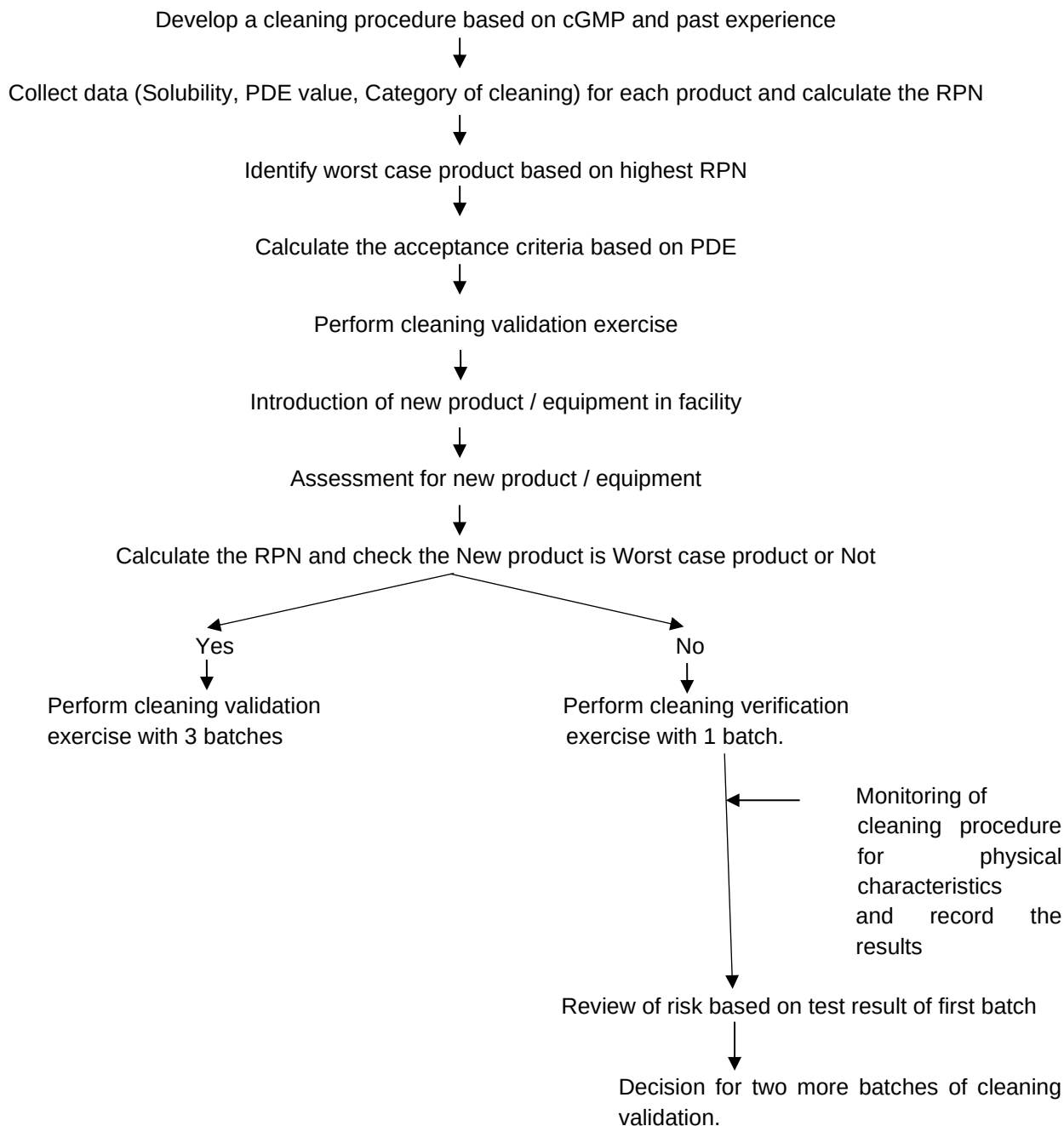
.....
.....

Section Head (Quality Assurance)
Date:

11.0 APPROVED BY:

Head Unit Quality Assurance
Date:

FLOW DIAGRAM OF VALIDATION AND IMPLEMENTATION OF VALIDATED CLEANING PROCEDURE



Note : If there is possibility of nitrosamine or Genotoxic impurity formation in manufacturing process, then evaluation to be done during cleaning of equipment.

ANNUAL CLEANING REVIEW REPORT : YEAR -----

1. **REVIEW OF TRAINING AND QUALIFICATION OF CONCERNED EMPLOYEE ON VISUAL INSPECTIONS:**
Details of training and qualification on visual inspection completed during this period
 - Number of new employees completed the Visual inspection training.
 - Number of periodic qualifications completed.
 - Total number of qualified persons are available for visual inspection.
2. **REVIEW OF TRAINING OF CONCERNED EMPLOYEE ON CLEANING OF EQUIPMENT:** Details of training on cleaning of equipment completed during this period
 - No of new employee completed the training on equipment cleaning.
 - No of periodic training completed.
 - Total number of qualified persons are available for cleaning of equipment.
3. **REVIEW OF TRAINING AND QUALIFICATION OF CONCERNED EMPLOYEE ON SWAB SAMPLING:**
Details of training and qualification on swab sampling completed during this period
 - No of new employee completed the training on swab sampling.
 - No of periodic training completed.
 - Total number of qualified persons are available for swab sampling.
4. **REVIEW OF ALL DEVIATIONS RELATED TO CLEANING AND LINE CLEARANCE. TRENDING OF DEVIATIONS SHALL BE DONE AND ADDITIONAL CAPA ACTIONS SHALL BE DERIVED BASED ON TRENDS:** Review of all the deviation related to cleaning and line clearance.
 - Conclusion of deviation.
 - CAPA proposed based on deviation trending.
5. **REVIEW OF CHANGES RELATED TO CLEANING PROCEDURES AND CLEANING MATRIX:**
 - Review of changes related to cleaning procedure and cleaning matrix, following points to be considered but not limited to,
 - Is there change in worst case product?
 - Is there change in cleaning procedure?
 - Is there change in cleaning agent composition and/or concentration of cleaning agent?
 - Is there change in cleaning validation sampling plan of the equipment?
 - Is there change in equipment design/ surface area?
 - Is there new product is introduced and cleaning procedure is verified?
 - Is there change in cleaning validation approach?
 - Is there change in acceptance criteria calculation?
6. **INCIDENCES / OOS / COMPLAINT/ OUT OF EXPECTATION) REPORTED RELATED TO EQUIPMENT CLEANING.** Details of incidences / OOS considering personnel involved in cleaning, SOP adherence, other contributing factors for non-compliance. Necessary action to be recommended based on assessment shall be specified. Details of complaint received related to contamination for Products which is manufactured using the equipment. *Action and alert limit established based on cleaning runs executed during review period. If results are below limit of detection or quantification in that case alert limit shall be 80% of action limit i.e. MACO limit* Details of investigation performed after identifying 'Out of Expectation' results and recommended action based on assessment shall be specified. Details of analytical Incidence reported related to extraneous peaks observed during analysis of cleaning validation / verification sample along with recommended action based on assessment shall be specified. **REVIEW OF CLEANING VALIDATION AND VERIFICATIONS RUNS:** Details of Worst-case cleaning validation details based on RPN calculation

SOLVENT GROUP	WORST CASE CLEANING VALIDATION BASED ON RPN CALCULATION					
	Worst case product	Initial validation done on	Initial validation batch number	Periodic verification done in last year		Periodic verification due on
				Batch number	Completed on	

ANNUAL CLEANING REVIEW REPORT : YEAR -----

B. SUBSEQUENT WORST-CASE PRODUCT IF WORST CASE IS NOT VALIDATED:

SOLVENT GROUP	WORST CASE CLEANING VALIDATION BASED ON RPN CALCULATION					Periodic verification due on
	Worst case product	Initial validation done on	Initial validation batch number	Periodic verification done in last year		
				Batch number	Completed on	

Note: Only completed validations/ Verifications along with details of group shall be captured.

8. Review of open change control and CAPA related to cleaning process:

- Details of open change control and CAPA related to cleaning process shall be specified.

9. Summary and conclusion:

10. Recommendation (Including requirements of any additional documentation):

11. Additional CAPA:

Based on review of data and trends, additional CAPA actions if any shall be proposed, and details shall be recorded.

12. Team approval:

Production
Date :

Engineering
Date:

Quality Control
Date:

Quality Assurance
Date:

13. Review (Inclusive of follow up action, if any) **and approval:**

**ANNEXURE 1035-A-0072/A14
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ANNUAL CLEANING REVIEW REPORT : YEAR -----

Reviewed by: _____
Unit Head
Date:

Approved by: _____
Head Unit Quality Assurance
Date:

**ANNEXURE 1035-A-0072/A15
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ASSESSMENT FOR CLEANING REQUIREMENT FOR PRODUCT CONTAINING GENOTOXIC / NITROSAMINE COMPOUND

1. Product Name: _____

2. Procedure for Assessment of Genotoxic / Nitrosamine compound:

Sr.	Identification of Nitrosamine/ Genotoxic compound in Manufacturing Process	High risk	Low risk	No Risk
2.1	R & D, Tech Transfer, MSTG, Site QA, QC, Manufacturing team shall review the Nitrosamine assessment reports of the product. Based on review identify the Raw materials, key starting materials, intermediates etc which are nitrosamine compound (General Nitrosamine / NDSRI) or containing nitrosamine or NDSRI impurity.	Any of the RM, KSM, Intermediate compound itself is Nitrosamine. (List the compound)	Any of the RM, KSM, Intermediate compound may contain Nitrosamine as impurity (List the impurities)	None of the RM, KSM, Intermediate compound contain Nitrosamine as impurity
2.2	R & D, Tech Transfer, MSTG, Site QA, QC, Manufacturing team shall review the genotoxic assessment reports of the product. Based on review identify the Raw materials, key starting materials, intermediates etc which are genotoxic or containing genotoxic impurity.	Any of the RM, KSM, Intermediate compound itself is Genotoxic. (List the compound)	Any of the RM, KSM, Intermediate, API compound may contain Genotoxic impurity (List the impurities)	None of the RM, KSM, Intermediate compound contain Genotoxic impurity.

3. Action plan:

Sr.	Risk category	Action plan
3.1	High Risk	Review solubility data of genotoxic compound and existing/ proposed cleaning procedure for the adequacy of the cleaning procedure. Based on review suggest modification to cleaning procedure. Evaluate the acceptable carryover of the compounds which are genotoxic or based on the health based (PDE / ADI / TTC etc) methods. Determine the cleaning limits based on the health based acceptable carryover. Evaluate / develop the cleaning method for the above determined limits. Perform cleaning validation for three batches with above established limits. Periodically evaluate the cleaning procedure SOP and results.

**ANNEXURE 1035-A-0072/A15
(SPECIMEN COPY)**



ASSESSMENT FOR CLEANING REQUIREMENT FOR PRODUCT CONTAINING GENOTOXIC / NITROSAMINE COMPOUND

Sr.	Risk category	Action plan
3.2	Low Risk	Evaluate the requirement of the controls stated in the cleaning for the impurities based on the control strategy of those impurities in the compounds. With the below considerations in general but not limited to. -The knowledge of the level and source of impurity formation. -The knowledge of the impurity carryover in the process. -Track the fate and purge of impurities throughout the process based on available data. -Estimate the potential carryover based on the control limits of the impurities and evaluate the risk. I. If the concerned impurity is not a release test requirement, then the likelihood of formation of impurity is very low and no action recommended for the cleaning. II. If the concerned impurity is part of the release test (routine/skip/ periodic etc.) further evaluate as below i. Calculate possible carryover of impurity based on Maximum Carryover of API (MACO of API) as per formula: $A \text{ (Carryover of impurity in mg)} = \text{MACO of API (mg)} \times \text{Impurity limit in the product (\%)}$ ii. Calculate the carryover of impurity based on PDE/TTC criteria as per formula: $B \text{ (Carryover of impurity in mg)} = \frac{\text{TTC/ PDE (mg/ day)} \times \text{MBS (mg)}}{\text{TDD (mg/ day) (next)}}$ where, MBS: Minimum batch size (in the product group). TDD: Maximum therapeutic daily dose (in the product group). TTC: Threshold of Toxicological concern

ANNEXURE 1035-A-0072/A15
(SPECIMEN COPY)



ASSESSMENT FOR CLEANING REQUIREMENT FOR PRODUCT CONTAINING GENOTOXIC / NITROSAMINE COMPOUND

Sr.	Risk category	Action plan
		Validation approach: 1. If $A < B$, then it is considered that if product is cleaned at MACO level, impurity will also get cleaned and no further action is recommended. 2. If $A > B$, then perform three run cleaning validation.
3.3	No Risk	No action is recommended.

4. Evaluation:

Sr.	Compound / Product	Identified Risk High / Low / No risk	Action recommended	Rationale of the decision on the Risk identification
4.1				

5. Conclusion:

ANNEXURE 1035-A-0072/A15
(SPECIMEN COPY)



ASSESSMENT FOR CLEANING REQUIREMENT FOR PRODUCT CONTAINING GENOTOXIC / NITROSAMINE COMPOUND

Action	Responsibility	Name, Sign and Date
Compiled by:	R &D	
	Tech Transfer	
	MSTG	
	Manufacturing	
Reviewed by:	QA	
	Unit Head	
Approved by:	Head QA	

STUDY PROTOCOL		
No.: SP/PLANT CODE/DEHT	DIRTY EQUIPMENT HOLD TIME STUDY	Date of Issue
Version No.:		Effective Date
Supersedes:		Page 1 of 4

1. OBJECTIVE:

To establish and validate the dirty equipment hold time period of equipment.

2. SITE OF STUDY:

Unit and Department to be recorded in the study report.

3. STUDY TEAM:

Individual Name to be recorded in the study report section

Responsibilities:

Production: Cleaning of equipment, coordinate to compile and review of the study protocol and report in coordination with the team members.

Quality Control: Ensuring analyst is trained, instruments used are calibrated and within validity, ensuring validated method are of current version, coordinate to compile and review of the study protocol and report in coordination with the team members.

Quality Assurance: Preparation of study protocol and report in coordination with the team members, coordinate for sampling plan, collection of samples along with production, compilation of study report after completion of analysis.

4. STANDARD OPERATING PROCEDURE (SOP) TO BE FOLLOWED:

Sr. No.	Document Name	Document No.
i.	SOP for cleaning validation.	
ii.	SOP for swab sampling of Equipment's and Accessories.	
iii.	SOP for limit of Bioburden of equipment - Microbiological environment monitoring of Air, surfaces and personnel'.	
iv.	SOP for labelling.	
v.	Cleaning specification of rinse and swab.	

5. CONTROLS:

The staff involved in the study exercise shall be trained for the same.

6. DESCRIPTION OF EQUIPMENT:

Equipment Name, Equipment code, Equipment capacity and equipment from which group details to be recorded in report section.

7. PROCEDURE:

- 7.1 List down the products manufactured at the site and grouped based on solubility. Worst case product selection shall be done as per below

STUDY PROTOCOL		
No.: SP/PLANT CODE/DEHT	DIRTY EQUIPMENT HOLD TIME STUDY	Date of Issue
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- For API: Product selection shall be done as per worst-case scenario used for cleaning validation, which is based on the RPN calculation (Less solubility, least PDE and cleanability). The product with the highest RPN shall be chosen as the worst-case product for the DEHT study.
- For Intermediate / API (before active moiety formed): Product selection shall be done as per worst-case scenario used for cleaning validation, which is based on the RPN calculation (Less solubility and cleanability). The product with the highest RPN shall be chosen as the worst-case product for the DEHT study.

7.2 DEHT study shall be performed on worst case product from each solvent group when common cleaning procedures are used based on solubility. Priority shall be given for the product where the water base reaction equipment studied for DEHT.

7.3 When the worst product equipment is not available for study then subsequent worst product equipment to be considered for study.

7.4 Rationale for selection of equipment: As equipment are cleaning with common cleaning procedure based on solubility, study shall be done for at least one equipment from each group with three runs.

Group	Equipment	Runs
Group A	Crystalliser Reactor	Three runs
Group B	Closed type equipment – E.g., Centrifuge, Blender, Dryers, RCVD, VTD, ANFD, RVPD etc.	Three runs
Group C	Portable equipment & equipment which shall be dismantled completely- E.g. Multi mill, Sifters, Jet mill, Cad mill, Co mill etc	Three runs

7.5 After completion of processing, store the equipment under prevailing condition for at least 3 days/ 72 hours (Additional study with extrapolated time shall be evaluated by site for higher DEHT). The equipment which is identified shall not be cleaned by Type B till holding period completed

7.6 Record the previous product name, product from which solubility group. stage of processing, batch number of previous product and date of study in report section.

7.7 After emptying out of the equipment update the status board of equipment as 'To be cleaned' and record the date and time of completion of manufacturing in report section.

7.8 After completion of holding period perform the Type B cleaning and take cleaning validation samples (rinse and swab) in same way as per regular cleaning validation study. Record time till when equipment was hold dirty.

7.9 Sampling plan, sample collection, analysis and result reporting shall follow same as that of the regular cleaning validation protocol.

7.10 Applicable bioload sample shall be collected prior to chemical swab sample collection and location of micro swab is different than the chemical swab sampling.

7.11 If there is holiday as per planned sampling days next consecutive working day shall be considered for sampling.

STUDY PROTOCOL		
No.: SP/PLANT CODE/DEHT	DIRTY EQUIPMENT HOLD TIME STUDY	Date of Issue
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7.12 A computerized pre-printed document shall be prepared for Dirty equipment hold time report according to SOP annexure 1035-A-0072/A17.

7.13 Interim report shall be prepared after 1st and 2nd run of study.

7.14 The Dirty equipment hold time study shall be performed on three batches / runs to conclude the study.

8. ACCEPTANCE CRITERIA:

8.1 After holding the equipment for 72 Hours, the results of the cleaning validation study should meet the acceptance criteria as established in the cleaning validation protocol.

9. NON-COMPLIANCE:

After holding the equipment for 72 Hours, if the results of the cleaning validation do not meet the acceptance criteria, then reduced hold time shall be attempted by repeating the study. In such cases, respective product BMR of particular solvent group shall mention the specific hold time

10. FREQUENCY:

10.1 One time study with three runs of worst-case product from each solubility group, considering at least one equipment from each equipment group.

10.2 If major change done in cleaning SOP.

11. SUMMARY AND CONCLUSION OF STUDY:

Summarize the findings of study to draw an inference of the result in report section and draw the conclusion of study.

12. RECOMMENDATIONS:

Record the recommendation based on the interpretation of the results of the Study Report section.

13. ATTACHMENTS:

Annexure-I: Technical information sheet of rinse and swab (chemical and micro) samples.

Annexure -II: photocopy of ECR

14. TEAM SIGNING:

The personnel who have performed the study, supervised the study, completed the records, performed the testing shall approve the study report section.

ANNEXURE 1035-A-0072/A16
(SPECIMEN COPY)



STUDY PROTOCOL		
No.: SP/PLANT CODE/DEHT	DIRTY EQUIPMENT HOLD TIME STUDY	Date of Issue
Version No.:		Effective Date
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15. REVIEW AND COMMENTS:

Quality Assurance shall review the study report and shall include the comments in report section.

16. APPROVED BY:

Head Unit Quality Assurance shall finally approve the study report section.

17. PROTOCOL APPROVAL:

Prepared By (Study Team) Sign and Date			Reviewed By Sign and Date	Approved By Sign and Date
Quality Assurance	Production	Quality Control	Quality Assurance	Head Unit Quality Assurance

**ANNEXURE 1035-A-0072/A17
(SPECIMEN COPY)**



STUDY REPORT		
No.: SR/PLANT CODE/DEHT	DIRTY EQUIPMENT HOLD TIME STUDY	Page 1 of 3

1. SITE OF STUDY:

Unit : Unit no. to be recorded

Department : Department number to be recorded.

2. STUDY TEAM:

Department	Name of person
Quality Assurance	Individual Name to be recorded
Production	Individual Name to be recorded
Quality Control / Micro	Individual Name to be recorded

3. STANDARD OPERATING PROCEDURE (SOP) TO BE FOLLOWED:

Sr. No.	Name of SOP	Document Actual Followed
i.	SOP for cleaning validation	SOP no. to be recorded
ii.	SOP for swab sampling of Equipment's and Accessories.	SOP no. to be recorded
iii.	SOP for limit of Bioburden of equipment - Microbiological environment monitoring of Air, surfaces and personnel'.	SOP no. to be recorded
iv.	SOP for labelling.	SOP no. to be recorded
v.	Cleaning specification for analysis rinse and swab.	Specification no. to be recorded

4. CONTROLS:

Training details of staff involved in study: Class ID :

5. DESCRIPTION OF EQUIPMENT:

Parameters	DEHT Runs		
	1 st Run	2 nd Run	3 rd Run
Equipment Name			
Equipment Code			
Equipment Capacity			
Equipment from which group	Group A / Group B / Group C	Group A / Group B / Group C	Group A / Group B / Group C

6. RESULTS AND OBSERVATION:

**ANNEXURE 1035-A-0072/A17
(SPECIMEN COPY)**

Cipla
Location

STUDY REPORT		
No.: SR/PLANT CODE/DEHT	DIRTY EQUIPMENT HOLD TIME STUDY	Page 2 of 3

Parameters	DEHT Runs		
	1 st Run	2 nd Run	3 rd Run
Previous Product name			
Batch Number of Previous Product			
Product from which solubility group			
Stage of Processing			
Date of Study			
Date and time of completion of manufacturing			
Date and time of commencement of cleaning			
On 4 th day clean the equipment as per SOP No. and version No.			
AR number for rinse and swab (chemical and micro) samples	Chemical :		
	Micro:		

6.1 The results shall be compiled in the Cleaning validation report. Reference CV report number_____.

7. NON-COMPLIANCE:

Sr. No.	Document No.	Description Non-compliance	Outcome of non-compliance	CAPA

8. SUMMARY AND CONCLUSION (INFERENCE):

Summarise the results and observation observed during study, conclusion shall be recorded.

9. RECOMMENDATION:

Based on overall review and results of study the dirty equipment hold time period recommendation shall be recorded and updated in the respective equipment cleaning record

10. ANNEXURES:

Annexure-I	Technical information sheet along with sampling plan of rinse and swab (chemical and micro) samples and COA of study.
Annexure-II	Copy of Equipment cleaning record (ECR) of type B cleaning.

ANNEXURE 1035-A-0072/A17
(SPECIMEN COPY)



STUDY REPORT		
No.: SR/PLANT CODE/DEHT	DIRTY EQUIPMENT HOLD TIME STUDY	Page 3 of 3

11. STUDY TEAM SIGNING:

Department	Name of person	Sign	Date
Quality Assurance			
Production			
Quality control (Chemical)			
Quality control (Micro)			

12. REVIEW AND COMMENTS:

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Quality Assurance sign and date

13. APPROVED BY :

Head Unit Quality Assurance sign and Date

STUDY PROTOCOL		
No.: SP/PLANT CODE/CEHT	CLEANED EQUIPMENT HOLD TIME STUDY	Date of Issue
Version No.:		Effective Date
Supersedes:		Page 1 of 4

1. OBJECTIVE:

To establish and validate the Cleaned Hold time period of equipment by ensuring that the cleaned equipment when stored under prescribed conditions, does not exceed the microbial contamination level of freshly cleaned equipment / equipment parts even after holding.

2. SITE OF STUDY:

Unit and Department to be recorded in the study report.

3. STUDY TEAM:

(Individual Name to be recorded in the study report)

Responsibilities:

Production: Cleaning of equipment, coordinate to compile and review of the study protocol and report in coordination with the team members.

Quality Control: Ensuring analyst is trained, instruments used are calibrated and within validity, collection of bioload sample, coordinate to compile and review of the study protocol and report in coordination with the team members.

Quality Assurance: Preparation of study protocol and report in coordination with the team members, coordinate for sampling plan, compilation of study report after completion of analysis and review of study report.

4. STANDARD OPERATING PROCEDURE (SOP) TO BE FOLLOWED:

Sr. No.	Document Name	Document No.
i.	Microbiological monitoring of surfaces in production area.	To be recorded in report section.
ii.	SOP for swab sampling of Equipment's and Accessories.	
iii.	SOP for labelling	

5. CONTROLS:

The staff involved in the study exercise shall be trained for the same.

6. DESCRIPTION OF EQUIPMENT:

Equipment name, Equipment code, Equipment capacity and equipment from which groups details to be recorded in report section.

7. PROCEDURE:

- 7.1 List down the products manufactured in site. Collect the solubility data of listed product and grouped the product based on solubility.

STUDY PROTOCOL		
No.: SP/PLANT CODE/CEHT	CLEANED EQUIPMENT HOLD TIME STUDY	Date of Issue
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- 7.2 The study shall be performed only for equipment used in controlled and classified or controlled but not classified (CNC) clean areas. Typically for manufacturing of non-sterile API environment is non controlled till crystallisation step hence no microbial limits applied to such equipment
- 7.3 **Rationale for selection of product:** CEHT study shall be done for any one product from each solvent group when common cleaning procedures are used based on solubility. Priority shall be given for the product where cleaning solvent is water.
- 7.4 **Rationale for selection of equipment:** As equipment are cleaning with common cleaning procedure and exposed to same environment, study shall be done for at least one equipment from each group with three runs.

Group	Equipment	Runs
Group A	Crystalliser Reactor	Three runs
Group B	Closed type equipment – E.g., Centrifuge, Blender, Dryers, RCVD, VTD, ANFD, RVPD etc.	Three runs
Group C	Portable equipment & equipment which shall be dismantled completely- E.g. Multi mill, Sifters, Jet mill, Cad mill, Co mill etc	Three runs

- 7.5 Target for CEHT study shall be maximum 7 days.
- 7.6 Clean and dry the selected equipment according to the Type B cleaning procedure (product changeover cleaning).
- 7.7 The equipment designated for the study shall not be used for manufacturing until the study is completed.
- 7.8 Record the previous product, batch number of previous product, previous product from which solubility group, stage of processing in equipment, cleaning of equipment done as per cleaning procedure with version no. and equipment cleaning date and time in report section.
- 7.9 Visually inspect the equipment parts for cleanliness and mention the observation in report section.
- 7.10 A sampling plan for microbial swab shall be prepared and approved separately based on equipment type and swab location after each interval prior to initiation of study. The approved sampling plan shall be attached with study report.
- 7.11 Microbiologist / QA shall collect swab (bioload) samples from identified product contact surface after cleaning and drying (Day 1),
- 7.12 Keep the equipment in closed conditions as per procedure.
- 7.13 Periodically bioload sample shall be collected on Day 4, Day 6 and Day 7 from identified product contact part.
- 7.14 Take swab samples from adjacent locations identified for swabbing each time or take the swab from different parts for swabbing each time.

STUDY PROTOCOL		
No.: SP/PLANT CODE/CEHT	CLEANED EQUIPMENT HOLD TIME STUDY	Date of Issue
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- 7.15 If there is holiday as per planned sampling days next consecutive working day shall be considered for sampling.
- 7.16 Send swab samples to the microbial lab along with the TI sheet.
- 7.17 Based on the results obtained, record the conclusion on findings on the maximum number of days the equipment shall be kept after cleaning before re-cleaning is required and further course of action in the recommendation section.
- 7.18 A computerized pre-printed document shall be prepared for Cleaned equipment hold time report according to SOP annexure 1035-A-0072/A19.
- 7.19 Interim report shall be prepared after 1st and 2nd run of study.
- 7.20 The Cleaned equipment hold time study shall be performed on three batches / runs to conclude the study.

8. ACCEPTANCE CRITERIA:

- 8.1 Microbiological results shall be comply with limit mentioned below.

Total count limit: 500 cfu / swab

Mold count limit: 50 cfu / swab

- 8.2 Equipment shall be visually clean after cleaning.

9. NON-COMPLIANCE:

Any deviation / changes if observed shall be recorded in the study report section and shall be investigated.

10. FREQUENCY:

- 10.1 Equipment which are available in controlled and classified or controlled but not classified (CNC) clean areas: One time study with three runs on any product of each solubility and each equipment group specified in procedure.
- 10.2 If major change done in cleaning SOP.

11. SUMMARY AND CONCLUSION OF STUDY:

Summarize the findings of study to draw an inference of the result in report section and draw the conclusion of study.

12. RECOMMENDATIONS:

Record the recommendation based on the interpretation of the results of the Study Report section.

13. ATTACHMENTS:

**ANNEXURE 1035-A-0072/A18
(SPECIMEN COPY)**



STUDY PROTOCOL		
No.: SP/PLANT CODE/CEHT	CLEANED EQUIPMENT HOLD TIME STUDY	Date of Issue
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Annexure-I Technical information sheet of micro samples along with sampling plan.

14. TEAM SIGNING:

The personnel who have performed the study, supervised the study, completed the records, performed the testing shall approve the study report section.

15. REVIEW AND COMMENTS:

Quality Assurance shall review the study report and shall include the comments in report section.

16. APPROVED BY:

Head Unit Quality Assurance shall finally approve the study report section.

17. PROTOCOL APPROVAL:

Prepared By (Study Team) Sign and Date			Reviewed By Sign and Date	Approved By Sign and Date
Quality Assurance	Production	Quality Control	Quality Assurance	Head Unit Quality Assurance

**ANNEXURE 1035-A-0072/A19
(SPECIMEN COPY)**



STUDY REPORT		
No.: SR/PLANT CODE/CEHT	CLEANED EQUIPMENT HOLD TIME STUDY	Page 1 of 3

1. SITE OF STUDY:

Unit : Unit no. to be recorded

Department : Department to be recorded

2. STUDY TEAM:

Department	Name of person
Quality Assurance	Individual Name to be recorded
Production	Individual Name to be recorded
Quality Control / Micro	Individual Name to be recorded

3. STANDARD OPERATING PROCEDURE (SOP) TO BE FOLLOWED:

Sr. No.	Name of SOP	Document Actual Followed
i.	Microbiological monitoring of surfaces in production area.	SOP no. to be recorded
ii.	SOP for swab sampling of Equipment's and Accessories.	SOP no. to be recorded
iii.	SOP for labelling	SOP no. to be recorded

4. CONTROLS:

Training details of staff involved in study: Class ID :

5. DESCRIPTION OF EQUIPMENT:

Parameter	CEHT Run		
	1 st Run	2 nd Run	3 rd Run
Equipment Name			
Equipment Code			
Equipment Capacity			
Equipment from which Group	Group A / Group B / Group C	Group A / Group B / Group C	Group A / Group B / Group C

6. RESULTS AND OBSERVATION:

Parameters	CEHT Run		
	1 st Run	2 nd Run	3 rd Run
Previous Product			
Batch Number of Previous Product			
Product from which solubility group			
Stage of Processing			

STUDY REPORT		
No.: SR/PLANT CODE/CEHT	CLEANED EQUIPMENT HOLD TIME STUDY	Page 2 of 3

Cleaning of equipment done as per cleaning procedure with version no.			
Equipment cleaning date and time			
After cleaning equipment visual inspection comments			

6.1 Bio load results:

CEHT Runs	Equipment parts swabbed	Interval	Date of micro swab sampling	AR number of micro swab	Total Count	Mold Count
					NMT: 500 cfu / swab	NMT: 50 cfu / swab
1st Run		Initial (after cleaning)				
		On 4 th day				
		On 6 th day				
		On 7 th day				
2nd Run		Initial (after cleaning)				
		On 4 th day				
		On 6 th day				
		On 7 th day				
3rd Run		Initial (after cleaning)				
		On 4 th day				
		On 6 th day				
		On 7 th day				

STUDY REPORT		
No.: SR/PLANT CODE/CEHT	CLEANED EQUIPMENT HOLD TIME STUDY	Page 3 of 3

7. NON-COMPLIANCE:

Sr. No.	Document No.	Description Non-compliance	Outcome of non-compliance	CAPA

8. SUMMARY AND CONCLUSION (INFERENCE):

Based on overall review and results of study recommendation shall be recorded.

9. RECOMMENDATION:

Based on overall review and results of study recommendation shall be recorded.

10. ANNEXURE:

Annexure-I : Technical information sheet of micro samples along with sampling plan.

11. STUDY TEAM SIGNING:

Department	Name of person	Sign	Date
Quality Assurance			
Production			
Quality control (Micro)			

12. REVIEW AND COMMENTS:

.....
.....

Quality Assurance sign and date

13. APPROVED BY:

Head Unit Quality Assurance sign and Date

STUDY PROTOCOL		
No.: SP/PLANT CODE/CAMPCL	CAMPAIGN CLEANING VALIDATION	Date of Issue
Version No.:		Effective Date
Supersedes:		Page 1 of 6

1. OBJECTIVE:

To establish and validate the frequency on number of batches (campaign length) where product to product changeover cleaning (Type B cleaning procedure) to be executed to prevent any potential risk of contamination that could impact on product quality and / or patient safety.

2. SCOPE:

Applicable to all equipment cleaning wherein the equipment is used for routine production of API's and / or its intermediates.

3. SITE OF STUDY:

Unit and Department to be recorded in the study report.

4. CAMPAIGN CLEANING VALIDATION STUDY TEAM:

Individual Name to be recorded in the study report.

Responsibilities:

Production: Cleaning of equipment, coordinate to compile and review of the study protocol and report in coordination with the team members.

Quality Control: Ensuring analyst is trained, instruments used are calibrated and within validity, coordinate to compile and review of the study protocol and report in coordination with the team members.

Quality Assurance: Preparation of study protocol and report in coordination with the team members, coordinate for sampling plan, compilation of study report after completion of analysis and review of study report.

5. CONTROLS:

The staff involved in the study exercise shall be trained for the same.

6. STANDARD OPERATING PROCEDURE AND SPECIFICATION:

SOP for cleaning of equipment, SOP for swab and rinse sampling of equipment and accessories, analytical specification of cleaning with version no. and SOP for microbiological swab sampling from equipment surface followed during study to be recorded in study report.

7. DESCRIPTION:

Equipment name, Equipment code, equipment used for Product name, Product from which solubility group, Date of campaign initiated, Date of Campaign completed, Campaign period, Number of batches manufactured in campaign, Manufactured batch numbers, Cleaning of equipment done on, Equipment cleaning done as per SOP with version no. and equipment cleaning solvent to be recorded in report.

STUDY PROTOCOL		
No.: SP/PLANT CODE/CAMPCL	CAMPAIGN CLEANING VALIDATION	Date of Issue
Version No.:		Effective Date
Supersedes:		Page 2 of 6

8. PROCEDURE:

- 8.1 List down the products manufactured at the site and grouped the products based of solubility. The selection of worst-case product for campaign cleaning validation study shall be done as follows.
- API selection: Campaign length cleaning validation shall be performed on the worst-case product of each solubility group. The product selection shall be done as per worst-case scenario used for cleaning validation, which is based on the RPN calculation (Less solubility, least PDE and cleanability). The product with the highest RPN shall be chosen as the worst-case product for the campaign cleaning validation study.
 - For Intermediate / API (before active moiety formed) selection: Campaign length cleaning validation shall be performed on the worst-case product of each solubility group. The product selection shall be done as per worst-case scenario used for cleaning validation, which is based on the RPN calculation (Less solubility and cleanability). The product with the highest RPN shall be chosen as the worst-case product for the campaign cleaning validation study.
 - If any API / Intermediate of group having highest campaign length than selected worst-case product then same product campaign length also be validated.
 - If the sufficient number of batches of identified worst case product are not in manufacturing plan, then campaign-based cleaning validation shall be performed on subsequent worst-case product
 - Campaign cleaning validation shall be performed considering total equipment chain used for the worst-case product. If any type of equipment is not part of the worst-case product equipment chain, then such equipment campaign length validation shall be performed on subsequent worst-case product.
- 8.2 Identified worst case or subsequent worst-case product shall be validated once campaign manufacturing plan is available.
- 8.3 Ensure batch to batch cleaning is done post every batch in campaign.
- 8.4 After completion of campaign, cleaning of equipment shall be done by using product-to-product changeover (B type) cleaning procedure.
- 8.5 *If annual preventive maintenance schedule date falls during the campaign length validation run, then annual preventive maintenance shall be performed prior to start of campaign length run. Such preponement of annual preventive maintenance shall be documented in campaign length validation protocol and on preventive maintenance schedule. Any preponement of annual preventive maintenance initiation date due to campaign length study shall be informed to the Engineering department by production department.*
- 8.6 Quality Assurance shall issue the required ECR (Equipment cleaning record) as per request from production, check the availability of approved cleaning validation sampling plan.
- 8.7 After issuance of ECR, production shall initiate the cleaning of equipment.

STUDY PROTOCOL		
No.: SP/PLANT CODE/CAMPCL	CAMPAIGN CLEANING VALIDATION	Date of Issue
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- 8.8 Quality Assurance shall collect 100 ml of cleaning solvent as blank for evaluation of interference of cleaning solvent.
- 8.9 Before rinse/ swab sample collection, TI sheet shall be raised in SAP by quality assurance as an intimation to QC for campaign cleaning sample and preparation of swab diluent to store the swab stick.
- 8.10 On rinse sample TI sheet, include details of Product name, batch number, equipment name, MOC of equipment, analysis code, rinse sample number, rinse quantity, blank fresh solvent name, quantity, limit / acceptance criteria for rinse and samples for campaign cleaning validation study.
- 8.11 On swab sample TI sheet include details of Product name, batch number, equipment name, MOC of equipment, analysis code, swab sample number, information related to MOC of equipment parts, acceptance criteria for swab and samples for campaign cleaning validation study.
- 8.12 According to the validation sampling plan, Quality Assurance collect the rinse and swab samples as per SOP 1035-A-0082.
- 8.13 During rinse sampling, collect the entire volume of rinse content in suitable tank / vessel and collect the portion of rinse sample (about 100 ml) into sample bottle. The rinse samples collected shall represent the entire rinse volume.
- 8.14 After collection of rinse sample, check clarity of rinse sample by QA and recorded the observation on sampling plan. Further labelled each rinse sample and sent immediately to QC for analysis of previous product content / cleaning agent.
- 8.15 As per quantity of swabs mention on TI sheet, labelled the swab sample collection test tube / bottle and handed over to Quality control for issuance of diluent (10 ml for each swab).
- 8.16 After the issuance of the diluent, the sample collection test tubes / bottles shall be taken for swab collection according to the sampling plan.
- 8.17 Equipment / equipment part shall be dried according to the instruction specified in ECR after the final rinse. The dried equipment and its part shall be visually inspected for cleanliness by production, followed by QA. The observation of visual inspection shall be recorded on sampling plan.
- 8.18 After a satisfactory visual inspection, swab samples shall be collected by Quality Assurance as per SOP 1035-A-0082. The collected swab samples along with the sampling plan shall be sent to QC for analysis of previous product content.
- 8.19 In case of water is used for cleaning of equipment for final rinse, Swab of product contact part to be sent to Quality control for Complete microbiological analysis.
- 8.20 Quality Control shall initiate the analysis as per cleaning specification of product.
- 8.21 If rinse or swab result observed more than limit, out of specification shall be initiated and handled as per SOP 1035-L-0143. Corrective and preventive action shall be taken based on investigation.

STUDY PROTOCOL		
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- 8.22 If extraneous peak observed during analysis, incidence shall be initiated and handled as per SOP 1035-L-0146. Corrective and preventive action shall be taken based on investigation.
- 8.23 After completion of analysis, Quality Control shall provide the result of samples analysed along with limit of detection value and limit of quantification (For rinse as well as swab technique) of analytical method used to analyse the cleaning validation samples to Quality Assurance.
- 8.24 Quality Assurance shall verify the compliance of all the results (like rinse and swab results observed within the rinse and swab limit, absence of cleaning agent etc.), if non-compliance observed investigation shall be opened. Record the results detail in report section and compared with cleaning validation results.
- 8.25 Check total number of batches manufactured in the campaign and collect impurity profile data (RRT wise) of each batch and compared. This comparison report attached with report.
- 8.26 The last batch of campaign shall be analysed for microbiological examination and details shall be recorded in report section.
- 8.27 Based on the results, campaign length (Number of batches) and duration of the campaign shall be decided. Same shall be summarized in the report. Based on summary and result, conclusion and recommendation of campaign cleaning validation shall be recorded in the report.
- 8.28 Campaign cleaning validation report shall be compiled by cross functional team.
- 8.29 Campaign cleaning validation report against the protocol shall be reviewed by second competent Quality Assurance person and final approval from Head Unit Quality assurance.
- 8.30 *Three run campaign cleaning validation to be performed on each solubility group worst-case API and intermediate.*
- 8.31 *One out of three campaign runs shall be extended to establish uncleaned equipment hold time.*
- 8.32 A computerized pre-printed document shall be prepared for campaign cleaning validation study report according to SOP annexure 1035-A-0072/A21.
- 8.33 Interim report shall be prepared after 1st and 2nd run of study.

9. ACCEPTANCE CRITERIA

- 9.1 Equipment shall be visually clean after cleaning.
- 9.2 Final rinse shall be visually clear.
- 9.3 **Rinse and swab limit for API:** The residue of product content in rinse and swab sample shall be less than rinse and swab limit based on the MACO calculated by Permitted Daily Exposure (PDE). If limits of residue of previous product obtained from the final rinse and swabs are more than 10 ppm, then acceptance criteria shall be 10 ppm and if less than 10 ppm then that value calculated based on PDE shall be considered as acceptable limit.

STUDY PROTOCOL		
No.: SP/PLANT CODE/CAMPCL	CAMPAIGN CLEANING VALIDATION	Date of Issue
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- 9.4 **Rinse and swab limit for Intermediates:** The residue of product content in rinse and swab sample shall be less than 10 ppm.
- 9.5 **Microbiological limit:** The equipment shall be meet the microbiological acceptance criteria specified in SOP 1035-L-0086 (In case of equipment used in controlled and classified or controlled but not classified (CNC) clean areas).
- 9.6 Impurity profile trend of campaign batches shall be comparable, there is no specific impurity increasing trend.
- 9.7 Microbiological examination of last batch manufactured in campaign shall be within the acceptance criteria specified in SOP / specification.

10. NON-COMPLIANCE:

Any deviation / changes if observed shall be recorded in the study report section and shall be investigated. Details of out of expectation result to be recorded (Results to be compared with the previously validated results and if any significant difference in results is noticed for any of the sampling location during comparison, same shall be thoroughly reviewed (investigated) and outcome shall be addressed in the report.

11. FREQUENCY:

- 11.1 One time study with three runs on selected worst-case product from each solubility group.
- 11.2 If major change done in A type or B type cleaning procedure.
- 11.3 After change in worst-case product.

12. SUMMARY AND CONCLUSION OF STUDY:

Summarize the findings of the campaign cleaning validation study and draw an inference and record in report.

13. RECOMMENDATIONS:

Record the recommendation based on the interpretation of the results of the campaign cleaning validation in report section.

14. ATTACHMENTS:

Annexure -I: Technical information sheet along with sampling plan of rinse and swab sampling.

Annexure -II: Comparison of impurity profile (RRT wise) of all the batches manufactured in the campaign

Annexure -III: Microbiological monitoring result of last batch manufactured in the campaign.

Annexure -IV: Copy of Equipment cleaning record (ECR) of type B cleaning.

STUDY PROTOCOL		
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15. TEAM SIGNING:

The personnel who have performed the study, supervised the study, completed the records, performed the testing shall approve the study report.

16. REVIEW AND COMMENTS:

Quality Assurance shall review the study report and shall include the comments in study report.

17. APPROVED BY:

Head Unit Quality Assurance shall finally approve the study report.

18. PROTOCOL APPROVAL:

Prepared By (Study Team) Sign and Date			Reviewed By Sign and Date	Approved By Sign and Date
Quality Assurance	Production	Quality Control	Quality Assurance	Head Unit Quality Assurance

**ANNEXURE 1035-A-0072/A21
(SPECIMEN COPY)**

Cipla
Location

STUDY REPORT		
No.: SR/PLANT CODE/CAMPCL	CAMPAIGN CLEANING VALIDATION	Page 1 of 5

1. SITE OF STUDY:

Unit : Unit No. to be recorded
Department : Department name to be recorded.

2. CAMPAIGN CLEANING VALIDATION TEAM:

Department	Name of person
Quality Assurance	Individual Name to be recorded
Production	Individual Name to be recorded
Quality Control	Individual Name to be recorded

3. CONTROLS:

Training details of staff involved in study on SOP 1035-A-0072: Class ID :

4. STANDARD OPERATING PROCEDURE AND SPECIFICATION:

Sr. No.	SOP / Specification	Document no. actual followed
i.	SOP for cleaning of equipment	SOP no. to be recorded
ii.	SOP for swab and rinse sampling of equipment and accessories	SOP No. to be recorded
iii.	Specification used for cleaning sample analysis.	Specification No. to be recorded
iv.	SOP for microbiological swab sampling from equipment surface	SOP to be recorded

5. DESCRIPTION:

Parameter	Campaign cleaning run		
	1 st Run	2 nd Run	3 rd Run
Equipment Name			
Equipment Code			
Equipment used for Product name			
Product from which solubility group			
Date of campaign initiated			
Date of Campaign completed.			
Campaign period			
Number of batches manufactured in campaign			
Manufactured batch numbers			
Cleaning of equipment done on			
Equipment cleaning done as per SOP with version no.			
Solvent used for cleaning			

STUDY REPORT		
No.: SR/PLANT CODE/CAMPCL	CAMPAIGN CLEANING VALIDATION	Page 2 of 5

6. RESULTS AND OBSERVATION:

6.1 Visual inspection of last rinse:

Campaign run	Test to be performed	Limit	Results
1 st run	Clarity for last rinse	Shall be clear	
2 nd run	Clarity for last rinse	Shall be clear	
3 rd run	Clarity for last rinse	Shall be clear	

6.2 Visual inspection of equipment parts:

Campaign run	Swab parts name	Limit	Results
1 st run		Shall be clean.	
		Shall be clean.	
		Shall be clean.	
2 nd run		Shall be clean.	
		Shall be clean.	
		Shall be clean.	
3 rd run		Shall be clean.	
		Shall be clean.	
		Shall be clean.	

6.3 Possible contamination by rinse:

Campaign run	AR number	Equipment part name	Last rinse value Rinse Limit: NMT _____ ppm
1 st run			
2 nd run			
3 rd run			

6.4 Possible contamination by swab:

Campaign run	AR number	Equipment code number	Swab value Swab Limit: NMT _____ ppm or µg/swab						
			S1	S2	S3	S4	S5	S6	S7
1 st run									
2 nd run									
3 rd run									

6.5 Results of microbial determination:

Campaign run	AR number	Name of equipment parts	Microbial Results	
			Total count	Fungal count
1 st run				
2 nd run				
3 rd run				

STUDY REPORT		
No.: SR/PLANT CODE/CAMPCL	CAMPAIGN CLEANING VALIDATION	Page 3 of 5

6.6 Results of cleaning agent determination:

Cleaning agent name: _____ Concentration: _____					
Campaign run	AR number	Name of rinse part	Rinse number	Quantity of rinse (Lit.)	Content of cleaning agent (Limit: _____)
1 st run					
2 nd run					
3 rd run					

6.7 Results of impurity profile (RRT wise):

Campaign run: 1st run

AR number:

Impurity	Batch number					
	Batch 1	Batch 2	Batch 3	Batch 4	Batch 5	Batch 6
Known impurity						
Known impurity						
Unknown impurity at RRT						
Unknown impurity at RRT						
Unknown impurity at RRT						

Inference: The impurity profile trend of all the batches manufactured in campaign found comparable

Campaign run: 2nd run

AR number:

Impurity	Batch number					
	Batch 1	Batch 2	Batch 3	Batch 4	Batch 5	Batch 6
Known impurity						
Known impurity						
Unknown impurity at RRT						
Unknown impurity at RRT						
Unknown impurity at RRT						

STUDY REPORT		
No.: SR/PLANT CODE/CAMPCL	CAMPAIGN CLEANING VALIDATION	Page 4 of 5

Inference: The impurity profile trend of all the batches manufactured in campaign found comparable

Campaign run: 3rd run

AR number :

Impurity	Batch number					
	Batch 1	Batch 2	Batch 3	Batch 4	Batch 5	Batch 6
Known impurity						
Known impurity						
Unknown impurity at RRT						
Unknown impurity at RRT						
Unknown impurity at RRT						

Inference: The impurity profile trend of all the batches manufactured in campaign found comparable

6.8 Microbial proliferation of last batch manufacture in campaign:

Campaign run	Last batch manufactured in campaign	AR number	Total aerobic microbial count	Total combine yeasts and molds counts
			Limit:	Limit:
1 st run				
2 nd run				
3 rd run				

7. NON-COMPLIANCE

Sr. No.	Document No.	Description Non-compliance	Outcome of non-compliance	CAPA

8. SUMMARY AND CONCLUSION (INFERENCE):

Summarise the results and observation observed during study, conclusion shall be recorded.

9. RECOMMENDATION:

Based on overall review and results of study recommendation shall be recorded.

ANNEXURE 1035-A-0072/A21
(SPECIMEN COPY)



STUDY REPORT		
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10. ANNEXURES:

Annexure -I: Technical information sheet along with sampling plan of rinse and swab sampling.

Annexure -II: Comparison of impurity profile (RRT wise) of all the batches manufactured in the campaign

Annexure -III: Microbiological monitoring result of last batch manufactured in the campaign.

Annexure -IV: Copy of Equipment cleaning record (ECR) of type B cleaning.

11. STUDY TEAM SIGNING:

Department	Name of person	Sign	Date
Quality Assurance			
Production			
Quality control (Micro)			

12. REVIEW AND COMMENTS:

.....
.....

Quality Assurance sign and date

13. APPROVED BY:

Head Unit Quality Assurance sign and Date

STUDY PROTOCOL		
No.: SP/1030/CLSOLTRA	VERIFICATION OF CLEANING SOLVENT TRACES	Date of Issue
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1. BACKGROUND:

After completion of batch / product manufacturing, change over cleaning is performed as part of respective product cleaning SOP to ensure previous product traces to permissible level into the next product. The cleaning solvent absence is ensured by the traces of final solvent detection and shall be reviewed as one time study.

2. OBJECTIVE:

This protocol is written, to demonstrates the effective removal of cleaning solvent used for product change over cleaning.

3. SCOPE:

Applicable to evaluate the absence of cleaning solvent used for product change over cleaning.

4. SITE OF STUDY:

Unit and Department to be recorded in the study report.

5. STUDY TEAM:

Individual Name to be recorded in the study report.

RESPONSIBILITIES:

Production: Cleaning of equipment, coordinate to compile and review of the study protocol and report in coordination with the team members.

Quality Control: Ensuring analyst is trained, instruments used are calibrated and within validity, coordinate to compile and review of the study protocol and report in coordination with the team members.

Quality Assurance: Preparation of study protocol and report in coordination with the team members, coordinate for sampling plan, compilation of study report after completion of analysis and review of study report.

6. REASON FOR STUDY:

To provide evidence of removal of cleaning solvent used for product change over cleaning.

7. CONTROLS:

Record the training details of staff involved in study exercise on SOP 1035-A-0072.

8. DESCRIPTION:

Equipment Name, Equipment code, Equipment capacity, equipment from which group, equipment used for product name, Equipment used for product batch no., date of cleaning of equipment, cleaning of equipment done as per cleaning SOP with version no and solvent used for cleaning to be recorded in report.

STUDY PROTOCOL		
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9. PROCEDURE:

- 9.1 List down the products manufactured in site. Collect the solubility data of listed product and grouped the product based on solubility.
- 9.2 Study shall be done for any product within the solvent group. Study shall be done for each solvent with three runs.
- 9.3 This study is not applicable for surfactant-based cleaning agents and Acids, Bases and when only water is used as solvent.
- 9.4 **Rationale for selection of equipment:** As equipment are cleaning with common cleaning procedure and exposed to same environment, study shall be done for at least one equipment from each group with three runs.

Group	Equipment	Runs
Group A	Crystalliser Reactor	Three runs
Group B	Closed type equipment – E.g., Centrifuge, Blender, Dryers, RCVD, VTD, ANFD, RVPD etc.	Three runs
Group C	Portable equipment & equipment which shall be dismantled completely- E.g. Multi mill, Sifters, Jet mill, Cad mill, Co mill etc	Three runs

- 9.5 Clean and dry the selected equipment according to the Type B cleaning procedure (product changeover cleaning).
- 9.6 Rinse the equipment with 3 to 5 litre Purified water depending upon the equipment capacity.
- 9.7 Collect the rinse in amber coloured glass bottle, label it and send to Quality Control to determine the content of the solvent in the rinse which shall be analysed as per General Analytical Method for Residual Solvent (Reference document no. 1035-GAM-054-INH).
- 9.8 After collection of water rinse sample, give the rinse of 3 to 5 Lit respective solvent used for cleaning to the equipment depending on the equipment capacity and dry the equipment as per relevant SOP immediately.
- 9.9 Content of solvents shall be less than 1/10th of their respective ICH limit as per ICH Q3C.
- 9.10 Test method to be followed shall be included in the TI sheet.
- 9.11 Based on the results obtained, record the conclusion on findings and further course of action in the recommendation section.
- 9.12 A computerized pre-printed document shall be prepared for verification of cleaning solvent study report according to SOP annexure 1035-A-0072/A23.
- 9.13 Interim report shall be prepared after 1st and 2nd run of study.

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9.14 The verification of cleaning solvent traces study shall be performed on three batches / runs to conclude the study.

10. ACCEPTANCE CRITERIA:

Residue of solvents shall be less than 1/10th of their respective ICH limit as per ICH Q3C.

11. NON-COMPLIANCE:

Any non-compliance observed during study shall be investigated, outcome along with CAPA shall be recorded in the study report.

12. FREQUENCY:

One time study with three runs from each group equipment specified in above procedure.

13. SUMMARY AND CONCLUSION OF STUDY:

Summarize the findings of study to draw an inference of the result in report section and draw the conclusion of study.

14. RECOMMENDATIONS:

Record the recommendation based on the interpretation of the results of the Study Report section.

15. ATTACHMENTS:

Technical information sheet of solvent content checking.

16. TEAM SIGNING:

The personnel who have performed the study, supervised the study, completed the records, performed the testing shall approve the study report section.

17. REVIEW AND COMMENTS:

Quality Assurance shall review the study report and shall include the comments if any in report section.

18. APPROVED BY:

Head Unit Quality Assurance shall finally approve the study report section.

ANNEXURE 1035-A-0072/A22
(SPECIMEN COPY)

Cipla
Location

STUDY PROTOCOL		
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19. PROTOCOL APPROVAL:

Prepared By (Study team) Sign and Date			Reviewed By Sign and Date	Approved By Sign and Date
Quality Assurance	Production	Quality Control	Quality Assurance	Head Unit Quality Assurance

ANNEXURE 1035-A-0072/A23
(SPECIMEN COPY)

Cipla
Location

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1. SITE OF STUDY:

Unit : Unit no. to be recorded.

Department : Unit no. to be recorded.

2. STUDY TEAM:

Department	Name of person
Quality Assurance	Individual Name to be recorded
Production	Individual Name to be recorded
Quality Control	Individual Name to be recorded

3. CONTROLS:

Training details of staff involved in study on SOP 1035-A-0072: Class ID :

4. DESCRIPTION:

Parameters	Runs		
	1 st run	2 nd run	3 rd run
Equipment name			
Equipment Code			
Equipment Capacity			
Equipment from which Group	Group A / Group B / Group C	Group A / Group B / Group C	Group A / Group B / Group C
Equipment previously used for product name			
Equipment previous product batch number			
Date of Cleaning of equipment			
Equipment cleaning done as per SOP with version no			
Solvent used for cleaning			

5. RESULTS AND OBSERVATION:

Runs	Purified water rinse quantity	Rinse to equipment / part name	Rinse number	Content of solvent	Limit of solvent
1 st run					
2 nd run					
3 rd run					

6. NON-COMPLIANCE:

STUDY REPORT		
No.: SR/1030/CLSOLTRA	VERIFICATION OF CLEANING SOLVENT TRACES	Page 2 of 2

Sr. No.	Document No.	Description Non-compliance	Outcome of non-compliance	CAPA

7. SUMMARY AND CONCLUSION (INFERENCE):

Summarise the results and observation observed during study, conclusion shall be recorded.

8. RECOMMENDATION:

Based on overall review and results of study recommendation shall be recorded.

9. ANNEXURE:

Annexure -I: Technical information sheet along with results.

10. STUDY TEAM SIGNING:

Department	Name of person	Sign	Date
Quality Assurance			
Production			
Quality control (Micro)			

11. REVIEW AND COMMENTS:

.....
.....

Quality Assurance sign and date

12. APPROVED BY:

Head Unit Quality Assurance sign and Date

ANNEXURE 1035-A-0072/A24
(SPECIMEN COPY)



OPERATOR INTERVIEW FOR PRODUCT CLEANABILITY EVALUATION

Product name	
Stage	
Operator / Officer name	
Date	

EVALUATION FACTORS

Sr. No.	Factors identified	Answer	Score (yes -1, No-0)
1.	Process creating sticky or caked on solid which is difficult to clean.	Yes (1) / No (0)	
2.	Highly viscous or shear thickening material resistant to cleaning.	Yes (1) / No (0)	
3.	Material reacting with water (cleaning solvent) to form solid or sticky gel-like material.	Yes (1) / No (0)	
4.	Product residue may rust MOC in presence of Water.	Yes (1) / No (0)	
5.	Residue get hardened when left for longer time?	Yes (1) / No (0)	
6.	Total Score		

Based on interview, give the rating for cleanability of product.

Category of cleaning	Score range	Cleanability Rating	Cleanability Rating to product
Easily cleanable	0-1	1	
Moderately Difficult to clean	2-3	2	
Difficult to clean	4-5	3	

Remark:

Evaluation team : Sign and date

Operator / officer	Department Head	R&D / MSTG	Quality assurance

IMPACT ASSESSMENT OF CHANGES AND REVALIDATION CRITERIA

The details of changes with impact assessment and criteria for requirement of Revalidation / verification is given below. Cleaning matrix shall be updated during each change.

1. **Whenever a new product is introduced:** cleaning matrix shall be updated and check new product is a worst product or not.
 - If new product is worst product, cleaning validation to be performed for complete chain of equipment: Three consecutive successful batches / runs.
 - If new product in non-worst product, check the impact on worst case limit / previous validation.
 - If no impact: Previous validation stands valid, cleaning verification to be performed for complete chain of equipment of newly introduced product: One batch verification / run.
 - If impact observed on worst case limits / previous validation: if new limits are stringent than the previous limits, check the cleaning validation result, impact on previous cleaning validation and method validation. Ensure new limits shall be higher than LOQ value of analytical method. In case the new limits are found less than the LOQ of the analytical method, one or more actions specified as per step no 2.20.9 to be initiated to achieve lower limit of detection. Ensure capability of cleaning procedure if needed update the cleaning procedure to achieve new stringent limit.: Three consecutive successful batches / runs cleaning validation to be performed on worst case product.
2. **Addition of equipment:**
 - Higher capacity equipment included in worst product equipment chain: Cleaning matrix to be updated and three consecutive successful batches / run cleaning validation to be performed on worst case product.
 - Lower capacity equipment included in worst product equipment chain.: cleaning matrix to be updated and check the equipment cleaning procedure of equipment, impact on worst case limits and previous validation.
 - If no impact on worst case limit and higher capacity equipment cleaning procedure with same solvent quantity will be used for cleaning of additional equipment: One batch / run cleaning verification to be performed on worst case product.
 - If no impact on worst case limit and equipment will be cleaned with same cleaning procedure with lesser solvent quantity: Three consecutive successful batches / runs cleaning validation to be performed on worst case product.
 - If impact on worst case limit / previous validation: if new limits are stringent than the previous limits, check the cleaning validation result, impact on previous cleaning validation and method validation. Ensure new limits shall be higher than LOQ value of analytical method. In case the new limits are found less than the LOQ of the analytical method, one or more actions specified as per step no 2.20.9 to be initiated to achieve lower limit of detection. Ensure capability of cleaning procedure if needed update the cleaning procedure to achieve new stringent limit.: Three consecutive successful batches / runs cleaning validation to be performed on worst case product
 - Higher capacity equipment included in non-worst product.: Cleaning matrix to be updated and check the impact worst case limits / previous validation.
 - If no impact on worst case limit / previous validation: Higher capacity equipment will be cleaned with same cleaning procedure: No requirement of cleaning validation / verification.
 - If impact on worst case limit / previous validation: if new limits are stringent than the previous limits, check the cleaning validation result, impact on previous cleaning validation and method validation. Ensure new limits shall be higher than LOQ value of analytical method. In case the new limits are found less than the LOQ of the analytical method, one or more actions specified as per step no 2.20.9 to be initiated to achieve lower limit of

IMPACT ASSESSMENT OF CHANGES AND REVALIDATION CRITERIA

detection. Ensure capability of cleaning procedure if needed update the cleaning procedure to achieve new stringent limit.: Three consecutive successful batches / runs cleaning validation to be performed on worst case product.

- Lower capacity equipment included in non-worst product.: Cleaning matrix shall be updated and check the impact worst case limits / previous validation
- If no impact on worst case limit / previous validation: Equipment will be cleaned with same cleaning procedure: No requirement of cleaning validation / verification
- If impact on worst case limit / previous validation: if new limits are stringent than the previous limits, check the cleaning validation result, impact on previous cleaning validation and method validation. Ensure new limits shall be higher than LOQ value of analytical method. In case the new limits are found less than the LOQ of the analytical method, one or more actions specified as per step no 2.20.9 to be initiated to achieve lower limit of detection. Ensure capability of cleaning procedure if needed update the cleaning procedure to achieve new stringent limit.: Three consecutive successful batches / runs cleaning validation to be performed on worst case product.
- Addition of equipment in worst case product, the equipment having same principle, same capacity, same internal designed and same cleaning procedure: Cleaning matrix shall be updated: Cleaning validation / verification not required.
- Addition of equipment in non-worst case product equipment chain, the equipment having same principle having same capacity, same internal designed and same cleaning procedure: Cleaning matrix shall be updated and check the impact worst case limits / previous validation
- If no impact on worst case limit / previous validation: Cleaning validation / verification not required.
- If impact on worst case limit / previous validation: if new limits are stringent than the previous limits, check the cleaning validation result, impact on previous cleaning validation and method validation. Ensure new limits shall be higher than LOQ value of analytical method. In case the new limits are found less than the LOQ of the analytical method, one or more actions specified as per step no 2.20.9 to be initiated to achieve lower limit of detection. Ensure capability of cleaning procedure if needed update the cleaning procedure to achieve new stringent limit.: Three consecutive successful batches / runs cleaning validation to be performed on worst case product.
- *For dedicated equipment: Cleaning validation to be performed on three batches initially followed by cleaning verification on one batch every three years to ensure effective cleaning process.*
- 3. **Addition of New type equipment.:** Perform the evaluation for introduction of new type of equipment, assessment and development of cleaning procedure as per annexure 1035-A-0072/A2. Ensure cleaning procedure is prepared prior to initiation of batch in equipment.
- New type equipment included in equipment chain of worst product: cleaning matrix shall be updated: Three consecutive successful batches / run cleaning validation to be performed on worst case product.
- New type equipment included in non-worst product: cleaning matrix shall be updated and check the availability of equipment in worst case product equipment chain.
- If new type of equipment available in equipment chain of worst-case product: check the impact on worst case limit / previous validation. if new limits are stringent than the previous limits, check the cleaning validation result, impact on previous cleaning validation and method validation. Ensure new limits shall be higher than LOQ value of analytical method. In case the new limits are found less than the LOQ of the analytical method, one or more actions specified as per step no 2.20.9 to be initiated to achieve lower limit of detection. Ensure capability of cleaning procedure if needed update the cleaning procedure to achieve new stringent limit.: Three consecutive successful batches / runs cleaning validation to be performed on worst case product.

IMPACT ASSESSMENT OF CHANGES AND REVALIDATION CRITERIA

- If new type of equipment not available in equipment chain of worst-case product and it is dedicated: Initially cleaning validation to be performed on three batches of non-worst product and then cleaning verification on one batch after every three years to ensure effective cleaning process.
- 4. Removal of equipment:**
 - **Removal of equipment from worst case / non-worst-case product:** Cleaning validation matrix shall be updated. Removal of equipment will not impact limits of cleaning validation / previous validation: No requirement of cleaning validation / verification. Previous validation stands valid.
- 5. Change in batch size:**
 - Increasing Batch size of worst-case product: Cleaning matrix shall be updated, increasing batch size will increase the MACO limit of other solubility group of worst-case products and no impact on increased batch size worst case product: No requirement of cleaning validation / verification.
 - Reducing batch size of worst-case product: Cleaning matrix shall be updated. Reduced batch size will reduce the limits of other solubility group worst case products. Check the impact on worst case limit / previous validation of other solubility group.
 - If no impact on worst case limit / previous validation: No requirement of cleaning validation / verification.
 - if new limits are stringent than the previous limits, check the cleaning validation result, impact on previous cleaning validation and method validation. Ensure new limits shall be higher than LOQ value of analytical method. In case the new limits are found less than the LOQ of the analytical method, one or more actions specified as per step no 2.20.9 to be initiated to achieve lower limit of detection. Ensure capability of cleaning procedure if needed update the cleaning procedure to achieve new stringent limit.: Three consecutive successful batches / runs cleaning validation to be performed on worst case product of other solubility group.
 - Increasing Batch size of non-worst-case product: Increasing batch size will increase the MACO limit of worst-case product: No requirement of cleaning validation / verification.
 - Reducing batch size of non-worst-case product: Cleaning matrix shall be updated. Reduced batch size will reduce the limits worst case products. Check the impact on worst case limit / previous validation.
 - If no impact on worst case limit / previous validation: No requirement of cleaning validation / verification.
 - If new limits are stringent than the previous limits, check the cleaning validation result, impact on previous cleaning validation and method validation. Ensure new limits shall be higher than LOQ value of analytical method. In case the new limits are found less than the LOQ of the analytical method, one or more actions specified as per step no 2.20.9 to be initiated to achieve lower limit of detection. Ensure capability of cleaning procedure if needed update the cleaning procedure to achieve new stringent limit.: Three consecutive successful batches / runs cleaning validation to be performed on worst case product.
- 6. Change in cleaning process:**
 - Change in CPP: will impact the equipment cleaning: Three batches/run cleaning validation to be performed on worst case product.
 - Change in non-CPP: Check the impact on cleaning of equipment and previous validation.
 - If no impact on cleaning of equipment and previous validation: No requirement of cleaning validation / verification.
 - If impact observed on cleaning of equipment and previous validation: Three consecutive successful batches / runs cleaning validation to be performed on worst case product

IMPACT ASSESSMENT OF CHANGES AND REVALIDATION CRITERIA

7. Change in cleaning solvent / quantity:

- Change in cleaning solvent: Cleaning matrix shall be updated and select the worst-case product: check the impact on method validation wrt analytical diluent and cleaning solvent, based on impact assessment analytical method validation to be decided. : Three consecutive successful batches / runs cleaning validation to be performed on worst case product.
- Change in cleaning quantity: will impact the previous cleaning validation: Cleaning matrix shall be updated and three consecutive successful batches / runs cleaning validation to be performed on worst case product.

8. Any change in master data of cleaning matrix (like, surface area of equipment, batch size, Last rinse solvent quantity and analytical method etc.): Cleaning matrix shall be updated and check the impact on worst case limit / previous validation:

- If no impact on worst case limits, cleaning of equipment and previous validation: No requirement of cleaning validation / verification.
- If new limits are stringent than the previous limits, check the cleaning validation result, impact on previous cleaning validation and method validation. Ensure new limits shall be higher than LOQ value of analytical method. In case the new limits are found less than the LOQ of the analytical method, one or more actions specified as per step no 2.20.9 to be initiated to achieve lower limit of detection. Ensure capability of cleaning procedure if needed update the cleaning procedure to achieve new stringent limit.: Three consecutive successful batches / runs cleaning validation to be performed on worst case product.

9. Change in analytical method:

- Change in analytical method w.r.t. LOD, LOQ and recovery: Ensure LOQ value is lower than the MACO limits and lower than the previous LOQ: cleaning validation not required.
- Change in analytical method: Change TLC to HPLC or UV to HPLC: Check the impact on MACO limits. Ensure revised LOQ value of analytical method is lower than the MACO limit. In case the revised LOQ is higher than the MACO limit, one or more actions specified as per step no 2.20.9 to be initiated to achieve higher MACO limit or lower LOQ value of analytical method: Cleaning validation not required
- If change in analytical diluent: impact assessment to be done w.r.t. recovery of previous and proposed analytical diluent: Cleaning validation not required.
- If MACO limit has been revised based on change, the LOQ value shall be at least 50% of revised limit. If LOQ is higher, develop and revalidate the analytical method. Cleaning validation not required.